QUALITY MANAGEMENT

Easy-to-understand study notes with every class problem solved step by step

What these notes cover	
Class slides	Day 1 (12 Sep) · Day 2 (13 Sep) · Day 3 (27 Sep) · Day 4 (11 Oct) · Day 5 (17 Oct) · Day 6 (18 Oct) · Day 7 (25 Oct)
Textbook chapters	Ch 1 Introduction to Quality · Ch 2 Foundations of Quality Management · Ch 3 Customer Focus · Ch 7 Design for Quality and Product Excellence · Ch 8 Measuring and Controlling Quality · Ch 9 Process Improvement and Six Sigma
Cases	Aravind Eye Hospital · Mumbai Dabbawalas · iD Fresh Food · Zara · Flyrock Tires · Gati

How each topic is laid out	
In simple words	A plain-language explanation at the start of every topic — read this before the details.
Everyday example	A familiar real-life example of the same idea (shops, buses, exams, hostels…) so the concept sticks.
Purple problem boxes	Every numerical problem / concept check from the slides (and the textbook's worked examples), solved step by step in the professor's method.
Answer	The final answer and what it means.
Exam tip	Shortcuts, common traps and rules of thumb.
Careful	Places where a slide has a typo or arithmetic slip — the correct working is shown so you aren't confused in the exam.

All numbers in the worked problems were re-computed. Excel functions used: NORM.DIST, NORM.S.DIST, NORM.S.INV, BINOM.DIST. Notation: 'x-bar' = sample mean, 'x-double-bar' = mean of sample means, 'R-bar', 'p-bar', 'c-bar', 'u-bar', 's-bar' = averages of those statistics.

Contents

1. What is Quality? Why does it matter? 5

1.1 The professor's opening questions 5
1.2 Six ways to define quality (Textbook Ch 1) 6
1.3 History of quality management (Textbook Ch 1) 6
1.4 Evolution of quality management (Slides Day 2) 7
1.5 Quality in manufacturing, services and support functions (Textbook Ch 1) 8
1.6 Quality, competitive advantage and business results 9

2. Quality as Strategy: Aravind, Wastage and Variation 10

2.1 Case: Aravind Eye Hospital 10
2.2 Identification of wastages — the A → B → C problem 11
2.3 Understanding process variation (Day 2 + Day 5) 12

3. The Quality Gurus: Deming, Shewhart, Juran, Crosby & others 14

3.1 W. Edwards Deming 14
3.2 Walter Shewhart and the PDCA cycle 19
3.3 Joseph M. Juran 20
3.4 Philip B. Crosby 23
3.5 Other quality philosophers (Textbook Ch 2) 24
3.6 Comparing Deming, Juran and Crosby 24
3.7 Principles, practices and techniques; ISO 9000 (Textbook Ch 2) 25

4. Translating Philosophy into Practice: Case Studies 27

4.1 Mumbai Dabbawalas 27
4.2 iD Fresh Food 27
4.3 Zara — impact of customer focus (Day 4) 28

5. Customer Focus — Hearing the Voice of the Customer 29

5.1 Customer satisfaction and engagement 29
5.2 Who is the customer? 29
5.3 Quality dimensions of goods and services 30
5.4 The Kano model 31
5.5 Gathering the voice of the customer (VOC) — Requirements Engineering 32
5.6 Building a customer-focused organisation 33
5.7 Measuring customer satisfaction, loyalty and value 33

6. Design for Quality: Lifecycle, QFD, Taguchi & Reliability 36

6.1 Developing a comprehensive quality system — lifecycle thinking 36
6.2 Design thinking — the story of Doug Dietz (GE Adventure Series) 36
6.3 Product development process (Textbook Ch 7) 37
6.4 Quality Function Deployment (QFD) & the House of Quality 37
6.5 Target and tolerance design 39
6.6 The Taguchi loss function 39

6.7 Design for reliability . . . 42
6.8 Design optimisation, FMEA, FTA, manufacturability, environment, verification . . . 44
7. Process Quality and the Six Sigma Idea . . . 47
7.1 Concept check: Sam vs John (accuracy vs precision) . . . 47
7.2 Process management basics (Day 4) . . . 47
7.3 The normal distribution — the language of variation . . . 48
7.4 Where do the 'magic numbers' come from? (3.4 DPMO) . . . 49
7.5 Six Sigma philosophy and critical elements . . . 50
8. Measuring Defects and Sigma Levels — all class problems solved . . . 52
8.1 The key formulas . . . 52
8.2 Solved problems . . . 53
8.3 Other quality measures (Textbook Ch 8) . . . 55
8.4 Cost of quality measurement (Textbook Ch 8) . . . 55
9. The Six Sigma Method: DMAIC / DMADV, Lean and Service Blueprints . . . 56
9.1 When do you start a Six Sigma project? . . . 56
9.2 DMAIC vs DMADV . . . 56
9.3 DEFINE . . . 57
9.4 MEASURE . . . 58
9.5 ANALYSE — probing for root causes . . . 58
9.6 Service blueprinting . . . 59
9.7 IMPROVE . . . 60
9.8 CONTROL . . . 61
9.9 More from the textbook (Chapter 9) . . . 61
10. Process Capability: Cp, Cpk and the Flyrock Tires Case . . . 64
10.1 Voice of the process vs voice of the customer . . . 64
10.2 Capability indices . . . 65
10.3 Types of processes . . . 67
10.4 More from the textbook: capability studies, performance indices and pre-control . . . 68
10.5 Case: Six Sigma Quality at Flyrock Tires (Day 6, slides 7–16) . . . 68
11. Measurement System Evaluation (Textbook Chapter 8) . . . 72
12. Statistical Process Control — Foundations . . . 74
12.1 Ensuring a process continuously meets requirements . . . 74
12.2 Statistics you need (Day 5 slides 63–71, Textbook Ch 6) . . . 74
12.3 Control limits . . . 75
12.4 Which control chart? (depends on the type of data) . . . 76
13. Attribute Control Charts: p, np, c and u — every class example solved . . . 77
13.1 The p-chart (fraction defective) . . . 77
13.2 Control chart fail points: rational subgrouping . . . 81
13.3 The np-chart (number defective) . . . 83

13.4 The c-chart (defects per unit) . . . 84
13.5 The u-chart (defects per unit, varying size) . . . 84
13.6 Reading control charts: zones and out-of-control rules . . . 85
14. Variable Control Charts: x-bar & R, x-bar & s, Individuals & MR . . . 87
14.1 The x-bar and R charts . . . 88
14.2 The x-bar and s charts . . . 89
14.3 Individuals and moving-range (I-MR) charts . . . 90
14.4 Special cases: periodic vs persistent disturbances . . . 91
14.5 Types of chart patterns and the Preston caselet . . . 93
15. The 7 QC Tools, 5W2H and the Gati Case . . . 95
15.1 Flow chart and check sheet . . . 95
15.2 Run chart and histogram . . . 96
15.3 Scatter diagram . . . 96
15.4 Pareto charts . . . 97
15.5 Cause-and-effect (fishbone / Ishikawa) diagrams . . . 98
15.6 5W2H and the is–is not matrix . . . 100
15.7 Case: Gati — achieving quality excellence in service delivery (Day 7) . . . 101
16. Acceptance Sampling and OC Curves . . . 103
16.1 Types of sampling plans . . . 103
16.2 The Operating Characteristic (OC) curve . . . 103
16.3 Building an OC curve and the buyer's risk . . . 104
17. One-page Formula Sheet & Quick Revision . . . 106
Quick revision: one-line answers to the concept checks . . . 106

PART A · FOUNDATIONS

1. What is Quality? Why does it matter?

Slides Day 1 (12 Sep) · Textbook Chapter 1 — Introduction to Quality

In simple words
Quality is not one fixed thing — it depends on **who** is looking. A customer says 'it feels premium', an engineer says 'it meets the drawing', an accountant says 'it gives value for money'. Quality management is the discipline of making all of these agree, so that the organisation delivers what customers want, every single time, without waste.

1.1 The professor's opening questions

In simple words
The course starts by challenging an old idea: that quality is just an expense (paying inspectors to catch bad products). The professor's view is that quality makes money — it cuts waste and wins customers — and that it only works when it becomes everyone's way of thinking, not just a set of tools.

Everyday example
A restaurant that throws away 10% of its food because of wrong orders is losing money. If it fixes its ordering process, food cost falls *and* customers come back. The fix (quality) paid for itself — it acted like a profit centre.

- **Is quality a cost centre or a profit centre?** Traditionally seen as a cost (inspectors, testing). The course argues it is a **profit centre**: removing defects and waste cuts cost and raises revenue (Aravind, iD Fresh, Dabbawalas all prove this).
- **How do you compete (excel) through quality?** By building quality into the *process* and the *culture*, not just inspecting the product.
- **Tool, technique or philosophy?** All three. Tools (charts, diagrams) and techniques (SPC, Six Sigma) only work when quality is a philosophy shared by everyone.

Concept check (Slido, Day 1) [Day 1, slide 5]

Two similar products: Product A carries the **ISI mark**; Product B is made by a company with **ISO certification**. Which product has better quality?

Know ISI	ISI mark (Bureau of Indian Standards) is a **product certification**: the product itself has been tested and conforms to a specific Indian Standard.
Know ISO	ISO 9001 is a **management-system (process) certification**: the company has a documented, audited quality system. It does **not** certify that a particular product is good.
Compare	ISI says something direct about the product; ISO says the company's processes are organised and controlled. Neither alone guarantees the 'best' product.

Answer: You cannot say outright that B is better. ISI = **product quality** assurance; ISO = **process/system quality** assurance. The lesson the professor draws: distinguish **product quality vs process quality** — good processes make good products consistently, but a certificate is not the same as quality.

1.2 Six ways to define quality (Textbook Ch 1)

In simple words
There is no single definition of quality. It depends on who is judging: the customer's feeling of excellence, the number of features, fitness for the job, value for money, matching the design drawing, or meeting the customer's expectations. Good companies satisfy all these viewpoints together.

Everyday example
A mobile phone: a buyer says 'iPhone feels premium' (transcendent); 'it has 3 cameras' (product); 'it lasts all day for my work calls' (user); 'great phone for ₹15,000' (value); the factory checks 'screen gap ≤ 0.2 mm' (manufacturing); the company asks 'did it meet or beat what buyers expected?' (customer).

Perspective	What 'quality' means	Everyday example	Who uses it most
Transcendent (judgmental)	Excellence you 'just know when you see it' — absolute, hard to measure	Rolex, Ritz-Carlton	Customers / marketing
Product-based	Quantity of some attribute (more = better)	Thread count of bedsheets	Customers
User-based	**Fitness for intended use** (Juran)	A small car is 'quality' for city driving	Marketing
Value-based	Benefits relative to price	Generic medicines	Design / product planning
Manufacturing-based	**Conformance to specifications** (target ± tolerance)	Part = 0.236 ± 0.003 cm	Production
Customer-based	**Meeting or exceeding customer expectations**	Hotel guest delighted	Whole value chain

- **Specifications** = target (nominal value) + tolerance (allowed deviation). E.g. 0.236 ± 0.003 cm → anything 0.233–0.239 conforms.
- **Consumers** are final users; **external customers** are other firms in the chain; **internal customers** are the next department / person who receives your work. Every employee has a customer.
- In the value chain (Fig 1.1): customers think transcendent/product; marketing thinks user; design thinks value; manufacturing thinks conformance; the customer perspective ties them together.

Another way to see it
Think of a pizza. The customer says 'tasty and hot' (transcendent/user). The chef's recipe says '200 g cheese, 12-inch base' (manufacturing). The owner asks 'is it worth ₹400?' (value). All are 'quality' — just from different seats.

1.3 History of quality management (Textbook Ch 1)

In simple words
Quality moved from the craftsman checking his own work, to factories hiring armies of inspectors, to using statistics to control processes, to making quality every manager's job (TQM), and finally to data-driven projects with financial targets (Six Sigma).

Everyday example
Think of a tailor. Long ago one tailor measured, cut, stitched and checked every shirt himself (craftsmanship). A garment factory later split the work and put inspectors at the end (inspection era). Today the factory measures stitch length during sewing (SPC) and runs projects to cut defects to near zero (Six Sigma).

Era	What happened	Key idea
Ancient	Egyptian pyramids (precise measurement); China's Zhou Dynasty had quality departments & banned sub-standard goods	Standards + inspection
Craftsmanship (Middle Ages)	Craftsman was maker and inspector; guilds trained apprentices	Pride in workmanship
Industrial Revolution	Honoré Blanc / Eli Whitney — interchangeable parts (10,000 muskets)	Parts must meet a standard; variation matters
Early 1900s	F. W. Taylor separated planning from doing → quality became the inspector's job; Henry Ford's practices; Bell System/Western Electric	Mass inspection
1920s–40s	Shewhart (control charts), Dodge, Edwards, Juran, Deming at Bell Labs; WWII military sampling (MIL-STD); ASQ formed	Statistical quality control (SQC); quality assurance
Post-WWII	Deming & Juran teach Japan's top managers; Deming Prize (1951); kaizen	Quality led by top management
1980s USA 'quality revolution'	NBC 'If Japan can… why can't we?' (1980); Xerox, Motorola; Baldrige Award (1987)	Quality = competitiveness
1990s	Total Quality Management (TQM); early failures due to poor implementation	'Big Q' (whole organisation) vs 'little q' (manufacturing)
Late 1990s →	Six Sigma (Motorola, GE); Lean Six Sigma; performance excellence; globalisation	Data-driven, bottom-line results

- **Quality assurance**: any planned, systematic activity giving confidence that products meet requirements.
- **Quality control**: evaluating a process to decide whether corrective action is needed (often via inspection/measurement).
- **Total Quality (TQ)**: a people-focused management system aiming at continually increasing customer satisfaction at continually lower real cost; involves everyone, every function, and the supply chain. Feigenbaum coined 'total quality control'.
- **Performance excellence**: an integrated approach that delivers ever-improving value to customers, improves organisational capability, and creates learning.
- **Six Sigma**: customer-focused, results-oriented approach using proven quality tools with a bottom-line focus (Chapter 9).

1.4 Evolution of quality management (Slides Day 2)

In simple words
Each stage builds on the previous one: first you **find** bad items (inspection), then you **control** the process with data (QC), then you build a **system** that prevents problems (QA), then you involve **everyone and every function** (TQM), and finally you run **measured improvement projects** (Six Sigma).

Everyday example
A bakery: Stage 1 — throw away burnt bread at the end. Stage 2 — record oven temperatures. Stage 3 — written recipes, trained bakers, FMEA on ovens. Stage 4 — suppliers, delivery staff and cashiers all part of quality. Stage 5 — a project that brings burnt loaves from 3% to 3 per million.

Stage	What it involves
1. Inspection	Salvage, sorting, grading, blending, corrective actions, identify sources of non-conformance (finding bad items after they are made)
2. Quality Control	Quality manual, process performance data, self-inspection, product testing, basic quality planning, basic statistics, paperwork control

Stage	What it involves
3. Quality Assurance	Quality systems, advanced quality planning, comprehensive manuals, quality costs, involvement of non-production operations, FMEA, SPC
4. TQM	Policy deployment, involve suppliers & customers, involve all operations, process management, performance measurement, teamwork, employee involvement
5. Six Sigma	Data-driven, project-based defect reduction to 3.4 defects per million

Quality paradigms (slide): Quality Control → Creation of Quality Systems (audits, certification: ISO 9000, GMP, ISO 14000, ISO 22000) → Total Quality Management (TQM/TQC) → Programs of quality improvement (Six Sigma) → organisation's success.

1.5 Quality in manufacturing, services and support functions (Textbook Ch 1)

In simple words

Every department affects quality, not just the factory floor. Services are harder to control because they are intangible, made and consumed at the same moment, and depend heavily on how employees behave. Even back-office work (billing, HR) has customers and can have 'defects'.

Everyday example

Manufacturing: a refrigerator can be tested before it leaves the factory. Service: a haircut cannot — if the barber cuts it badly, the customer is already sitting there. Support function: a salary slip with a wrong deduction is a quality defect in HR's process.

Manufacturing system (Fig 1.2) — every function has a quality role: marketing & sales (learn what customers want), product design & engineering (avoid under- and over-engineering), purchasing & receiving (good suppliers), production planning, manufacturing & assembly, tool engineering, industrial engineering & process design, finished-goods inspection & testing, packaging/shipping/warehousing, installation & service.

Services differ from manufacturing:

- Customer needs and performance standards are harder to identify and measure (customer defines quality).
- Customisation is higher; output is intangible (cannot be stored or inspected beforehand).
- Produced and consumed at the same time, often with the customer present (no chance to 'fix it before shipping').
- Customers often take part in the process; high volume of transactions; labour-intensive (employee behaviour = quality).

Components of service quality: the *human* component (behaviour of front-line staff) and the *technical* component (systems, equipment). Key service dimensions: time, timeliness, completeness, courtesy, consistency, accessibility & convenience, accuracy, responsiveness.

Business support functions (finance, HR, IT, legal) also have internal customers and processes — billing errors are quality problems too. **The quality function** has shifted from 'policing' (inspecting) to supporting, training and coaching everyone else.

1.6 Quality, competitive advantage and business results

In simple words
Better quality helps a firm win in two ways: customers are willing to pay more and buy more (revenue goes up), and there is less scrap, rework and warranty cost (costs go down). Poor quality can destroy years of reputation very quickly.

Everyday example
Maruti built a reputation for reliable, low-maintenance cars, so customers trust the brand and resale value is high (revenue side), while fewer warranty repairs keep costs low (cost side). In contrast, a single big recall can wipe out trust overnight.

- **Competitive advantage** = a firm's ability to achieve market superiority. Quality supports both *differentiation* (better product) and *low cost* (less waste).
- Two routes to profit (Fig 1.3 in book): better **design quality** → higher perceived value → higher price & market share (revenue up); better **conformance quality** → less scrap, rework, warranty → lower cost.
- Evidence: PIMS studies, Baldrige award winners outperforming the stock market, J.D. Power ratings, etc. Quality cannot guarantee success, but poor quality can destroy it (e.g., Toyota's 2009 recalls cost $2 bn and its reputation).
- **Personal quality**: quality starts with individuals — personal values, doing one's own job right.

PART A · FOUNDATIONS

2. Quality as Strategy: Aravind, Wastage and Variation

Slides Day 1 (Aravind Eye Hospital case, identification of wastages) · Day 2 (process variation)

2.1 Case: Aravind Eye Hospital

In simple words
Aravind shows that a well-designed process can deliver high quality, low cost and high volume at the same time. By removing waste from every step, each doctor can do far more surgeries without compromising outcomes, so the hospital can treat poor patients free and still make a profit.

Everyday example
Like a well-run car service centre where the mechanic only does the skilled repair while helpers handle paperwork, washing and parts — the mechanic finishes many more cars a day. At Aravind, nurses and counsellors do the preparation so surgeons spend their time only on surgery.

Another way to see it
Aravind treats huge numbers of poor patients — more than half free — yet earns profit margins higher than corporate hospitals. The secret is not 'better doctors' but a superior **operations model** that removes waste, so each doctor does many more surgeries at consistent quality.

- **The problem they solve:** bringing eyesight to the masses. Backlog ≈ 20 million cataract cases, ≈ 2 million new cases every year, India's capacity ≈ 1.2 million surgeries/year — an 'infinite queue' (the queue grows every year).
- **Why it is hard:** affordability and availability of resources (doctors, beds).
- **Aravind model:** free care alongside paid care (more than 50% free), assembly-line style surgery, standardised processes, paramedical staff do non-surgical tasks, high volume → low cost per surgery.
- **Result:** despite >50% free and below-market prices, profit margins above 50% — compare Apollo (EBITDA ~17%, net 7.6–9.1%), Fortis (~18.4% / 10.5%), Max (~23.5% / 14–15%).

Worked example: productivity of doctors (Day 1, slide 17) [Day 1]

National: 8,000 eye surgeons perform 1.2 million surgeries a year. Aravind: 30 doctors performed 31,167 surgeries. Compare productivity and find how long India would take to clear the backlog at Aravind's efficiency.

National rate	1,200,000 ÷ 8,000 = **150 surgeries per doctor per year**.
Aravind rate	31,167 ÷ 30 ≈ 1,039 → roughly **1,000 surgeries per doctor per year**.
Ratio	1,000 ÷ 150 ≈ 6.7 → about **7 times** the national average.
Capacity at Aravind efficiency	8,000 doctors × 1,000 = **8 million surgeries/year**.
Years to clear backlog	Backlog 20 million; every year 2 million new cases arrive, so the backlog falls by only (8 − 2) = 6 million/year. Years = 20 ÷ (8 − 2) = 3.3 → **less than 4 years**.

Answer: Aravind doctors are ~7× more productive; at that efficiency India could eliminate the cataract backlog in under 4 years. The reason is a superior process (waste removed), not superior doctors.

Key takeaways

Excellence comes from **identifying wastage and removing it**, and embedding this in the **culture** — that is Quality Management.

Pyramid of excellence: **Technology & technical systems** → remove wastages; **Purpose & behaviours** → culture.

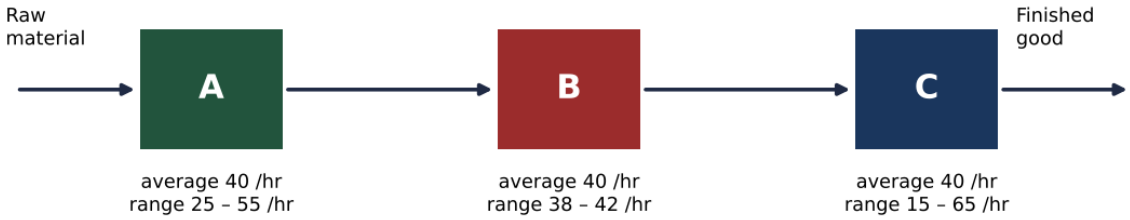
2.2 Identification of wastages — the A → B → C problem

In simple words

Average capacity is misleading when each stage varies. In a chain, the slowest stage at any moment decides the output, and time lost when one stage waits for another can never be recovered. So variation, not average capacity, is the real cause of waste.

Everyday example

Three friends pass bricks in a line, each averaging 40 bricks/hour. If the first friend is sometimes slow, the second stands idle; if the third is slow, the second has to stop. By evening they have moved fewer bricks than 40 × hours, even though each 'averages' 40.



Problem (Day 1, slides 20–21) [Day 1]

Raw material passes through three stages A → B → C. Each stage has an **average** capacity of 40 units/hour. In an 8-hour day at least 300 units must be processed. Are the capacities enough? Will your answer change if the hourly capacities vary (A: 25–55, B: 38–42, C: 15–65, all averaging 40)?

Average view	Capacity per day = 40 units/hr × 8 hr = **320 units ≥ 300** needed → looks sufficient (6.7% spare).
Add variability	Each stage only averages 40. In any hour A might make 25, C only 15. A line can only move as fast as its slowest stage *at that moment*. If B is idle because A was slow (**starving**), or A has to stop because C is slow (**blocking**), that lost time is **never recovered**.
Rough estimate	Without buffers, hourly output ≈ min(A, B, C). A quick simulation with the given ranges gives roughly 250–275 units/day — **below 300**.
Conclusion	Average capacity is not enough. Managing a process is hard because **synchronising flows** in a multi-stage system is very difficult when there is variability.

Answer: On averages, yes (320 ≥ 300). With variability, no — the line will fall short because of starving/blocking. The real enemy is **variation**.

Classical solutions vs the alternative

In simple words
The traditional fix for variation is to keep extra stock or extra capacity 'just in case'. That works but is expensive. The better fix is to reduce the variation itself — then you need much smaller buffers for the same level of service.

Everyday example
A household that never knows when the water tanker will come keeps 10 buckets filled (buffer). If water came at a fixed time every day (low variation), 2 buckets would do.

- **Classical solutions:** inventory/time buffers (stock between stages) and capacity buffers (extra machines/people).
- **Effect:** huge inventory and wastage → **organisational inefficiency**.
- 'Inventory is the only insurance against uncertainty' — e.g., keep 140 units so you rarely enter the **danger zone (stock-out)**.
- **Exploding the myth:** can better customer service be obtained with *lower* buffers? Yes — if you **reduce variability**. With less variation, a buffer of 120 (or 100) gives the same protection as 140 did before.

Key takeaways
Underlying principle: wastages are created by (1) process imperfection and (2) lack of synchronisation — both are caused by **PROCESS VARIATION**. So the whole of quality management is about understanding and reducing variation.

2.3 Understanding process variation (Day 2 + Day 5)

In simple words
Everything varies. **Common-cause** variation is the normal 'noise' built into the system; only changing the system reduces it. **Special-cause** variation comes from something unusual that can be identified and removed. The first job is always to tell the two apart.

Everyday example
Your commute takes 35–45 minutes every day because of normal traffic lights and traffic (common cause). One day it takes 90 minutes because of an accident on the road (special cause). Leaving earlier or changing route (system change) helps the first; the second needs no permanent change.

Type	Also called	Example (burger shop)	How to handle
Controllable	**Assignable / special cause**	A poorly trained chef makes uneven burgers	Find the specific cause and remove it (train the chef)
Inherent	**Common / random / chance cause**	Natural variation in burger quality; random arrival of customers	Built into the system — only management can change the system

- The **first step** is to decide whether variability is due to assignable causes or not; then whether the causes are of common or special nature.
- Assignable variation is controlled by addressing the factors that caused it. Common causes are addressed by looking at their severity relative to customer requirements (capability).
- Controlling variation = **Measuring** (quantitative methods) → **Improving** (system improvement models) → **Correction** and **Prevention** (quality philosophies).

Textbook Ch 2 — sources of variation: operators, materials, machines/methods, measurement instruments, human inspection performance, environment. Deming estimated that ~85–94% of

problems are due to common causes (the system), which only management can fix.

> **Exam tip**
> Two mistakes are possible: (1) treating a common-cause variation as special (tampering — reacting to noise, which makes things worse); (2) treating a special cause as common (missing a real problem). Control charts help avoid both.

PART A · FOUNDATIONS

3. The Quality Gurus: Deming, Shewhart, Juran, Crosby & others

Slides Day 2 (13 Sep) & Day 3 (27 Sep) · Textbook Chapter 2 — Foundations of Quality Management

In simple words
Five people shaped modern quality thinking. **Shewhart** invented the control chart and the Plan-Do-Check-Act loop. **Deming** said 'the system, not the worker, causes most problems — so management must fix the system'. **Juran** said 'quality is fitness for use — plan it, control it, improve it, and talk to managers in money'. **Crosby** said 'quality is free — aim for zero defects'. **Ishikawa** gave everyone simple tools like the fishbone diagram.

3.1 W. Edwards Deming

In simple words
Deming taught that most problems come from the system that management designs, not from careless workers. Improve the system and you get better quality, lower cost, more customers and more jobs — a chain reaction.

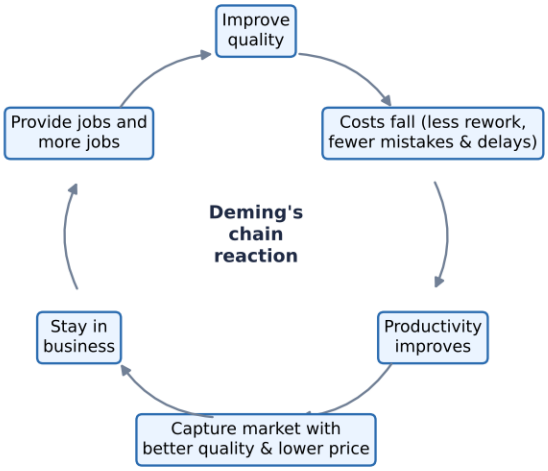
Everyday example
If a call centre's software keeps freezing, agents will make customers wait no matter how hard they try. Scolding agents won't help; fixing the software (management's job) will.

Deming defined quality as **continual improvement of a stable system**. A product or service possesses quality if it helps somebody and enjoys a good and sustainable market. He held that most quality problems (**about 90% or more**) are caused by poor **systems**, not by workers — so **management is responsible**.

Deming's chain reaction

In simple words
Improving quality starts a cycle: fewer mistakes mean lower cost, lower cost raises productivity, better quality at a lower price wins the market, which keeps the business alive and creates jobs.

Everyday example
A sweet shop that stops burning batches saves ingredients (cost ↓), makes more sellable sweets per hour (productivity ↑), can price lower with better taste (market ↑), and hires more staff for its new branch (jobs ↑).



Improve quality → costs decrease (less rework, fewer mistakes, delays, snags; better use of machine time and materials) → productivity improves → capture the market with better quality and lower price → stay in business → provide jobs and more jobs. The question is: **how** to improve? → the 14 points.

Deming's 14 Points (Textbook Table 2.1)

In simple words
The 14 points are Deming's instructions for managers. In short: have a long-term purpose, build quality in instead of inspecting it, choose suppliers for quality, keep improving, train and lead people, remove fear and barriers, drop slogans and quotas, and make the change everyone's job.

Everyday example
Point 11 (no numerical quotas): if a bank teller must open 20 accounts a day, she may skip KYC checks to hit the number. Remove the quota, train her, and give her a simpler process — accounts are opened correctly.

#	Point	What it means in plain words
1	Create and publish the aims and purposes of the organisation (constancy of purpose)	Have a long-term vision; don't chase only next quarter's numbers
2	Learn the new philosophy	Everyone, top management first, must adopt quality thinking
3	Understand the purpose of inspection	Stop depending on mass inspection; build quality in
4	Stop awarding business based on price alone	Choose suppliers on quality & long-term relationship (fewer, better suppliers)
5	Improve constantly and forever the system	Continuous improvement (kaizen) of every process
6	Institute training	Train people properly for their jobs
7	Teach and institute leadership	Supervisors should help people do a better job, not police them
8	Drive out fear, create trust, create a climate for innovation	People must feel safe to report problems and suggest ideas
9	Optimise the efforts of teams, groups and staff areas	Break down barriers between departments
10	Eliminate exhortations for the workforce	Slogans/posters ('Zero defects!') don't change a bad system
11a	Eliminate numerical quotas for the workforce	Quotas push quantity over quality

#	Point	What it means in plain words
11b	Eliminate management by objective (numerical goals)	Manage the process, not just targets
12	Remove barriers that rob people of pride of workmanship	e.g., annual merit ratings, poor tools, bad materials
13	Encourage education and self-improvement for everyone	Lifelong learning
14	Take action to accomplish the transformation	Everyone's job — make it happen

Exam tip

The 14 points must be applied **as a system** — you can't pick and choose (the slide's '15th principle').

Why are they not fully practised? They require a deep cultural change, long-term commitment, giving up quotas/MBO/appraisals, and top-management patience; results are slow and hard to attribute.

Deming and the myths of management (Day 2, slide 13)

In simple words

These are common management beliefs that Deming said were false — for example, that workers cause most problems or that fire-fighting improves an organisation. Believing them leads managers to blame people instead of fixing systems.

Everyday example

Myth 'optimising the parts optimises the whole': the purchase department saves money by buying cheaper cloth, but the stitching department now has more rejects and customers return shirts — the company as a whole loses.

Deming called all of the following **myths** (false beliefs): most quality problems are caused by workers · rewards and punishments are the most effective motivators · results are achieved by setting objectives and stating them clearly · quality is inversely related to quantity · rational decisions can be made on 'gut feel' · organisations can be improved by firefighting · competition is a necessary aspect of business · optimising the parts leads to optimisation of the whole.

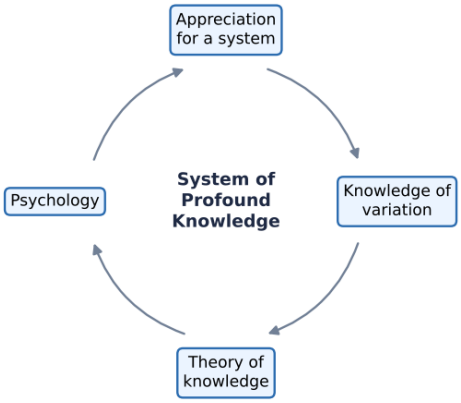
System of Profound Knowledge (SoPK)

In simple words

Deming's four lenses for managing: see the organisation as one system; understand variation; learn by predicting and testing (theory of knowledge); and understand what motivates people (psychology).

Everyday example

A school wanting better results: *system* — teachers, timetable and parents all interact; *variation* — one bad test score isn't a trend; *knowledge* — try a new teaching method in one class first and measure; *psychology* — students learn better when curious than when afraid of punishment.



Element	Meaning	Implication (slide)
Appreciation for a system	An organisation is a system of interdependent parts with an aim. Optimising one part can hurt the whole (sub-optimisation).	**Optimise the entire system, not just parts**
Knowledge of variation	All processes vary. Distinguish common causes (system) from special causes. Reacting to common-cause variation as if special = tampering.	**Manage using process orientation, not just result orientation**
Theory of knowledge	Management is prediction; learning needs theory + testing (PDSA). Copying an example without understanding the theory is risky.	Use data and experiments to learn
Psychology	People differ; intrinsic motivation (joy in work) matters; extrinsic rewards/ranking can destroy it.	**People are inspired by intrinsic motivation; replace competition by collaboration**

Red Bead experiment

In simple words
The red bead game proves that when defects come from the system (the box of beads), workers' results differ only by chance. Rewarding or punishing them changes nothing; only management changing the system (better raw material) reduces defects.

Everyday example
Students scoop 50 beads from a box with 20% red beads. One gets 4 red, another 14 red — pure luck. Praising the first and warning the second is like blaming a driver for potholes on a road the municipality hasn't repaired.

A box contains white beads (good) and red beads (defects, ~20%). 'Willing workers' dip a paddle with 50 holes; the foreman rewards those with few red beads and punishes those with many, posts slogans and quotas. The number of red beads varies purely by chance.

Key takeaways
Variation is part of the **system**; workers cannot do better than the system allows.

Rewarding/punishing workers for system variation is meaningless and demoralising; slogans and quotas don't help.

Only management can change the system (e.g., better raw material = fewer red beads).

Slide picture (eat an elephant): big problems are solved one bite at a time — by improving the system step by step.

Later (Section 10) we draw a p-chart of the class red-bead data and confirm the process is **stable** — all variation is common-cause.

Funnel experiment

In simple words
If a process is stable, adjusting it after every result (reacting to random noise) makes things worse. This over-adjusting is called **tampering**. Leave a stable process alone and improve the system instead.

Everyday example
If your bathroom scale shows 70.2 kg one morning and 69.8 the next, and you change your diet every day based on that tiny change, your weight becomes more erratic, not less.

A marble is dropped through a funnel onto a target. Where it lands varies. Four 'rules' for adjusting the funnel were tested (simulated below with 300 drops each):



Rule	What you do after each drop	Result	Real-life example
1	Leave the funnel over the target	**Smallest variation** — stable	Don't tamper with a stable process
2	Move funnel from its current spot to compensate the last error	Variance doubles (spread ≈ 1.4×)	Operator adjusting a machine after every part
3	Set funnel at the mirror image of the last error, measured from the target	Oscillates and explodes — unstable	Over-reacting budgets, over-correcting
4	Set funnel over the last landing spot	Random walk — drifts away	Worker training worker; copying the previous copy

Answer: Slido question: the chance of hitting the target is highest (variation lowest) under **Rule 1 — no adjustment**. Learning: adjusting a stable process in response to common-cause variation (tampering) **increases** variation.

Human psychology experiment (Day 2, slides 23–28)

In simple words
In the points game, each person gains a little by keeping their points, but if everyone does that, the whole group ends up much worse off. Organisations full of departments optimising their own targets behave the same way.

Everyday example
A housing society: each family saves money by not paying for common-area cleaning, but then the stairs are dirty for everyone. If all pay a little, everyone gets a clean building.

The 'common pool' game [Day 2]

Each participant has 2 points. You may keep both, contribute 1, or contribute both to a common pool. **Your pay-off = points you keep + ½ × (total points in the common pool)**. Objective: maximise your pay-off. What will you do?

Your own view	Every point you contribute costs you 1 point but returns only ½ point to you (the other ½ is shared). So for you alone, keeping is better: contributing a point changes your pay-off by −1 + ½ = **−½**.
Everyone keeps	Each person ends with 2 + ½ × 0 = **2 points**.
Everyone contributes	With N people, pool = 2N, so each gets 0 + ½ × 2N = **N points** (e.g., 60 students → 60 points each, 30× better).
Insight	What is best for 'me' individually (keep) leads to the worst outcome for everyone. This is the logic of silos.

Answer: The best strategy for (you) and for (others) separately is NOT the best strategy overall. Organisations must go beyond 'silo' thinking — **replace competition with collaboration** (win/win: 'not your way or my way, but the best way'). This is Deming's psychology + systems element.

Summary: Deming's contribution (slide)

In simple words
Deming's legacy in one line: build a stable system, keep improving it, understand variation, and remember that management owns the system.

Everyday example
When a train runs late every day, the fix is the timetable and signals (system), not shouting at the driver.

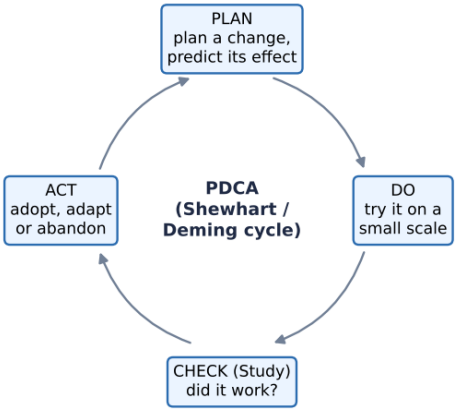
- Quality = continual improvement of a stable system.
- Most problems (≈90%) are system problems → management's responsibility.
- Common vs special causes; variation and consistency.
- Chain reaction; 'quality is a never-ending process'.
- 14 points; System of Profound Knowledge.

3.2 Walter Shewhart and the PDCA cycle

In simple words
PDCA is a simple learning loop: plan a small change, try it, check whether it worked, then adopt it or drop it — and repeat. Shewhart also showed that every process varies, which led to control charts.

Everyday example
A tea stall wants shorter queues. **Plan**: pre-boil milk at 8 am. **Do**: try it for a week. **Check**: average wait fell from 6 to 3 minutes. **Act**: make it permanent, then plan the next change (pre-cut ginger).

Shewhart (Bell Labs, 1920s) — **control theory: any process will display variation**. He invented the control chart and the Plan-Do-Check-Act cycle (Deming later called it Plan-Do-**Study**-Act). The slide calls it the **FOCUS-PDCA** model.



Stage	What you do
PLAN	Plan a change to the process. Predict the effect it will have and plan how the effects will be measured.
DO	Implement the change on a small scale and measure the effects.
CHECK / STUDY	Study the results to learn what effect the change had, if any.
ACT	Adopt the change as a permanent modification to the process, or abandon it (and start again).

3.3 Joseph M. Juran

In simple words
Juran defined quality as **fitness for use** and believed in improving step by step within the existing system. He spoke to top managers in money terms (cost of poor quality) and asked them to focus on the 'vital few' problems.

Everyday example
A school bag is fit for use if it carries books comfortably for years — not if it has 12 pockets the student never uses.

- **Evolutionary** theory of quality — *adapt* rather than replace the existing system (unlike Deming's revolution).
- Quality = **fitness for use**.
- Talk to top managers in the **language of money** (cost of poor quality); to workers in the language of things.
- Pareto principle: focus on the **vital few** problems rather than the useful (trivial) many.

Fitness for use — its elements (Day 3, slide 7)

In simple words
A product is fit for use only if it is well designed, made as designed, available when needed, reliable and easy to maintain, safe, and supported after sale.

Everyday example
An air-conditioner: right capacity for the room (design), correctly assembled (conformance), in stock in summer (availability), runs for years and is easy to service (reliability & maintainability), no electrical hazard (safety), installer and service centre nearby (field use).

Quality of **design** · quality of **conformance** · **availability** · **reliability & maintainability** · **safety** · **field use**. How to achieve: a **product-life-cycle (PLC) approach** and a **cost-of-quality**

accounting system.

Cost of Quality (COQ) — Juran's four categories

In simple words

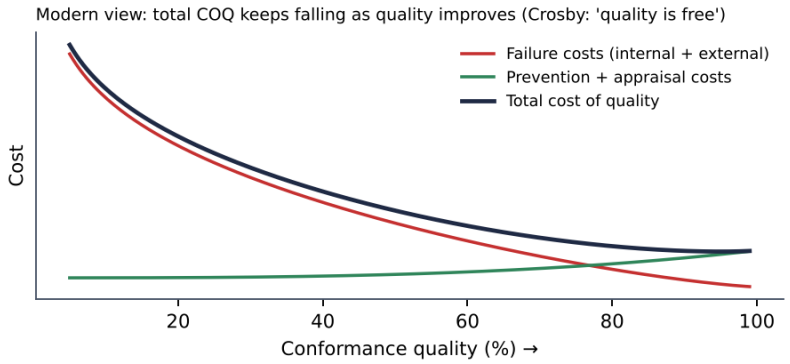
Quality costs fall into four buckets: money spent to **prevent** defects, to **check** for them (appraisal), and money lost when defects are found **inside** the company or, worse, **by the customer**. Spending more on prevention usually saves far more on failures.

Everyday example

A printing press: training operators (prevention); proof-reading (appraisal); reprinting a wrong batch before dispatch (internal failure); a client returning 10,000 wrongly printed wedding cards and never coming back (external failure — the most expensive).

Category	Meaning	Examples (from slide)
Prevention costs	Spent to stop defects happening	Quality planning, new-product review, training, process control, quality data analysis, quality reporting, improvement projects
Appraisal costs	Spent to find defects (measuring/checking)	Incoming material inspection, inspection & testing, maintaining test equipment, material consumed in test, evaluation of stock
Internal failure costs	Defects found **before** the customer gets the product	Scrap, rework, retest, downtime, yield loss, disposition (deciding what to do with bad items)
External failure costs	Defects found **after** delivery to the customer	Warranty charges, complaint adjustment, returned material, allowances (markdowns), lost goodwill

- Failure costs (internal + external) are typically **50–80%** of total quality cost — the 'iceberg' below the surface.
- Spending a little more on prevention (and some appraisal) reduces failure costs by much more → total cost falls. Rule of thumb ('1-10-100'): a defect costs ~1 to prevent, ~10 to catch inside, ~100 once it reaches the customer.



Juran's Quality Trilogy — achieving low cost of quality

In simple words

Three management processes: **plan** quality into the product, **control** it during operations so it doesn't get worse, and **improve** it to a new, better level.

Everyday example

Diet analogy: plan a healthy meal chart (planning), weigh yourself weekly to stay on track (control), and later change to a better routine to drop to a new weight (improvement).

Quality Planning	Quality Control	Quality Improvement
Establish quality goals; identify customers & their needs; translate needs into our language; develop a product for these needs; optimise product features	Prove the process can produce under operating conditions; transfer the process to operations; measure performance, compare with goals, act on differences	Seek to optimise the process via tools of diagnosis; reduce chronic waste to a new, better level ('breakthrough')

Juran's Quality Planning Road Map (8 steps)

In simple words
An eight-step recipe that starts with the customer and ends with a process proven to work in real conditions: who are the customers → what do they need → put that in technical language → design product → optimise → design process → optimise → prove it.

Everyday example
Designing a new school canteen: identify customers (students, teachers), learn needs (healthy, quick, cheap), translate ('meal served in < 5 min, < ₹50'), design menu, design kitchen workflow, trial run for a week, then launch.

1. Identify who the customers are.
2. Determine the customers' needs.
3. Translate the needs into our language.
4. Develop a product (service) to meet those needs.
5. Optimise the product (service) to meet our needs as well as the customer's.
6. Develop a process able to produce the product.
7. Optimise the process.
8. Prove the process can make the product under operating conditions.

Juran's Breakthrough Sequence

In simple words
A step-by-step way to break out of chronic problems: convince people change is needed, pick the few biggest problems, form teams to diagnose them, find root causes, deal with resistance, make the change and lock it in with controls.

Everyday example
A hospital with long OPD waits: management agrees it is unacceptable (attitude), Pareto shows 70% of delay is at billing (vital few), a team studies billing, finds the root cause, introduces pre-payment kiosks, and sets daily waiting-time monitoring (controls).

1. **Breakthrough in attitudes** — create a climate for change.
2. **Identify the vital few projects** — Pareto analysis.
3. **Organise for breakthrough in knowledge** — a steering group (direction) and a diagnostic group (analysis).
4. **Conduct the analysis** — identify symptoms, develop hypotheses, conduct experiments; ascertain the nature of the defects.
5. **Plan strategies to overcome resistance to change.**
6. **Institute the change.**
7. **Institute controls** to hold the gains.

3.4 Philip B. Crosby

In simple words
Crosby's message: define quality as meeting requirements, prevent problems instead of inspecting, expect zero defects, and measure quality in money — because doing things right the first time is cheaper ('quality is free').

Everyday example
A courier company that delivers every parcel right the first time saves the cost of re-delivery, phone complaints and refunds — so the 'extra' effort costs nothing overall.

- Quality is **conformance to requirements**.
- **Quality is FREE** — if quality improves, total cost falls (what costs money is doing things wrong).
- The only performance standard that makes sense is **Zero Defects** — perfection is possible if it is expected.
- **Prevention**, not after-the-fact inspection.
- Zero-defects programme: **Determination → Education → Implementation**.

Crosby's Four Absolutes

In simple words
Four answers to four questions: quality means conformance to requirements; it is achieved by prevention; the standard is zero defects; and it is measured by the cost of not conforming.

Everyday example
For a pharmacy: requirement = right medicine and dose; prevention = barcode scanning; standard = zero dispensing errors; measure = money lost on returns, re-work and complaints.

Question	Absolute
What is quality?	**Definition:** conformance to requirements
What system is needed to cause quality?	**System:** prevention
What performance standard should be used?	**Performance standard:** zero defects
What measurement system is required?	**Measurement:** the price of non-conformance (PONC)

Crosby's tools: cost of quality & the Management Maturity Grid

In simple words
The maturity grid is a self-check that places a company on five stages, from 'we don't know why we have problems' (Uncertainty) to 'we know why we don't have problems' (Certainty). As maturity rises, the cost of quality falls from about 20% of sales to about 2.5%.

Everyday example
A start-up that fixes every customer complaint by fire-fighting is at Uncertainty/Awakening. A mature firm like a top airline has prevention built into routines — Wisdom/Certainty.

The Maturity Grid is a **self-assessment tool** describing five stages of quality awareness:

Stage	Management understanding & attitude	Problem handling	COQ as % of sales	Summation of quality posture
1. Uncertainty	No relevance for quality; failure to recognise quality as a management tool	Fire-fighting	≈20% (often 20–40%)	'We don't know why we have problems'

Stage	Management understanding & attitude	Problem handling	COQ as % of sales	Summation of quality posture
2. Awakening	Quality recognised as important but no action taken	Teams set up for major problems	≈18%	'Is it absolutely necessary to always have problems?'
3. Enlightenment	Address problems through a formal quality programme	Problems faced openly, resolved in order	≈12%	'Through commitment we are identifying and resolving problems'
4. Wisdom	Prevention, early identification, routine corrective action	Problems identified early	≈8%	'Defect prevention is a routine part of our operation'
5. Certainty	Quality is a way of life; quality manager on the board	Problems are prevented (except a few)	≈2.5–3%	'We know why we don't have problems'

Crosby's 14 steps (textbook): management commitment; quality improvement team; quality measurement; cost of quality evaluation; quality awareness; corrective action; ad hoc committee for the zero-defects programme; supervisor training; Zero Defects Day; goal setting; error-cause removal; recognition; quality councils; do it over again.

3.5 Other quality philosophers (Textbook Ch 2)

In simple words
Feigenbaum said quality must cover every function (total quality control) and warned of the 'hidden plant' — capacity wasted on rework. Ishikawa made quality everyone's job through quality circles and simple tools like the fishbone diagram.

Everyday example
Hidden plant: if a factory spends one day a week re-doing faulty work, 20% of the factory is effectively a 'hidden plant' producing nothing new. Quality circle: a group of 6 machine operators meeting weekly to solve their own line's problems.

- **A. V. Feigenbaum** — coined **Total Quality Control**: quality is everybody's job and must span all functions. The **'hidden plant'**: 15–40% of capacity is wasted on rework. Quality and cost are a sum, not a difference. Three steps: quality leadership, modern quality technology, organisational commitment.
- **Kaoru Ishikawa** — **company-wide quality control (CWQC)**; father of **quality circles**; developed the **cause-and-effect (fishbone) diagram**; 'the next process is your customer'; promoted the **7 basic QC tools** (he said ~95% of problems can be solved with them); 'quality begins and ends with education'.

3.6 Comparing Deming, Juran and Crosby

In simple words
All three agree that management must lead quality and that prevention beats inspection. They differ in style: Deming wants a complete transformation, Juran works step by step with projects and money, Crosby motivates with zero defects and 'quality is free'.

Everyday example
Fixing a messy kitchen: Deming would redesign the whole kitchen and how the family works together; Juran would tackle the biggest mess first and show how much money it saves; Crosby would set the rule 'no dirty dish left behind — ever'.

Aspect	Deming	Juran	Crosby
Definition of quality	Continual improvement; predictable degree of uniformity at low cost suited to the market	Fitness for use	Conformance to requirements
Approach	Revolutionary — transform the system (14 points, SoPK)	Evolutionary — work within existing system (trilogy, projects)	Motivational — zero defects, quality is free
Role of management	Responsible for ~94% of problems	Less than 20% of problems due to workers	Management commitment first
Statistics / SPC	Essential (statistical thinking)	Recommended, but beware 'tool-driven'	Rejects statistically acceptable levels — ZD
Cost of quality	No optimum — continuous improvement	Quality is not free; there is an optimum	Quality is free
Goal setting	Avoid targets/slogans	Annual improvement goals	Zero defects

Summary of core principles (Day 3, slide 19)

In simple words

The common ground of the gurus: quality is not just inspection, it needs everyone, it needs long-term top-management commitment and training, and quality must come before schedule. Their weakness: they don't tell you exactly where to start or how much benefit you will get — Six Sigma fills that gap.

Everyday example

Telling a student 'study hard, be disciplined' is a philosophy; giving a timetable with a target score and weekly tests is a method. Six Sigma is the timetable.

- Quality improvement is not just inspection.
- It requires organisation-wide efforts.
- Long-term commitment with major investment in training and wholehearted participation of top management.
- A fundamental shift in priorities: from 'schedule first' to 'quality first'.
- Need tolerance for new programmes and techniques.
- BUT: Are there differences between gurus? Are they comprehensive? Do they forecast the benefits? What is the quantifiable improvement? Where should we focus first? (These gaps lead to Six Sigma — measurable, project-based.)

3.7 Principles, practices and techniques; ISO 9000 (Textbook Ch 2)

In simple words

Principles are beliefs (focus on the customer), **practices** are what you do to apply them (run customer surveys), and **techniques** are the tools you use (a Pareto chart of complaints). ISO 9000 is an international standard for setting up a documented quality management system.

Everyday example

Principle: 'customers matter'. Practice: a monthly feedback call to key clients. Technique: an affinity diagram to group their comments.

Three core principles of TQ: customer focus; participation & teamwork (engagement of people); process focus with continuous improvement & learning. **Practices** are activities that put principles into action (e.g., collecting customer data, training, benchmarking). **Techniques/tools** are the specific methods (Pareto charts, control charts, QFD). Principles → practices → techniques.

ISO 9000 family and the seven quality management principles

In simple words
ISO 9001 lists what a company's quality system must contain; an outside auditor checks it and issues a certificate. It proves the company has organised, controlled processes — it does not by itself prove every product is excellent.

Everyday example
A school with ISO 9001 has written procedures for admissions, exams and complaints, and follows them. Whether its teaching is excellent is a separate question.

ISO's seven quality management principles (2015): 1 customer focus · 2 leadership · 3 engagement of people · 4 process approach · 5 improvement · 6 evidence-based decision making · 7 relationship management.

Standard	What it is
ISO 9000	Fundamentals and vocabulary
ISO 9001	Requirements of a quality management system — the **only one you get certified (registered) to**
ISO 9004	Guidance for sustained success / performance improvement beyond 9001
Related	ISO 14000 (environment), ISO 22000 (food safety), GMP; sector versions (e.g., IATF 16949 automotive, AS9100 aerospace)

- ISO 9001:2015 clauses 4–10: context of the organisation, leadership, planning (incl. **risk-based thinking**), support, operation, performance evaluation, improvement.
- Certification is done by accredited third-party **registrars** through audits; surveillance audits follow.
- Benefits: discipline, documentation, consistent processes, market access. Limits: can become paperwork; certifies the **system**, not the product (link back to ISI vs ISO).
- Building an effective QMS: align it with customers and strategy, keep documentation useful, use audits for improvement, integrate with Baldrige-type excellence.

PART A · FOUNDATIONS

4. Translating Philosophy into Practice: Case Studies

Aravind (Day 1) · Dabbawalas of Mumbai (Day 2) · iD Fresh Food (Day 3) · Zara (Day 4)

In simple words
The professor uses video cases to show that quality is a **business strategy**. In every case a simple, disciplined, low-variation process beats expensive technology. The common thread: consistency, waste removal, customer focus, and culture.

4.1 Mumbai Dabbawalas

In simple words
The dabbawalas deliver lakhs of lunch boxes daily with almost no errors using a simple colour/number code, fixed hand-over points, teamwork and strong values — no computers. Quality comes from process design and culture.

Everyday example
The code on a dabba works like a PIN code on a letter: it tells every person in the relay exactly where the box goes, so even a new member can't send it to the wrong office.

- About 5,000 dabbawalas deliver ~2 lakh lunch boxes daily from homes to offices and back, using trains, bicycles and hand-carts.
- Reported error rate of about **1 in 16 million** deliveries — better than Six Sigma (3.4 per million) — with almost no technology and mostly semi-literate staff.
- **Why the quality is superior:** a simple **colour/number coding** system on each box (mistake-proofing); standardised relay process with fixed handover points & times; teamwork and self-organised groups; strong values (punctuality, discipline, pride in work, 'customer is God'); equal sharing; low cost; no fear-driven supervision.
- **Lessons for your organisation:** design simple, visual, error-proof processes; standardise; build culture and ownership; measure punctuality; collaboration over competition (Deming).

4.2 iD Fresh Food

In simple words
iD Fresh succeeded with a simple product (batter) by making it taste the same every time, keeping it fresh through a fast supply chain, and earning trust through transparency — quality across the whole chain, not just in the factory.

Everyday example
If you buy a batter packet that makes perfect idlis on Monday and sour ones on Thursday, you switch brands. Consistency is what made customers stay with iD without heavy advertising.

- Started (2005, Bengaluru) selling ready-to-cook idli/dosa batter — a commodity market full of local players with inconsistent quality.
- **Business problem:** make fresh, natural (no preservatives) batter that tastes the same every time and reaches shelves fresh despite very short shelf life.

- **How quality helped:** consistency of taste and texture (process control), hygiene and automation, a fast cold-chain logistics system with daily delivery, taking back unsold stock — **quality extends across the whole supply chain**, not just the product.
- **Why show the factory publicly?** Transparency builds trust ('we have nothing to hide'); real advantage is the hard-to-copy system and culture, not secret steps.
- **Little marketing spend yet big share:** consistent quality → word of mouth and loyalty; customer trust (e.g., honesty-based self-service shops).
- **Key learnings (slide):** relevance of consistency in product success; essence of quality; role of technology and logistics; moving from product quality to the entire supply chain; designing a product that meets customer requirements and expectations.

4.3 Zara — impact of customer focus (Day 4)

In simple words
Zara's quality is speed and responsiveness: it listens to customers through its stores and puts new designs on shelves in weeks, with little leftover stock.

Everyday example
A college fest trend (say, a particular jacket) appears in Zara stores within 2–3 weeks, while a traditional brand would bring it next season — after the trend is over.

- Fashion industry problems: long lead times (6–9 months), forecasting errors, large unsold inventory, heavy markdowns.
- Zara's model: store managers feed daily customer feedback (voice of the customer) to designers; in-house design + vertically integrated, nearby production; small batches; twice-weekly replenishment; design-to-shelf in ~2–3 weeks; limited stock creates urgency; few markdowns.
- Is Zara's product 'superior quality'? Not in durability — its quality is **responsiveness**: giving customers what they want, when they want it. Quality = meeting customer expectations.
- Competitors with long supply chains struggled to copy it because it is a whole operating system, not one practice.
- **Learning:** customer focus + fast, low-variation processes = competitive advantage.

Key takeaways
Aravind, Dabbawalas, iD, Zara: **consistency, waste removal, process design, culture and customer focus** drive performance — not just product features.

Quality is a strategic, organisation-wide function (pyramid of excellence: technical systems remove waste; purpose & behaviour build culture).

PART B · CUSTOMER & DESIGN

5. Customer Focus — Hearing the Voice of the Customer

Slides Day 3 (voice of the customer, requirements engineering), Day 4 & Day 5 (Define phase) · Textbook Chapter 3

In simple words
Before you can make something 'high quality' you must know what the customer actually wants — in their words — and then translate it into the engineer's language. Customers and engineers speak different languages; most quality failures start as misunderstood requirements.

5.1 Customer satisfaction and engagement

In simple words
Satisfaction means the customer's expectations were met. Engagement goes further — the customer feels attached, keeps coming back and recommends you. Satisfied customers may still switch; engaged ones rarely do.

Everyday example
You may be satisfied with a petrol pump but go to any pump on the way. You are *engaged* with your favourite tea stall — you walk an extra 5 minutes and bring friends.

- **Customer satisfaction**: the customer's judgement that a product met (or exceeded) expectations. **Customer engagement**: emotional commitment — loyalty, willingness to recommend, actively choosing you. Satisfied customers can still leave; engaged customers stay and advocate.
- **American Customer Satisfaction Index (ACSI)**: customer expectations + perceived quality → perceived value → **customer satisfaction** → fewer complaints and higher **loyalty**.
- Loyal customers spend more, cost less to serve, and bring referrals; acquiring a new customer costs several times more than keeping one.

5.2 Who is the customer?

In simple words
A customer is anyone who receives your output — the final consumer, other companies in the chain, and colleagues inside your own organisation (internal customers).

Everyday example
In a restaurant, the diner is the external customer; the waiter is the chef's internal customer; the cashier is the waiter's internal customer (needs the correct bill details).

- Consumers (end users), external customers (other firms in the chain), internal customers (next process). The slide uses **AICP** — Associate, Investor, Customer, Process — as the set of stakeholders whose 'voice' must be heard.
- **Customer segmentation**: grouping customers with similar needs (e.g., by demographics, usage, value) so you can design for each segment.
- **Census vs sample**: you rarely ask every customer; you draw a representative sample.

5.3 Quality dimensions of goods and services

In simple words
Customers judge goods on dimensions like performance, features, reliability, durability, serviceability and looks. They judge services on reliability, assurance, tangibles, empathy and responsiveness (RATER). Knowing the dimensions tells you what to measure.

Everyday example
A bank: reliability — no wrong debits; assurance — polite, knowledgeable staff; tangibles — clean branch; empathy — a manager who understands your loan worry; responsiveness — complaint solved the same day.

Garvin's dimensions (goods)	Meaning	Guitar amplifier example
Performance	Primary operating characteristics	Signal-to-noise ratio, wattage
Features	'Bells and whistles' — secondary characteristics	SD card slot, drum kits
Reliability	Probability of not failing over time	Mean time to failure
Conformance	Meeting specifications	Accurate tuner
Durability	Amount of use before replacement	Withstands frequent handling
Serviceability	Ease, speed, courtesy of repair	Ease of repair
Aesthetics	Look, feel, sound	Location and size of knobs
Perceived quality	Reputation, brand image	Brand reputation

Service quality dimensions (SERVQUAL — remember 'RATER'), used by the professor in the Six Sigma Define phase:

Dimension	Meaning	Example
Reliability	Perform the promised service dependably and accurately	Receive the product at the same time each day
Assurance	Knowledge and courtesy of staff; ability to convey trust and confidence	Showing respect for the customer's needs
Tangibles	Physical facilities, equipment, appearance of staff, facilitating goods	Cleanliness
Empathy	Caring, individual attention; being approachable	Easy to talk to
Responsiveness	Willingness to help customers promptly	Avoid keeping customers waiting for no apparent reason

Textbook Example 3.1 — classify needs on SERVQUAL [Ch 3]

A car-rental agency asks about: cleanliness of the rental facility; courtesy of staff; efficiency of vehicle pick-up/return; cleanliness of vehicle.

Answer: Cleanliness of facility → **Tangibles**; courtesy of staff → **Assurance** (and Empathy); efficiency of pick-up/return → **Responsiveness**; cleanliness of vehicle → **Tangibles** (and Reliability, since a clean car is part of the promised service).

Perceived service quality (Day 5, slide 15)

In simple words
Customers compare what they expected with what they experienced. Below expectation = poor quality, equal = satisfactory, above = delight. Expectations come from word of mouth, personal needs and past experience.

Everyday example
If friends told you a café serves coffee in 2 minutes and it takes 10, you're unhappy — even though 10 minutes might be fine at another café you expected less from.

Expected service (ES) is shaped by word of mouth, personal needs and past experience. Perceived service (PS) is what the customer experiences on the five dimensions.

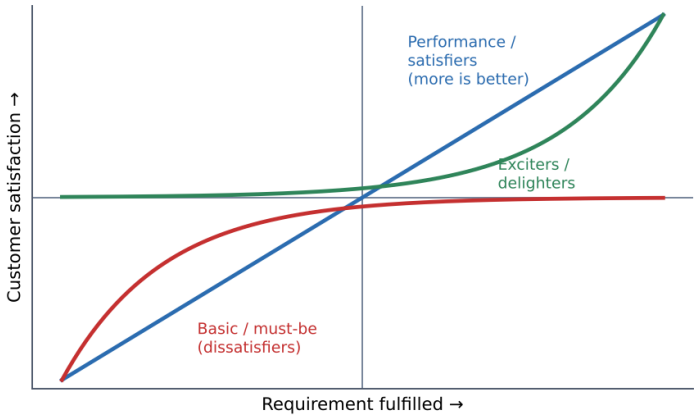
Comparison	Customer's verdict
ES > PS — expectations not met	Unacceptable quality
ES ≈ PS — expectations met	Satisfactory quality
ES < PS — expectations exceeded	Quality surprise (delight)

Gap model (textbook): gaps can arise between (1) customer expectations and what managers think customers want; (2) management perceptions and the specs written; (3) specs and actual delivery; (4) delivery and what is communicated/promised; (5) expected vs perceived service — the sum of the others.

5.4 The Kano model

In simple words
Not all features affect satisfaction the same way. Basic features only cause anger when missing; performance features please more the better they are; delighters surprise and impress. Over time delighters become basics.

Everyday example
Car: seat belts (basic — nobody praises them, but missing = unacceptable); mileage (performance — the more km/l, the happier you are); a free 3-year service you didn't expect (delighter).
Air-conditioning was once a delighter, now it is basic.



Category	Behaviour	Example (hotel)
Basic / must-be (dissatisfiers)	Expected; absence causes anger, presence gives no extra delight	Clean sheets, hot water

Category	Behaviour	Example (hotel)
Performance / one-dimensional (satisfiers)	More is better — satisfaction rises in proportion	Wi-Fi speed, check-in time
Exciters / delighters	Unexpected; presence delights, absence not noticed	Free upgrade, handwritten welcome note
(Indifferent)	Customer doesn't care either way	Brand of soap

Exam tip
Over time delighters become performance needs, then basic needs (e.g., in-room Wi-Fi). Kano analysis is listed on the slides as an elicitation technique to understand features vs satisfaction.

5.5 Gathering the voice of the customer (VOC) — Requirements Engineering

In simple words
Requirements engineering is the careful process of finding out what customers really need, using surveys, interviews, focus groups, observation and similar methods, because customers and designers often mean different things by the same words.

Everyday example
A customer asks for a 'fast' website. To her it means pages load in under 2 seconds on mobile data; the developer assumed 'fast' meant a fast server. Observing her actual use reveals the real requirement.

The slide's 'Where is the problem?' cartoon: what the customer explained, what the analyst understood, what the designer drew, what was built, and what the customer really needed are all different. Requirements engineering = systematically eliciting, analysing and agreeing requirements.

Elicitation technique (slide)	Examples
Traditional	Questionnaires, surveys, analysis of existing documentation, organisational charts, process models, literature, interviews, recorded calls, live chats
Group elicitation	Focus groups, JAD/RAD (joint/rapid application development) workshops
Kano analysis	Understand product/service features and customer satisfaction
Cognitive techniques	Protocol analysis (think-aloud), laddering, card sorting, repertory grid
Contextual techniques	Ethnomethodology (observing people in their real setting), conversational analysis

Textbook adds: comment cards & formal surveys, focus groups, direct customer contact, field intelligence, complaint analysis, internet & social-media monitoring (sentiment analysis), 'big data' analytics. Inputs (surveys, interviews, focus groups, suggestions, observations) → **voice of AICP** → **Y's** (key output requirements, the CTQs).

Analysing VOC: affinity diagram (Textbook Example 3.2)

In simple words
An affinity diagram groups dozens of scattered customer comments into a few themes, so you can see the main needs clearly.

Everyday example
Hostel feedback slips — 'cold food', 'late dinner', 'same menu', 'slow Wi-Fi', 'router in corridor only' — grouped into two themes: **Mess** and **Internet**.

An **affinity diagram** groups many raw customer statements into natural themes. In Example 3.2 a bank's mortgage customers wanted 'timely closings'; statements like expeditious processes, prior approvals, competitive rates, timely status reports, flexibility of programmes, etc. were grouped under headings such as **Communication, Effective service, Loan products, Reliability, Innovation, Flexibility**. Those headings become the WHATs for design (QFD).

5.6 Building a customer-focused organisation

In simple words
Customer focus is built through clear promises, good handling of every customer contact (moments of truth), hiring and empowering front-line staff, and recovering well when things go wrong.

Everyday example
Service recovery: a hotel that loses your booking upgrades you to a better room and sends a dessert. You leave more loyal than if nothing had gone wrong.

- **Customer commitments** — explicit promises/guarantees (e.g., service guarantees) that build trust.
- **Customer contact & moments of truth** — every contact is a 'moment of truth' where you can satisfy or dissatisfy (Jan Carlzon, SAS).
- **Selecting and developing customer-contact employees** — hire for attitude, train, empower them to solve problems on the spot.
- **Service recovery & complaint management** — a service recovery turns a dissatisfied customer into a loyal one; complaints are free information. Process: record complaint → immediate action → root-cause analysis → corrective action → verify → close the loop with the customer.
- **Managing relationships** — CRM, strategic partnerships and alliances, customer-focused technology and analytics.

5.7 Measuring customer satisfaction, loyalty and value

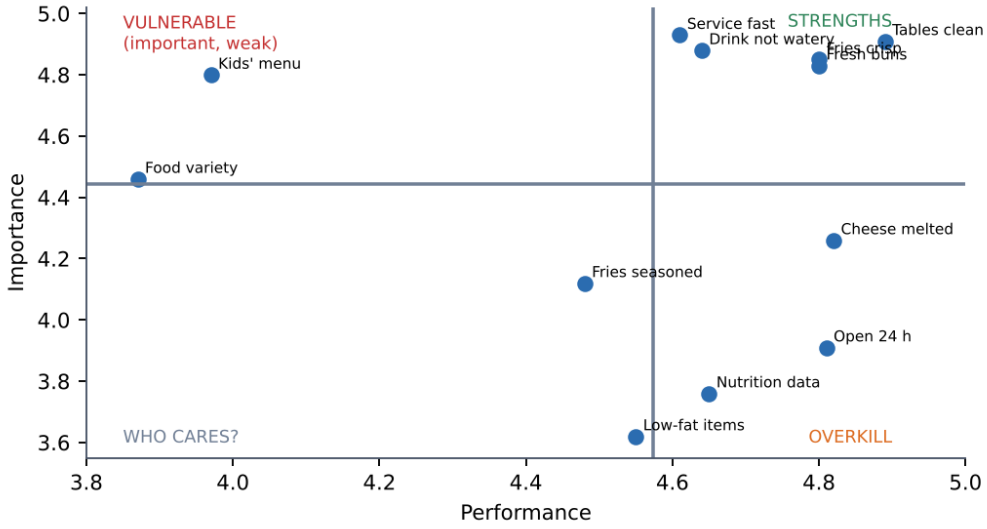
In simple words
Measure what customers think through surveys, and act on it. Useful tools: a performance–importance grid (what to fix first), Net Promoter Score (loyalty), and customer perceived value (how you compare with competitors on what matters most).

Everyday example
NPS in one line: ask 'Would you recommend us, 0–10?'. If 50% say 9–10 and 20% say 0–6, NPS = 50 − 20 = 30.

- Survey design: clear purpose, right questions, Likert scales (1–5), representative sample, act on results. Why efforts fail: poor questions, measuring satisfaction but not loyalty, ignoring results, no link to processes.

Textbook Example 3.3 — Performance–importance grid [Ch 3]

A fast-food restaurant measured importance and performance (5-point scale) for 12 attributes. What should managers do?



Average lines	Plot each attribute (x = performance, y = importance); draw lines at the mean importance and mean performance → four quadrants.
Read quadrants	**Strengths** (high importance, high performance): fresh buns, fries crisp, fast service, drink not watery, tables clean. **Vulnerable** (high importance, low performance): good food variety, children's menu. **Overkill** (low importance, high performance): cheese melted, open 24 hours, nutritional data displayed. **Who cares?** (low, low): fries seasoned, low-fat items.

Answer: Improve food variety and add a children's menu (vulnerable); consider cutting 24-hour opening (overkill, saves cost); maintain strengths.

Textbook Example 3.4 — Net Promoter Score (NPS) [Ch 3]

300 customers rated 'How likely are you to recommend us?' (0–10): 10→63, 9→82, 8→64, 7→41, 6→21, 5→12, 4→6, 3→7, 2→3, 1→0, 0→1.

Promoters (9–10)	63 + 82 = 145 → 145/300 = 48.3%
Passives (7–8)	64 + 41 = 105 (ignored)
Detractors (0–6)	21 + 12 + 6 + 7 + 3 + 0 + 1 = 50 → 50/300 = 16.7%
NPS	% promoters − % detractors = 48.3 − 16.7 = 31.6

Answer: NPS = **31.6** (above 50 is considered good; top firms score 70–90).

Textbook Example 3.5 — Customer perceived value (competitive comparison) [Ch 3]

Importance weights: menu variety 30%, food quality 20%, atmosphere 10%, value 40%. Ratings (A, B): menu variety (8, 10), food quality (7, 4), atmosphere (8, 8), value (7, 6).

Weighted score A	0.3×8 + 0.2×7 + 0.1×8 + 0.4×7 = 2.4 + 1.4 + 0.8 + 2.8 = **7.4**
Weighted score B	0.3×10 + 0.2×4 + 0.1×8 + 0.4×6 = 3.0 + 0.8 + 0.8 + 2.4 = **7.0**
Relative performance	A − B per attribute: −2, +3, 0, +1 → weighted difference = 0.4 in A's favour.

Answer: Restaurant A has higher perceived value (7.4 vs 7.0) but should improve menu variety, where B is stronger.

PART B · CUSTOMER & DESIGN

6. Design for Quality: Lifecycle, QFD, Taguchi & Reliability

Slides Day 3 (Design for Quality, QFD) & Day 4 (lifecycle approach) · Textbook Chapter 7 — Design for Quality and Product Excellence

In simple words
Most quality is decided on the drawing board. If you design a product that customers don't want, or that the factory can't make consistently, no amount of inspection will save you. Design for quality = link what customers want (WHATs) with how we'll deliver it (HOWs), set smart targets and tolerances, and make the product reliable.

6.1 Developing a comprehensive quality system — lifecycle thinking

In simple words
Quality must be managed over the product's whole life — design, making, delivery, use, repair and disposal. The biggest lever is the design stage, where most cost and quality are decided.

Everyday example
A mobile phone with a glued-in battery may be cheap to make but hard to repair and recycle. Thinking about the full life at design time leads to a replaceable battery.

- Achieve quality in the **entire product life cycle**: design → manufacturing → distribution & retail → use → maintenance & repair → end-of-life (recycle, reuse, take-back).
- Where do we start? **Design** (circled on the slide) — it fixes most of the cost and quality.
- When is a design 'high quality'? When it **meets the customer requirement** → so the first step is to **hear the customer**.

6.2 Design thinking — the story of Doug Dietz (GE Adventure Series)

In simple words
Design thinking starts with empathy for the real user and their experience, not just the technical specification.

Everyday example
The MRI machine worked perfectly, but children were terrified. Turning the room into a pirate-ship adventure solved a quality problem no engineering change could have.

Doug Dietz designed GE's MRI scanners. He saw a frightened child crying before a scan — about 80% of children needed sedation. Using design thinking (empathy with the user), he redesigned the experience: the MRI room became a pirate ship or space adventure, with a story the child takes part in. Sedation fell dramatically and satisfaction rose. **Learning:** the technical product was fine; quality failed in the **user experience**. Design thinking = empathise → define → ideate → prototype → test.

6.3 Product development process (Textbook Ch 7)

In simple words
New products go through stages — idea, concept, design and testing, production, launch, review. Doing these stages in parallel with all departments involved (concurrent engineering) saves time and late changes.

Everyday example
Designing a new scooter: if the factory team joins at the concept stage, they can say 'this frame shape can't be welded cheaply' before the design is frozen, not after.

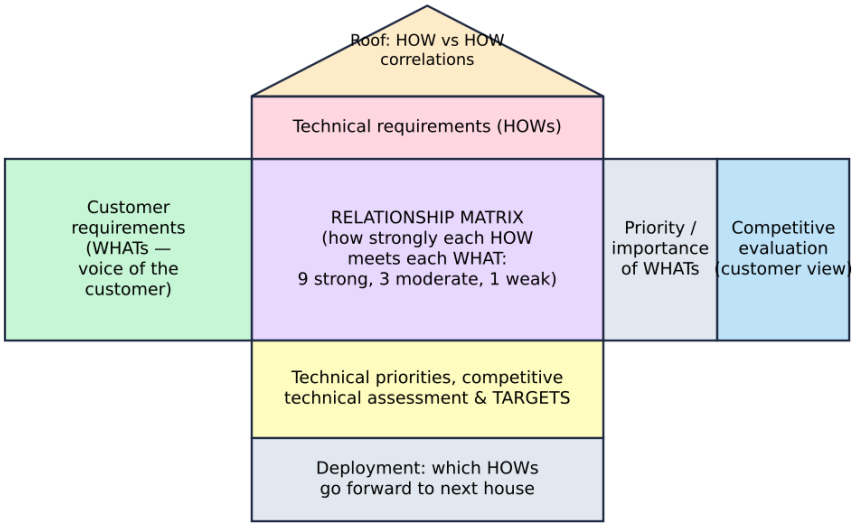
- Typical stages: idea generation → preliminary concept development → product/process development (design, prototype, testing) → full-scale production → product introduction → market evaluation.
- **Concurrent engineering**: design, manufacturing, marketing, suppliers work together from the start (not 'over the wall') → shorter time, fewer late changes.
- **Design for Six Sigma (DFSS)**: concept development → design development → design optimisation → design verification (also written IDOV or DMADV).
- **Concept development & innovation**: creativity, generating and screening ideas; innovation can be incremental or breakthrough.

6.4 Quality Function Deployment (QFD) & the House of Quality

In simple words
QFD is a matrix that translates what customers want (WHATs, in their words) into measurable technical features (HOWs), shows how strongly they are linked, compares you with competitors, and tells you which technical features deserve the most attention.

Everyday example
Customer says 'the bag should be comfortable'. HOWs: strap width (cm), padding thickness (mm), weight (kg). The House of Quality shows 'weight' is strongly linked to comfort, so you set a target like ≤ 0.8 kg.

Design for Quality (slide): link customer requirements with technical feasibility — but that is not enough; link customer requirements (**WHATs**) with technical requirements (**HOWs**) *strategically*, i.e., in comparison with competitors. The tool is the **House of Quality (QFD)**.



The six steps of QFD

In simple words
Six steps: list customer needs → list technical measures → link them in a matrix → compare with competitors → set technical targets → decide which measures to carry into production planning.

Everyday example
For a coffee shop: needs (hot coffee, quick service) → measures (serving temperature, minutes to serve) → link (strong) → competitor serves in 4 min → target 3 min → deploy into staff training and machine layout.

Six steps in QFD (slide & textbook):

1. Identify customer requirements (voice of the customer — the WHATs, with importance ratings).
2. Identify technical requirements (measurable design characteristics — the HOWs).
3. Relate customer requirements to technical requirements (relationship matrix: ⊙ strong = 9, ○ moderate = 3, Δ weak = 1, blank = 0).
4. Conduct an evaluation of competing products or services (customer view of us vs competitors A, B).
5. Evaluate technical requirements and develop targets (technical priorities and target values).
6. Determine which technical requirements to deploy in the remainder of the production/delivery process (the 'deployment' row).

Roof (interactions of HOWs): ++ strong positive, + positive, − negative, − − strong negative. A negative roof entry signals a trade-off (improving one HOW hurts another). The first house links to three more: part deployment → process planning → production/quality control planning.

Worked example (illustration) — computing technical priorities [Illustrative]

A pizza chain lists WHATs with importance: Hot pizza (5), Fast delivery (4), Correct order (3). HOWs: (a) delivery time in minutes, (b) insulated-bag temperature, (c) order read-back step. Relationships: Hot–a = 3, Hot–b = 9; Fast–a = 9; Correct–c = 9, Correct–a = 1. Which HOW matters most?

Rule	Technical importance of each HOW = Σ (importance of WHAT × relationship score).
HOW a (delivery time)	5×3 + 4×9 + 3×1 = 15 + 36 + 3 = **54**

HOW b (bag temperature)	5×9 = **45**
HOW c (read-back)	3×9 = **27**
Relative weights	Total 126 → a 43%, b 36%, c 21%.

Answer: Delivery time is the top technical priority, then bag temperature; these get targets (e.g., ≤ 25 min, ≥ 65 °C) and are 'deployed' to process design and control plans.

Textbook example (Figs 7.6–7.8): a fitness centre: WHATs such as 'has programs I want', 'family activities', 'clean locker rooms', 'well-maintained equipment'; HOWs such as program offerings, maintenance schedule, staff training, facility hours, fee structure. Customer importance + weak competitive position (e.g., programs and family activities) → higher targets; deployed HOWs: program offerings, amount/types of equipment, facility hours, fee structure, Internet access.

6.5 Target and tolerance design

In simple words
Every specification has a target (ideal value) and a tolerance (how far you may be from it). Tight tolerances cost more to achieve; loose ones may hurt performance. Set them based on what the customer and function really need.

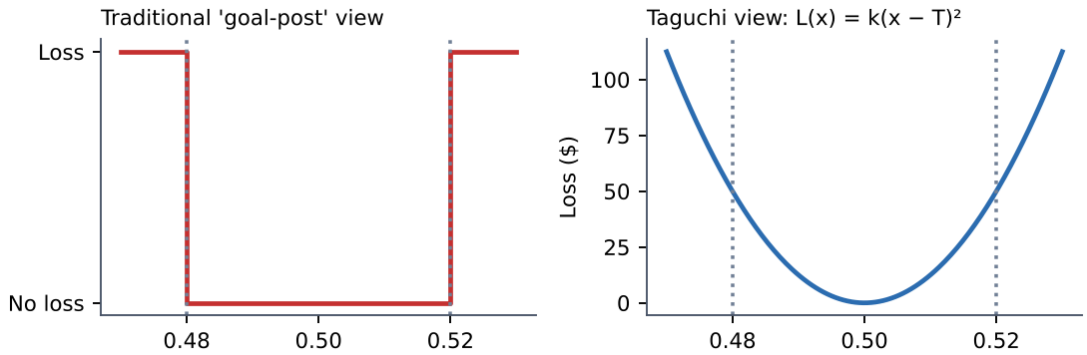
Everyday example
Bottled water: target 1000 ml, tolerance ± 10 ml. Holding ± 1 ml would need very expensive filling machines and the customer wouldn't notice the difference.

- **Nominal** = ideal/target value; **tolerance** = permissible variation. Needed because of natural variation from the '5 Ms': men & women, materials, machines, methods, measurement.
- Example 7.1: a microprocessor dimension '0.514/0.588' means nominal 0.551 ± 0.037.
- Narrow tolerances → higher cost but better interchangeability, performance, appearance. Wide tolerances → cheaper but poorer product. Tolerances are often set by convention; setting them scientifically (e.g., with the Taguchi loss function) can save a lot (a $1,000 part was being scrapped needlessly — relaxing the tolerance saved $20,000/yr).

6.6 The Taguchi loss function

In simple words
Taguchi argued that any deviation from the target creates some loss, and the loss grows with the square of the deviation — so being 'just inside the spec' is not good enough. Aim for the target and reduce variation.

Everyday example
Room temperature target 24 °C. At 25 °C people are slightly uncomfortable; at 27 °C much more so (loss grows fast). A thermostat that keeps 24 ± 0.5 °C is far better than one that wanders between 22 and 26 °C, even if both are 'within 22–26'.



Another way to see it

The traditional 'goal-post' view says any part inside the tolerance is perfect and any part outside is junk — so 0.519 is 'fine' and 0.521 is 'bad', though they are nearly identical. Taguchi says every deviation from the target costs something (to the customer, to society), and the cost grows with the **square** of the deviation. So aim for the target, and reduce variation, even inside the spec.

```
Nominal-is-best: L(x) = k (x − T)²
Smaller-is-better: L(x) = k x²  Larger-is-better: L(x) = k (1/x)²
Expected loss per unit: EL = k (σ² + D²), where D = (mean − T)
```

Sony TV example: San Diego plant — all units inside spec but spread uniformly ($\sigma^2 = 8.33$); Japanese plant — some outside spec but concentrated at the target ($\sigma^2 = 2.78$). With k = 0.16: loss = 0.16 × 8.33 = **$1.33** vs 0.16 × 2.78 = **$0.44** per unit → the Japanese plant was $0.89/unit better even with some out-of-spec units.

Expected loss and tolerance design with the loss function

In simple words

Find the constant k from a known cost at a known deviation, then compute the loss for any value or the average (expected) loss for a whole process. Use break-even: fix a unit in the factory only when its expected loss exceeds the cost of fixing it.

Everyday example

If shipping a slightly-off product costs ₹50 in warranty at a deviation of 2 mm, $k = 50/2^2 = 12.5$ per mm². At 1 mm the loss is ₹12.50. If adjusting in the factory costs ₹20, adjust when $12.5 \cdot d^2 > 20$, i.e., when the deviation exceeds about 1.26 mm.

Example 7.2 — Finding the loss function [Ch 7]

Spec 0.500 ± 0.020. If the characteristic is off by the tolerance (0.020) the product needs a $50 warranty repair. Find k and the loss at x = 0.510.

Set up	At deviation 0.020 the loss is $50: $50 = k (0.020)^2$.
Solve k	$k = 50 / 0.0004 =$ **125,000** → $L(x) = 125{,}000\ (x − 0.500)^2$.
Loss at 0.510	$L = 125{,}000 \times (0.010)^2 = 125{,}000 \times 0.0001 =$ **$12.50**.

Answer: k = 125,000; a unit at 0.510 'costs' $12.50 even though it is within spec.

Example 7.3 — Expected loss of two processes [Ch 7]

Spec 0.50 ± 0.02, L(x) = 125,000 (x − 0.50)². Process A: 0.48, 0.49, 0.50, 0.51, 0.52 each with probability 0.2. Process B: 0.47 (0.02), 0.48 (0.03), 0.49 (0.15), 0.50 (0.60), 0.51 (0.15), 0.52 (0.03), 0.53 (0.02).

x	Loss L(x)	P(A)	A: weighted	P(B)	B: weighted
0.47	$112.50	0	$0.00	0.02	$2.25
0.48	$50.00	0.2	$10.00	0.03	$1.50
0.49	$12.50	0.2	$2.50	0.15	$1.875
0.50	$0.00	0.2	$0.00	0.60	$0.00
0.51	$12.50	0.2	$2.50	0.15	$1.875
0.52	$50.00	0.2	$10.00	0.03	$1.50
0.53	$112.50	0	$0.00	0.02	$2.25
Total			**$25.00**		**$11.25**

Step 1	Compute the loss at each value, e.g., L(0.48) = 125,000 × 0.02² = $50; L(0.47) = 125,000 × 0.03² = $112.50.
Step 2	Multiply each loss by its probability and add up (expected value).
Step 3	Compare: A = $25.00, B = $11.25.

Answer: Process B has less than half the expected loss, even though 4% of its output is outside specification — concentration near the target matters more than 'all within spec'.

Example 7.5 — Expected loss with the formula [Ch 7]

Spec 0.6500 ± 0.0275, L(x) = 99,174 (x − T)². Process mean 0.6620, σ = 0.0087.

Variance	σ² = 0.0087² = 0.00007569
Off-target squared	D² = (0.6620 − 0.6500)² = 0.012² = 0.000144
Expected loss	EL = 99,174 × (0.00007569 + 0.000144) = 99,174 × 0.00021969 = **$21.79 per unit**

Example 7.6 — Tolerance design by break-even [Ch 7]

Target tape speed 1.875 in/s. A customer returns a unit if off by ≥ 0.15 in/s; warranty repair = $20. A factory adjustment costs $5. What factory tolerance should trigger an adjustment?

Find k	20 = k (0.15)² → k = 20 / 0.0225 = 888.9 → L(x) = 888.9 (x − 1.875)².
Break-even	Adjust at the factory only when the expected loss of shipping exceeds $5: 888.9 d² = 5 → d² = 0.005625 → d = **0.075**.
Tolerance	1.875 ± 0.075 → 1.800 to 1.950 in/s.

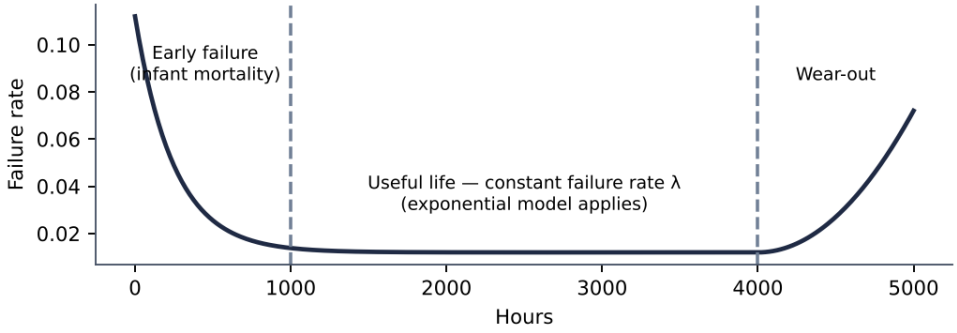
Answer: Ship as-is if speed is between 1.800 and 1.950; outside these limits it is cheaper to adjust ($5) than to risk the loss (> $5). General rule: economic tolerance = $\Delta_0 \times \sqrt{(A/A_0)} = 0.15 \times \sqrt{(5/20)} = 0.075$.

6.7 Design for reliability

In simple words
Reliability is the chance that something works for a given time under given conditions. Products usually fail more at the start (weak units), then rarely for most of their life, then more again as they wear out — the 'bathtub curve'.

Everyday example
A new LED bulb either fails in the first few days (manufacturing defect) or lasts for years; after many years it starts to dim and fail (wear-out).

Reliability = the probability that a product performs its intended function for a stated period of time under specified operating conditions (four elements: probability, time, performance, operating conditions). **Inherent** reliability = predicted by design; **achieved** = actually observed in use. Failures: *functional* (at start of life due to defects) vs *reliability* failures (after some use).




```
Failure rate λ = number of failures ÷ total unit-operating hours
MTTF (non-repairable) or MTBF (repairable): θ = 1/λ
Exponential (useful life): R(T) = e^(−λT) = e^(−T/θ), F(T) = 1 − e^(−λT)
Series: Rs = R₁ × R₂ × … × Rₙ Parallel: Rs = 1 − (1−R₁)(1−R₂)…(1−Rₙ)
```

Failure rate, MTTF and MTBF

In simple words
Failure rate λ = failures per hour of operation. Its reciprocal is the average life: MTTF for items you throw away, MTBF for items you repair.

Everyday example
A printer that breaks down 4 times in 2,000 hours has λ = 4/2000 = 0.002 per hour and MTBF = 500 hours — on average one breakdown every 500 hours of use.

Example 7.7 — Failure rate [Ch 7]

10 units tested for 100 hours. 4 failed at 6, 35, 65 and 70 hours; 6 survived the full 100 hours.

Operating hours	6 + 35 + 65 + 70 + (6 × 100) = 176 + 600 = **776 unit-hours**.
λ	4 ÷ 776 = **0.00515 failures/hour**.
Interpret	≈ 0.5% of units fail per hour; over 100 hours ≈ 51.5% expected (actual test: 40%).

The reliability function R(T)

In simple words
R(T) gives the probability of surviving up to time T. During the useful life it follows R(T) = e^(−λT). For wear-out items (like tyres) a normal distribution can be used.

Everyday example
If a fan's λ = 0.0001 per hour, the chance it runs 5,000 hours without failing is e^(−0.5) = 0.61, i.e., about 61% of fans survive that long.

Example 7.8 — Reliability function (normal) [Ch 7]

Tyre life ~ Normal(mean 50,000 miles, σ 1,500). Probability a tyre wears out before 48,000 miles?

z	z = (48,000 − 50,000)/1,500 = −1.33
Probability	P(z < −1.33) = NORM.DIST(48000, 50000, 1500, TRUE) = **0.0912**
Use	A 48,000-mile warranty → ≈9% of tyres replaced; 45,000 miles → virtually none (z = −3.33). R(51,000) ≈ 0.25.

Example 7.9 — Exponential reliability [Ch 7]

A component has R = 0.97 for 100 hours. Find λ.

Equation	0.97 = e^(−100λ)
Take ln	ln 0.97 = −0.0305 = −100λ → λ ≈ **0.0003 per hour**; R(T) = e^(−0.0003T); MTTF = 1/0.0003 ≈ 3,333 h.

MTBF example: a machine runs 10,000 h with 4 failures (repaired) → θ = 10,000/4 = 2,500 h; λ = 0.0004/h.

Example 7.10 — Using MTTF [Ch 7]

λ = 0.0001/h. Probability the component survives 15,000 hours?

MTTF	θ = 1/0.0001 = 10,000 h
Reliability	R(15,000) = e^(−15,000/10,000) = e^(−1.5) = **0.223**

System reliability: series, parallel and mixed

In simple words
In **series** every part must work, so system reliability is the product of parts' reliabilities (it drops as parts are added). In **parallel** (backups) the system fails only if all backups fail, so reliability rises sharply.

Everyday example
Series: festival lights where one bulb failing kills the string. Parallel: a hospital with mains power + generator + UPS — power fails only if all three fail. If each is 0.9 reliable, the parallel system is $1 - 0.1^3 = 0.999$.

Examples 7.11–7.14 — Series and parallel systems [Ch 7]

(a) PC with processor 0.997, graphics board 0.980, keyboard 0.975 in series. (b) Two components in series with λ = 0.004 and 0.001 — survival to 100 h. (c) Five shuttle computers in parallel, each 0.99. (d) Two parallel components each λ = 0.01 — survival to 100 h.

(a) Series	Rs = 0.997 × 0.980 × 0.975 = **0.953** (adding series parts lowers reliability).
(b) Series, exponential	Rs = e^(−(0.004 + 0.001)×100) = e^(−0.5) = **0.6065**.
(c) Parallel	Rs = 1 − (1 − 0.99)⁵ = 1 − 0.01⁵ = **0.9999999999**.
(d) Parallel, exponential	Each: e^(−1) = 0.3679. Rs = 1 − (1 − 0.3679)² = 2(0.3679) − 0.1353 = **0.6004**.

Example 7.15 — Mixed systems [Ch 7]

(a) A (0.99) in series with three B's in parallel (0.9 each), then C (0.96) and D (0.98). (b) Two parallel branches: A–B–C in series (0.95, 0.98, 0.99) and D–E in series (0.99, 0.97).

(a) collapse parallel	R_B = 1 − (1 − 0.9)³ = 0.999 → Rs = 0.99 × 0.999 × 0.96 × 0.98 = **0.930**.
(b) collapse series	R_ABC = 0.95 × 0.98 × 0.99 = 0.92169; R_DE = 0.99 × 0.97 = 0.9603.
(b) then parallel	Rs = 1 − (1 − 0.92169)(1 − 0.9603) = 1 − 0.07831 × 0.0397 = **0.9969**.

Example 7.16 — Comparing alternative designs [Ch 7]

Two components (0.9 and 0.95) in series. Option 1: duplicate the whole series chain in parallel. Option 2: duplicate each component (two parallel pairs in series).

Current	0.9 × 0.95 = 0.855
Option 1	1 − (1 − 0.855)² = 1 − 0.145² = **0.979**
Option 2	Pair 1: 1 − 0.1² = 0.99; Pair 2: 1 − 0.05² = 0.9975; series: 0.99 × 0.9975 = **0.9875**

Answer: Option 2 (redundancy at the component level) is more reliable than duplicating the whole chain.

6.8 Design optimisation, FMEA, FTA, manufacturability, environment, verification

In simple words
A set of design tools to make products robust, safe, easy to make, environmentally responsible, and proven before launch.

Everyday example
Before launching a new pressure cooker, the company lists what could fail (FMEA), traces how an explosion could happen (FTA), designs the lid so it can only lock one way (mistake-proofing), and tests it at high pressure (verification).

Design optimisation and robust design

In simple words
Robust design makes a product perform well even when conditions vary, instead of trying to control every condition.

Everyday example
A cake recipe that turns out well whether your oven runs 10 °C hot or cold is robust; one that only works at exactly 180 °C is not.

- **Robust design** (Taguchi): choose design parameters so the product performs well even when 'noise' factors (environment, wear, material variation, how customers use it) vary.
- Uses designed experiments (DOE) and signal-to-noise ratios to find the best settings of controllable factors.

Design Failure Mode and Effects Analysis (DFMEA)

In simple words
FMEA is a structured 'what could go wrong?' exercise. Each possible failure is scored for how serious it is, how often it may happen and how hard it is to detect; the product of the three (RPN) ranks what to fix first.

Everyday example
Phone charger: failure 'overheats' — severity 9, occurrence 3, detection 5 → RPN 135; failure 'cable frays' — 4 × 6 × 2 = 48. Fix overheating first.

- For each component list its possible **failure modes** → their **effects** on the customer → their **causes** → current controls.
- Rate **Severity (S)**, **Occurrence (O)** and **Detection (D)** on 1–10 scales (10 = worst; for D, 10 = almost impossible to detect).
- **Risk Priority Number RPN = S × O × D**; act on the highest RPNs first, then re-score after the fix.

Worked example (illustration) — FMEA for a pressure cooker [Illustrative]

Failure modes: (A) safety valve blocks: S 10, O 3, D 4. (B) handle becomes loose: S 6, O 5, D 2. (C) gasket wears out early: S 4, O 7, D 3. Which to fix first?

RPN A	10 × 3 × 4 = **120**
RPN B	6 × 5 × 2 = **60**
RPN C	4 × 7 × 3 = **84**
Decision	Fix A first (highest RPN, and severity 10 = safety). Even if an RPN were lower, any severity 9–10 item deserves attention.

Fault Tree Analysis (FTA)

In simple words
FTA starts from one bad outcome and works backwards through AND/OR logic to find combinations of causes.

Everyday example
'Car won't start' happens if (battery dead) OR (no fuel) OR (starter faulty). 'Battery dead' happens if (lights left on) AND (alternator weak). Probabilities combine through the tree.

- Top-down: start from an undesirable **top event** and trace back to basic causes using **AND** gates (all inputs needed) and **OR** gates (any input is enough).
- Probabilities: AND gate → multiply; OR gate → $1 - \Pi(1 - p_i)$.

Design for manufacturability (DFM) and mistake-proofing

In simple words
Design the product so it is easy and cheap to make correctly — fewer parts, standard parts, and parts that physically cannot be assembled the wrong way.

Everyday example
A SIM card has one cut corner so it can only go in one way; USB-C works either way up. Both prevent assembly/usage errors by design.

- Fewer parts, standard parts, modular design, parts that can only fit one way (**poka-yoke / mistake-proofing**), avoid tight tolerances unless needed, design for easy assembly and testing.

Design for environment and Design for Excellence (DfX)

In simple words
Design with the environment and every later stage in mind: recycling, disassembly, energy use, servicing, testing and cost.

Everyday example
Furniture designed with screws instead of glue can be repaired, moved and recycled; a laptop with a standard screw type is easier for service centres.

- **Design for environment**: recyclable materials, design for disassembly, energy efficiency, take-back programmes.
- **Design for Excellence (DfX)**: design for X — cost, assembly, service, testability, reliability, etc.

Design verification: reviews and reliability testing

In simple words
Before mass production, check the design formally (design reviews) and test it hard (life testing, accelerated testing, environmental testing) so failures are found in the lab, not by customers.

Everyday example
Accelerated testing: opening and closing a fridge door 100,000 times in a week with a machine instead of waiting 10 years of real use.

- **Design reviews**: preliminary (concept), intermediate (detailed design), final (before production) — cross-functional teams check the design against requirements.
- **Reliability testing**: life testing, **accelerated life testing** (stress the product harder to get failures faster), environmental testing (heat, humidity), vibration & shock, burn-in (run new units for a while to weed out infant mortality).

PART C · SIX SIGMA

7. Process Quality and the Six Sigma Idea

Slides Day 4 (11 Oct) & Day 5 (17 Oct) · Textbook Chapter 9 — Process Improvement and Six Sigma

In simple words
Six Sigma is a way of running improvement projects with data. The name comes from statistics: if your process is so consistent that six standard deviations fit between the target and the nearest specification limit, you will make only about 3.4 defects per million chances — even when the process drifts a little.

7.1 Concept check: Sam vs John (accuracy vs precision)

In simple words
Accuracy = hitting the target on average; precision = being consistent. A consistent but off-target process is easy to fix by shifting it; an inconsistent one is hard to fix.

Everyday example
Two cooks: one always adds a little too much salt (consistent — just tell him to use less), the other's salt varies wildly (sometimes perfect, sometimes inedible). The first is the better hire.

Day 4, slides 4–6 [Day 4]

Sam and John are expert shooters. Of 10 shots, Sam hit the target 5 times; John only once. Should we select Sam?

Look at the pattern	John's 10 shots form a **tight cluster** just to the left of the bull's-eye (very consistent, but off-centre). Sam's shots are **scattered** all over; 5 happened to land on target.
Which is easier to fix?	John's error is a **shift in the mean** — adjust the sight once and he will hit almost every time. Sam's problem is **high variation** — much harder to fix.
Quality language	John = precise (low σ) but not accurate; Sam = accurate on average but imprecise (high σ).

Answer: Select **John**. Learning: **consistency (low variation) is more valuable than occasional hits**; centring a consistent process is easy, reducing variation is hard. This is exactly what Six Sigma targets.

7.2 Process management basics (Day 4)

In simple words
A process turns inputs into outputs for a customer. To manage it you need an owner, clear boundaries, a known flow, objectives and measures such as cycle time, takt time and lead time.

Everyday example
Takt time for a sandwich shop: 8 hours (480 min) of opening and 240 customers a day → one sandwich every 2 minutes. If making one takes 3 minutes, the queue will keep growing.

- Why study processes? Service systems fail because of process failures (hand-offs, waits, errors). Improve the process and quality follows.

- **Elements of a process**: inputs, activities, outputs, customers, suppliers (SIPOC). Ask: Who **owns** the process? What are its **boundaries**? What is the **flow**? What are its **objectives**? What are the **process measurements**?
- **Methods of improvement**: process blueprints (mapping), process improvement (incremental), process redesign (radical/re-engineering), process management (ongoing control).

Process measure	Definition	Example
Cycle time	Time between successive outputs (or time a unit spends at a stage)	A burger every 2 minutes
Takt time	**Available time ÷ customer demand** — the pace needed to meet demand	480 min ÷ 240 burgers = 2 min/burger
Throughput time (lead time)	Total time a unit spends in the system from start to finish	Order to delivery = 25 min
Customer service measures	On-time %, waiting time, complaints, satisfaction	% orders delivered < 30 min

Exam tip
If cycle time > takt time, the process cannot keep up with demand — the 'voice of the process' is worse than the 'voice of the customer' → improvement needed.

Limitations of the quality movement (philosophies) — why Six Sigma emerged

In simple words
The gurus' philosophies were inspiring but vague about measurement, responsibility and expected benefits. Six Sigma added hard numbers, financial targets and a project method.

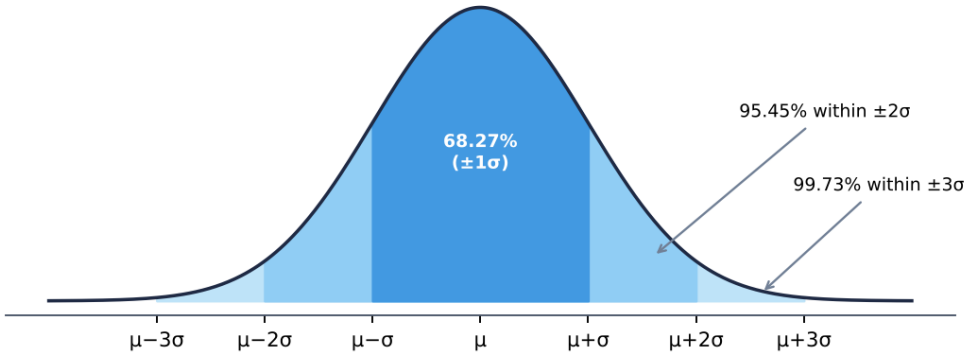
Everyday example
'Exercise regularly' (philosophy) vs 'walk 8,000 steps a day, measured by your phone, target −3 kg in 3 months' (Six Sigma style).

- Who is responsible for quality? Process vs systems approach unclear.
- Cost of quality (and cost of poor quality, COPQ) — many costs are non-visible.
- Complex to implement; a 'black box' with no explainable model; benefits not predictable.
- What do stakeholders really care about?
- **Lack of measurement** → Lord Kelvin: 'When you can measure what you are speaking about and express it in numbers, you know something about it…'

7.3 The normal distribution — the language of variation

In simple words
Most natural measurements cluster around an average in a bell shape. About 68% lie within 1 standard deviation of the mean, 95% within 2, and 99.73% within 3. This lets us predict how many items will fall outside any limit.

Everyday example
Heights of adult men in a city: average 170 cm, standard deviation 7 cm. About 95% are between 156 and 184 cm; a man of 191 cm (3σ above) is very rare.

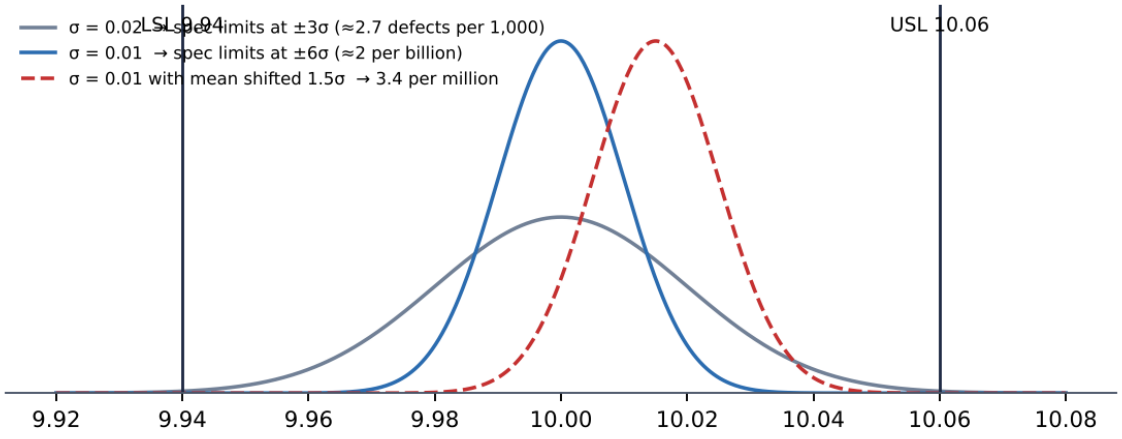


- Bell-shaped, symmetric, described by mean μ (centre) and standard deviation σ (spread). Many measurements are normal because of the Central Limit Theorem.
- Excel: NORM.DIST(x, μ, σ, TRUE) = P(X ≤ x). Standardise with **z = (x − μ)/σ**; NORM.S.DIST(z, TRUE) = P(Z ≤ z); NORM.S.INV(p) gives z for a probability.
- Within ±1σ: **68.27%** (0.682689); ±2σ: **95.45%**; ±3σ: **99.73%** (0.9973).

7.4 Where do the 'magic numbers' come from? (3.4 DPMO)

In simple words
If the process is so tight that 6 standard deviations fit between the target and each spec limit, defects are almost impossible. Allowing for a realistic drift of 1.5σ, you still get only 3.4 defects per million — that is 'Six Sigma'.

Everyday example
A road 12 lanes wide for a car that wobbles only half a lane: even if the driver drifts a bit to one side, the car almost never goes off the road.



3-sigma process	Target 10, specs 9.94–10.06 (±0.06). If σ = 0.02, the spec is ±3σ → outside area = 1 − 0.9973 = 0.0027 → **2.7 defects per 1,000**.
6-sigma process	Halve the spread: σ = 0.01 → specs are ±6σ → outside area ≈ 0.000000002 → **about 2 defects per billion**. Six Sigma 'essentially limits the spread to within 12 standard deviations' (USL − LSL = 12σ).
Allow for drift	Real processes drift over time (tool wear, new material lots, different operators, temperature). Motorola observed long-run shifts of about **1.5σ**. If the mean shifts 1.5σ, the nearest limit is only 6 − 1.5 = 4.5σ away → tail area beyond 4.5σ ≈ **3.4 per million**.
So	**Six Sigma = no more than 3.4 defects per million opportunities (DPMO)** — 'defect-free 999,997 times out of 1,000,000' (slide's rounding of 999,996.6).

Sigma level	DPMO (with 1.5σ shift)	% good	Cost of poor quality (COPQ)	Capability
6	3.4	99.99966%	< 10% of sales	World class
5	233	99.977%	10–15% of sales	
4	6,210	99.38%	15–20% of sales	Industry average
3	66,807	93.3%	20–30% of sales	
2	308,537	69.1%	30–40% of sales	Non-competitive
1	691,462 (~700,000)	30.9%	—	

- Motorola study: a 4σ company spends **> 10% of revenue** on internal and external repair; a 6σ company **< 1%**.
- Four-sigma (6,210 DPMO = 99.38% right) still means 20,000 lost articles of mail every hour, 7 hours of power outage each month, 5,000 incorrect surgical procedures a week, unsafe drinking water 9 minutes a day, two long/short landings a day.
- Officer clearing 1,000 files/year: at 2σ, ≥ 940 on time (slide); 3σ: 990; 4σ: all but 6; 5σ: at most 1 late every 4 years; 6σ: never late in a 271-year career.
- Most organisations operate around **3–3.5 sigma** because they 'hit the wall': 1%, 2%, 3% error levels in marketing, sales, ordering, billing add up to 6–12–18% enterprise-wide, and each error costs far more to fix than to create.

7.5 Six Sigma philosophy and critical elements

In simple words
Six Sigma focuses on the customer, uses data for every decision, fixes processes rather than people, works across departments, and treats the output (Y) as a result of input causes (X's) that must be found and controlled.

Everyday example
Y = late pizza deliveries. X's = order-taking time, oven capacity, driver availability, distance. Watching Y alone doesn't help; controlling the X's does.

Snowball effect: defect reduction → yield improvement → customer satisfaction → better organisation.

Critical element	In plain words
Genuine focus on the customer	Defects are defined by what the customer cares about (CTQs)
Data- and fact-driven management	Decisions based on measurements, not opinion
Process focus	Fix the process (the X's), not the people
Proactive management	Prevent problems rather than fire-fight
Boundary-less collaboration	Break down silos across departments, suppliers, customers
Drive for perfection; tolerance for failure	Aim high, but allow people to experiment and learn

$$Y = f(X_1, X_2, \ldots, X_n)$$

Y: dependent · output · effect · symptom → MONITOR
X: independent · input/process · cause · problem → CONTROL

The focus of Six Sigma is to identify and control the X's. What makes Six Sigma different: leadership commitment, competence & involvement; a methodology & tools that are data-driven and statistically validated; the best people 100% dedicated to defect reduction; and a project focus.

Textbook principles of Six Sigma: think in terms of key business processes and customer requirements; focus on strategic objectives; focus on corporate sponsors (champions); emphasise

quantifiable measures such as DPMO that apply to every part of the business; ensure appropriate metrics; provide extensive training followed by project-team deployment; create highly qualified process-improvement experts ('belts'); set stretch objectives.

PART C · SIX SIGMA

8. Measuring Defects and Sigma Levels — all class problems solved

Slides Day 5 (Measure phase) · Textbook Chapter 8 (quality measures) & Chapter 9 (sigma level)

In simple words
To know how good a process is, count defects relative to the number of chances (opportunities) to make one. Convert that rate into a 'sigma level' using the normal table. Higher sigma = fewer defects.

8.1 The key formulas

In simple words
Count defects, divide by the number of chances to make a defect, and multiply by one million to get DPMO. Then use the normal table (or NORM.S.INV) to convert DPMO into a sigma level.

Everyday example
An online store ships 5,000 orders; each order has 4 things that could go wrong (item, size, address, packaging). 100 mistakes found → DPMO = 100/(5,000 × 4) × 10^6 = 5,000 → about 4.1σ (shifted scale).

$$\text{DPU (defects per unit)} = \text{number of defects} \div \text{number of units}$$

$$\text{DPMO} = \text{number of defects} \div (\text{number of units} \times \text{opportunities per unit}) \times 1{,}000{,}000$$

$$\text{Sigma level (short-term, professor's method)} = \text{NORM.S.INV}(1 - \text{DPMO}/10^6) + 1.5$$

$$\text{Sigma level without the shift ('long-term')} = \text{NORM.S.INV}(1 - \text{DPMO}/10^6)$$

$$\text{DPMO for a sigma level} = (1 - \text{NORM.DIST}(\text{sigma}, 1.5, 1, \text{TRUE})) \times 10^6$$

Exam tip
Defect vs defective: a *defective* unit has at least one defect; one unit can have many defects. DPMO counts **defects** (errors detected), not defective items.

Opportunities = critical-to-quality characteristics that could go wrong (CTx / CTQs) for each unit.

Always state whether your sigma includes the 1.5σ shift. Six Sigma tables quote the shifted ('short-term') sigma: 3.4 DPMO = 6σ.

8.2 Solved problems

In simple words
The class problems follow one pattern: (1) find the number of opportunities, (2) compute DPU/DPMO, (3) convert to sigma with NORM.S.INV, (4) add 1.5 if using the Six Sigma (shifted) scale, (5) interpret which process is worst.

Everyday example
Think of it like converting marks into a grade: DPMO is the raw score, the sigma level is the grade you report.

Problem 1 — Hospital complaints (Day 5, slides 6 & 17) [Day 5]

A hospital received 1,434 complaints from 1,38,600 patients treated between March 2024 and December 2024. What DPMO and sigma level?

DPU	1,434 ÷ 1,38,600 = **0.0103463** (here each patient = 1 opportunity).
DPMO	0.0103463 × 10^6 = **10,346.32**.
Sigma (no shift)	NORM.S.INV(1 − 0.0103463) = NORM.S.INV(0.9896537) = **2.315**.
Sigma (with 1.5 shift)	2.315 + 1.5 = **3.815**.

Answer: DPMO ≈ 10,346; the hospital operates at about **2.3σ** without the shift, i.e., **3.8σ** on the usual Six Sigma (shifted) scale. Next questions: *why* improve and *how*?

Problem 2 — Pharmacy with four critical characteristics (Day 5, slides 17–18) [Day 5]

Four critical characteristics — delivery time, medicine formula, formula potency, dosage — were checked on a sample of 100 prescriptions. 12 prescriptions had an error in at least one characteristic. Find DPMO.

Opportunities	Each prescription has 4 chances to be wrong → opportunities = 4 × 100 = 400.
DPMO	12 ÷ 400 × 10^6 = **30,000**.
Sigma (optional)	NORM.S.INV(1 − 0.03) + 1.5 = 1.881 + 1.5 ≈ **3.38σ**.

Careful — check against the slide
The slide writes 12/(4 × **1000**) × 10^6 = 3,000, but the problem states a sample of **100** prescriptions, which gives **30,000**. In the exam, use the sample size actually given. (Strictly, 12 is the count of prescriptions with ≥ 1 error, so 30,000 is a lower bound on true DPMO.)

Problem 3 — Flybynight Airlines (textbook, Day 5 slides 19–21) [Day 5 / Ch 8]

130 bags lost per 10,000 customers in a month; average 1.3 bags per customer. What is the sigma level?

Opportunities	10,000 customers × 1.3 bags = 13,000 bags.
DPU	130 ÷ 13,000 = **0.01**.
DPMO	0.01 × 10^6 = **10,000**.

Method A — Sigma conversion chart (interpolate)	From the chart: 2.3σ ↔ 10,724 DPMO and 2.4σ ↔ 8,198 DPMO. 10,000 lies between, closer to 2.3: σ = 2.3 + (10,724 − 10,000)/(10,724 − 8,198) × 0.1 = 2.3 + 724/2,526 × 0.1 = 2.3 + 0.0287 = **2.328**.
Method B — Normal table (Appendix A)	We need z with right-tail area 0.01 → cumulative 0.99 → z between 2.32 and 2.33 → **2.33** (= NORM.S.INV(0.99) = 2.326).
Shifted scale	2.33 + 1.5 ≈ **3.83σ**.

Answer: Flybynight operates at ≈ **2.33σ** (long-term, no shift) ≈ 3.83σ on the Six Sigma scale. (Textbook Example 8.5 is similar: 3 lost bags, 8,000 passengers × 1.6 bags → epmo = 3/12,800 × 10^6 = 234.4.)

Problem 4 — Class exercise: 600 software changes through 5 serial steps (Day 5, slides 22–28) [Day 5]

600 items pass through 5 process steps. Items with errors are reworked. For each step find DPMO and sigma level.

Step	Critical characteristics (opportunities/item)	Errors detected	Items with errors	DPMO = errors/(CC × 600) × 10^6	z = NORM.S.INV(1 − DPMO/10^6)	Sigma = z + 1.5
1	4	74	51	74/2,400 → **30,833**	1.869	**3.37**
2	3	103	98	103/1,800 → **57,222**	1.579	**3.08**
3	1	67	66	67/600 → **111,667**	1.218	**2.72**
4	3	157	86	157/1,800 → **87,222**	1.358	**2.86**
5	1	44	43	44/600 → **73,333**	1.451	**2.95**

Step 1	Opportunities for step 1 = 4 characteristics × 600 items = 2,400.
Step 2	Use **errors detected** (defects), not 'items with errors': DPMO = 74/2,400 × 10^6 = 30,833.
Step 3	Sigma = NORMSINV(1 − 0.030833) + 1.5 = 1.87 + 1.5 = 3.37. Repeat for each row.
Step 4 — interpret	Step 3 is the worst (2.72σ) → start the improvement project there; step 1 is best (3.37σ).

Exam tip
Why 'items with errors' is given: it is a distractor (and useful for a yield calculation: e.g., step 1 first-pass yield = (600 − 51)/600 = 91.5%). Rolled throughput yield = product of step yields.

Textbook Examples 9.2–9.4 [Ch 9]

(a) LSL 0.02, USL 0.10, nominal 0.06, σ = 0.01 — sigma level? (b) DPMO for 6σ and 4σ? (c) Sigma level for epmo = 35,256?

(a)	k × σ = half the tolerance → k × 0.01 = (0.10 − 0.02)/2 = 0.04 → **k = 4 → 4-sigma process** (if it stays centred and drifts no more than 1.5 × 0.01 = 0.015).
(b)	(1 − NORM.DIST(6, 1.5, 1, TRUE)) × 10^6 = **3.4**; for 4σ: (1 − NORM.DIST(4, 1.5, 1, TRUE)) × 10^6 = **6,210**.
(c)	NORM.S.INV(1 − 35,256/10^6) + 1.5 = 1.81 + 1.5 = **3.31σ**.

8.3 Other quality measures (Textbook Ch 8)

In simple words
Besides DPMO, quality can be measured as the fraction of bad units (p), the average defects per unit (NPU), the chance a unit has no defects (throughput yield), and the share of units that pass every step first time (rolled throughput yield).

Everyday example
A form passes three desks with first-pass yields of 90%, 95% and 98%. Only 0.90 × 0.95 × 0.98 = 84% of forms get through all three without any rework.

```
Fraction nonconforming p = nonconforming units ÷ units inspected
Nonconformances per unit NPU = total nonconformances ÷ units inspected
Throughput yield TY = e^(−NPU) Rolled throughput yield RTY = Y₁ × Y₂ × … × Yₙ
```

Textbook Examples 8.2–8.4 [Ch 8]

A hotel inspects 35 rooms on 12 checklist items; 3 rooms had nonconformances (1, 3 and 2 of them). Then: a 3-step process has yields 97%, 95%, 99%.

p	3 ÷ 35 = **0.086** (8.6% of rooms nonconforming)
NPU	(1 + 3 + 2) ÷ 35 = 6/35 = **0.171**
TY	e^(−0.171) = **0.843** → ~84% of rooms expected to be defect-free
dpmo	6 ÷ (35 × 12) × 10^6 = **14,286**
RTY	0.97 × 0.95 × 0.99 = **0.912** → only 91% of units pass all three steps first time

Defect classification: *critical* (unsafe/prevents function), *major* (likely failure, reduced usability), *minor* (little effect). Firms weight them in composite indices — e.g., FedEx's Service Quality Indicator (lost or damaged packages weighted 10, wrong-day late 5, complaints reopened 3, right-day late 1, etc.).

8.4 Cost of quality measurement (Textbook Ch 8)

In simple words
Turning quality problems into money gets top management's attention. Track prevention, appraisal, internal and external failure costs as a percentage of sales.

Everyday example
Telling the CEO '2,000 defects' means little; telling her 'defects cost us ₹4 crore last year, 8% of sales' starts action.

- Quality-related costs are typically **20–40% of sales**; most arise from poor quality and are avoidable; they occur in all functions, not just manufacturing.
- Juran: workers speak the 'language of things', top managers the 'language of money' — COQ translates quality problems into money.
- COQ = prevention + appraisal + internal failure + external failure (detailed in Section 3.3). Report as % of sales, by category, and track trends; hidden costs (lost customers, goodwill) are usually much larger than recorded ones.
- Six Sigma calls the failure part **COPQ — cost of poor quality**.

PART C · SIX SIGMA

9. The Six Sigma Method: DMAIC / DMADV, Lean and Service Blueprints

Slides Day 5 (17 Oct) · Textbook Chapter 9 — Process Improvement and Six Sigma

In simple words
DMAIC is a five-step recipe for fixing an existing process: **Define** the problem, **Measure** how bad it is, **Analyse** why it happens, **Improve** it, and **Control** it so it stays fixed. DMADV is the recipe for designing a new process or product from scratch.

9.1 When do you start a Six Sigma project?

In simple words
Start a project when the process cannot deliver what the customer needs (voice of the process worse than voice of the customer) and there is a real chance to improve it.

Everyday example
A bank promises loan approval in 3 days, but its process averages 6 days with big variation. That gap justifies a Six Sigma project.

- Not just 'when improvement is required' (the slide strikes this out) — but when improvement is **possible and justified**, i.e., when the **voice of the process** (its capability) is **inferior** to the **voice of the customer** (CTx requirements).
- So you need three inputs: **Voice of Customer**, **CTx** (critical-to-quality / cost / delivery characteristics) and **Process Capability** (voice of the process).
- Example: cycle time (voice of process) > takt time (voice of customer) → a project is warranted.
- **Slido answer:** a Six Sigma exercise is carried out to **reduce variation/defects in a process so it meets customer requirements** (not merely to inspect or to certify).

9.2 DMAIC vs DMADV

In simple words
DMAIC improves a process that already exists; DMADV (Design for Six Sigma) is used when you must design a new product or process, or the old one is beyond repair.

Everyday example
Reducing waiting time at an existing clinic → DMAIC. Designing a brand-new online appointment system → DMADV.

	DMAIC (improve an existing process)	DMADV (design a new process/product — DFSS)
D	**Define** — what is important (customer, CTQs, project scope)	**Define** — what is important
M	**Measure** — the current DPMO / sigma level; verify measurement system	**Measure** — the opportunities/customer needs (QFD)
A	**Analyse** — why? find the vital few root causes	**Analyse** — why? causes → vital causes; design options

	DMAIC (improve an existing process)	DMADV (design a new process/product — DFSS)
I / D	**Improve** — change the process to remove root causes	**Design** — create the new design
C / V	**Control** — hold the gains (SPC, standard procedures)	**Verify** — pilot, test, validate against requirements

Phase	Description (slide)
Define	Define areas of focus for organisational Six Sigma implementation; justify Six Sigma project selection.
Measure	Verify reliability of the measurement system; measure process capability; set improvement targets.
Analyse	Identify root causes of process variation (defects); understand variable relationships.
Improve	Improve process capability using the findings of the Analyse phase.
Control	Monitor the process after improvement to verify lasting change; document process changes and verify project benefits.

9.3 DEFINE

In simple words
Decide which process to improve, who the customer is, what they consider a defect (CTQs) and the project's scope and goal.

Everyday example
Define for a food-delivery app: problem — 18% of orders arrive late; customer CTQ — delivery within the promised 30 minutes; scope — Bengaluru, dinner hours; goal — late orders below 5%.

- Define **where to focus** (process selection) and **what to measure** (the defects — including service defects).
- **Service defect**: a flaw in the process that results in lower customer satisfaction or a lost customer. Services are harder: defects are intangible and judged by the customer.
- **CTx factors**: Critical-to-Quality (CTQ), Critical-to-Cost, Critical-to-Delivery — characteristics the customer cares about.
- **Moments of truth**: each customer contact is a moment of truth — a chance to satisfy or dissatisfy. **Service recovery** = satisfying a previously dissatisfied customer and making them loyal.
- Use the SERVQUAL dimensions (reliability, responsiveness, assurance, empathy, tangibles) and expected vs perceived service (Section 5.3) to define defects.
- Textbook tools: project charter, SIPOC diagram (Suppliers–Inputs–Process–Outputs–Customers), VOC/CTQ trees, high-level process map, cost-of-quality analysis.

9.4 MEASURE

In simple words
Collect data on the current process, make sure the measurements are reliable, and calculate the current DPMO and sigma level as a baseline.

Everyday example
Track 1,000 dinner orders for two weeks; 180 late → DPMO = 180,000 → about 2.4σ (shifted). That is the starting point to beat.

- Observe the process for a few days; record non-compliances (service quality failures = defects); convert them into **DPMO**; assess the **sigma level** (worked problems in Section 8).
- Tools: process capability and customer-satisfaction surveys; customer-service turnaround time; mapping service and information flow; check sheets; measurement-system evaluation (Section 11).
- **Process Cycle Efficiency (PCE) = value-added time ÷ total lead time.**

9.5 ANALYSE — probing for root causes

In simple words
Use data and tools (Pareto, fishbone, scatter plots, queue calculations) to find the few causes that create most of the defects.

Everyday example
Analysis of late orders shows 60% are delayed at the restaurant (food not ready), not on the road. Root cause: restaurants get the order only after the rider is assigned.

Analyse why the sigma level is low using QC tools — **Pareto charts** and **fishbone diagrams** (Section 13) — and probe for root causes such as **time traps**, capacity mismatch, the time effect of quality, process complexity, rule complexity and inconsistency.

Problem — Capacity matching in a vaccination department (Day 5, slides 30–31) [Day 5]

A hospital vaccination department receives on average 130 patients per 8-hour day. A nurse takes on average 3.5 minutes per patient (including preparation). The hospital estimates a nurse can serve ~138 patients/day and posts only one nurse, believing the 6% spare capacity will absorb variation. Is this right?

Arrival rate λ	130 ÷ 8 = **16.25 patients/hour**.
Service rate μ	60 ÷ 3.5 = **17.14 patients/hour** (= 137 per 8 h — the '138' estimate).
Utilisation	ρ = λ/μ = 16.25/17.14 = **0.948** — the nurse is busy 95% of the time.
Time in system (queueing, M/M/1)	W = 1 ÷ (μ − λ) = 1 ÷ (17.14 − 16.25) = 1 ÷ 0.893 = **1.12 hours ≈ 67 minutes**.
Process cycle efficiency	Value-added time ÷ total time = 3.5 min ÷ 67.2 min = **≈ 5.2%** (slide: 5.28%) → about **94–95% of the patient's time is non-value-adding waiting**.
Queue length (Little's law)	Patients in the system L = λ × W = 16.25 × 1.12 ≈ **18**; waiting in line Lq = λ²/[μ(μ − λ)] ≈ **17**.

Answer: No. A 6% capacity cushion is far too small when arrivals and service times vary — for a 3.5-minute vaccination a patient waits over an hour and the waiting room holds 15–18 people at any time. This is a classic **time trap** caused by capacity matching too tightly with variability.

Careful — check against the slide
The slide writes '16.25/(1.12) = 15 patients waiting'. By Little's law the number in the system is λ × W = 16.25 × 1.12 ≈ 18 (≈ 17 actually in the queue). The conclusion is the same: a large queue. Use multiplication (L = λW) in the exam.

Addressing time traps

In simple words
A time trap is any step where work sits waiting. Little's Law links work in progress, throughput and lead time: more items waiting means longer lead time. Lean removes waiting and non-value steps; Six Sigma reduces variation.

Everyday example
Passport office: 300 files waiting and 100 completed per day → lead time = 300/100 = 3 days, even though each file needs only 20 minutes of actual work (PCE is tiny).

- Process inefficiencies → **Lean**; variability and quality → **Six Sigma** (together: Lean Six Sigma).
- Pinpoint time traps using WIP and lead time — **Little's Law: Lead time = WIP ÷ throughput (exit) rate** (equivalently WIP × average cycle time per unit). (The slide writes 'Lead time = WIP/Cycle time'; read 'cycle time' there as the completion *rate*.)
- **Time-trap analysis**: map the process (value stream map / blueprint), classify every step as **value-added** (customer would pay for it, changes the product, done right first time), **business non-value-added** (needed for business/regulation, e.g., billing, approvals) or **non-value-added** (waiting, rework, moving) — then look for complexity and bottlenecks.

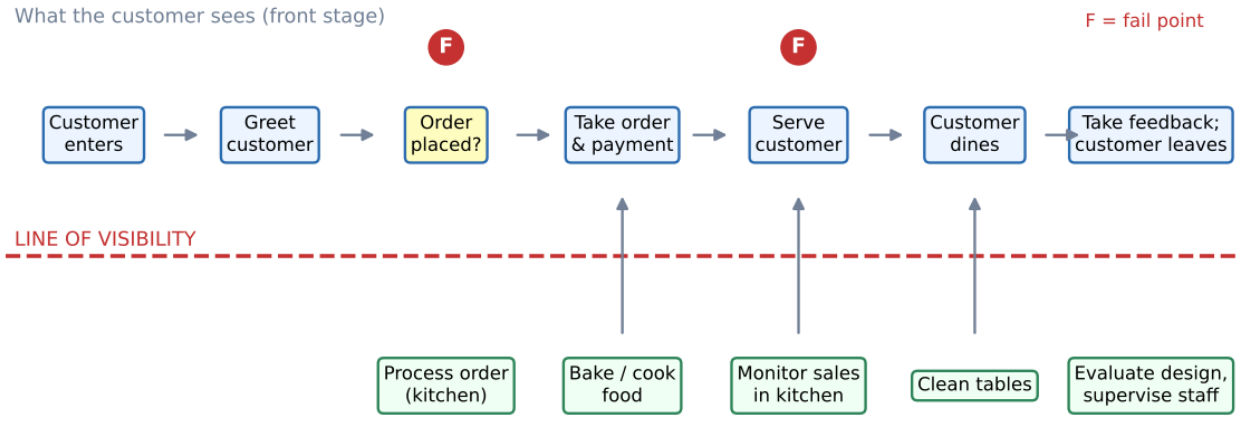
9.6 Service blueprinting

In simple words
A service blueprint is a flow chart of a service that also shows what the customer sees (above the line of visibility) and what happens behind the scenes, and marks fail points.

Everyday example
Movie theatre: customer sees ticket counter, entry check, seating (front stage); behind the scenes are projection, cleaning between shows, snack preparation. Fail point: ticket scanner not reading the QR code.

Flow charts are used in manufacturing; **blueprints** are used in services. A blueprint shows the customer's actions, front-stage employee actions, the **line of visibility** (above it the customer sees; below it is back-stage) and support processes.

Symbol (slide)	Meaning
Rectangle	Operation / task
Triangle / inverted triangle	Storage / queue (waiting)
Diamond	Decision point
Arrow	Movement / flow
Oval / rounded	Start / termination
Circled F	Fail point — where the service is likely to fail



Restaurant blueprint (simplified from the slide): front-stage steps above the line of visibility, kitchen and management steps below; F marks fail points.

- **Restaurant (slide):** customer enters → greet → order placed? (decision) → take order & payment → serve → customer dines → take feedback → return? → leaves / end service. Below the line: process order, bake food, monitor sales in kitchen, clean tables; management: revise strategy, allocate resources, evaluate service process design, supervise staff.
- **Automobile service blueprint (slide):** customer enters → greet & obtain vehicle info → customer specifies problem → preliminary diagnosis (F) → cause clear? → if not, detailed diagnosis → prepare cost & time estimate → customer approves → perform work → work verified (F) → vehicle cleaned → invoice → customer notified → pays → departs.

Key takeaways
Uses of blueprinting: identify **fail points**; make the process **fail-safe** (poka-yoke); **measure the opportunities** and hence DPMO; identify constructive opportunities for improvement.

9.7 IMPROVE

In simple words
Design and test solutions that remove the root causes, starting with those giving the biggest gain, using simple, sustainable changes.

Everyday example
Food-delivery fix: send the order to the restaurant first and assign the rider only when food is 5 minutes from ready. Pilot in one area, measure, then roll out.

- Look at what makes the maximum improvement (vital few causes); use simple improvement schemes; measure; control.
- Improvement must be **sustainable**. Three levels: **process/system level** (e.g., 5S, redesign), **individual level** (training, incentives), **lean principles** (reduce service cost, improve delivery time, expand capacity).
- Methods of improvement: process blueprints → process improvement → process redesign → process management.

Class exercise — Lean ideas for a typical restaurant (Day 5, slide 42) [Day 5]

Find ways to reduce service cost, increase capacity and improve service delivery time WITHOUT compromising quality and market focus.

Goal	Example ideas
Reduce service cost	Standardise recipes and portioning; cross-train staff; reduce menu complexity; self-ordering kiosks/QR menus; prep in batches during idle time; cut food waste (FIFO, demand forecasting)
Increase capacity	Re-layout kitchen to reduce motion; flexible seating (2-tops that combine); move non-value work off-peak; parallel stations; remove bottleneck (e.g., add a second fryer)
Improve delivery time	Pre-prepare common components; limit choices at peak; separate takeaway line; visual order boards (kitchen display); pay-at-table; 5S work stations

9.8 CONTROL

In simple words
Make sure the improvement stays: standard procedures, training, control charts and regular review of key metrics, so the process doesn't slip back.

Everyday example
After fixing the delivery process, a daily dashboard tracks % late orders on a p-chart; if it rises above the control limit, the area manager investigates the same day.

- In manufacturing: SPC control charts (Section 12) — 'management by exception'.
- **EMBED it**: make it a habit; 5S; key leaders' support; watch key metrics; sustain the transformational change.
- Also: standard operating procedures, documentation, training, mistake-proofing, control plans, process owners. It is not easy — the biggest risk is sliding back to the old way.

9.9 More from the textbook (Chapter 9)

In simple words
The textbook adds other improvement methods (PDCA, creative problem solving), how Six Sigma is organised (belts and champions), how projects are chosen, the tool kit for each DMAIC phase, and lean tools.

Everyday example
Like a cricket team: the champion is the team owner who funds and backs the project, the black belt is the captain, green belts are players, and DMAIC is the game plan.

Process-improvement methodologies

In simple words
PDCA and creative problem solving are general-purpose improvement loops; many companies build their own versions of these steps.

Everyday example
Creative problem solving for 'students skip 8 am lectures': understand the mess → gather facts (attendance data) → define the real problem (bus timings) → brainstorm ideas → choose the best (shift class to 9 am) → implement.

- **Deming cycle (PDSA)**. Example 9.1: a diner with 15-minute lines. *Plan*: collect data (people in line, empty tables, time to get food) and test three changes (e.g., expand self-serve, separate order-taking from food preparation, add a waiting/order area). *Do*: try them. *Study*: waiting time fell sharply and more customers were served. *Act*: make the changes permanent and look for the next improvement.
- **Creative problem solving** (Osborn–Parnes): understanding the 'mess' → finding facts → identifying specific problems → generating ideas → developing solutions → implementing solutions.
- **Custom methodologies**: many firms adapt PDCA into their own step model (review situation, describe process, explore causes, collect & analyse data, generate solutions, remeasure).

Implementing Six Sigma — people and projects

In simple words
Six Sigma needs trained people in defined roles and carefully selected projects that matter to customers and the bottom line and can be finished in a few months.

Everyday example
Choosing between 'reduce billing errors (₹50 lakh savings, 80% chance, 4 months)' and 'redesign the whole ERP (₹2 crore, 30% chance, 2 years)': the first usually wins as a Six Sigma project.

Role	Responsibility
Champions / sponsors	Senior managers who select projects, provide resources, remove barriers
Master Black Belts	Full-time experts; train and coach Black Belts; statistical authorities
Black Belts	Full-time project leaders, ~4 weeks of training, lead teams through DMAIC
Green Belts	Part-time project members/leaders with basic training (do Six Sigma alongside their jobs)
Team members / Yellow Belts	Process experts who contribute knowledge and data

- **Project selection criteria**: financial return (savings − cost), impact on customers and organisational effectiveness, probability of success, impact on employees, fit to strategy and competitive advantage. A common score is the Pareto Priority Index PPI = (savings × probability of success) ÷ (cost × time to complete).
- Good projects are scoped to finish in ~4–6 months with clear, measurable CTQs and a sponsor.

DMAIC tools (Textbook Table 9.3)

In simple words
Each DMAIC phase has its typical tools — charters and SIPOC to define, check sheets and capability to measure, fishbone and regression to analyse, experiments and mistake-proofing to improve, control charts and SOPs to control.

Everyday example
Like a doctor: define (patient's complaint), measure (tests), analyse (diagnosis), improve (treatment), control (follow-up check-ups).

Define	Measure	Analyse	Improve	Control
Project charter, SIPOC, VOC/CTQ, cost-of-quality analysis, Pareto analysis, high-level process map	Check sheets, detailed process maps, descriptive statistics, run charts, measurement-system evaluation, process-capability analysis, benchmarking, DPMO/sigma	Cause-and-effect diagrams, scatter diagrams, statistical inference / hypothesis tests, regression, FMEA, root-cause analysis (5 Whys)	Brainstorming, design of experiments, mistake-proofing (poka-yoke), lean production tools, Deming cycle, seven management & planning tools	Control charts, standard operating procedures, control plans, training, mistake-proofing

A3 report (Toyota): a one-page story — theme, background, current condition, root-cause analysis (5 Whys), target condition, countermeasures, implementation plan, follow-up.

Lean tools for process improvement

In simple words

Lean removes waste — anything the customer wouldn't pay for: waiting, extra movement, overproduction, excess stock, defects, over-processing and unused talent. Tools include 5S, visual controls, pull systems, quick changeovers and kaizen.

Everyday example

5S in your study desk: **Sort** (throw old papers), **Set in order** (pens in one holder), **Shine** (clean the desk), **Standardise** (same place every day), **Sustain** (make it a habit).

- **Lean** = eliminate waste (muda). The 8 wastes: overproduction, waiting, transportation, over-processing, inventory, motion, defects, unused talent.
- **5S**: Sort (Seiri), Set in order (Seiton), Shine (Seiso), Standardise (Seiketsu), Sustain (Shitsuke) — an organised, clean, visual workplace.
- **Visual controls**; **efficient layout & standardised work**; **pull production / kanban** (make only what is consumed); **single-minute exchange of dies (SMED)** — quick changeovers; **Total Productive Maintenance (TPM)**; **kaizen** and kaizen blitz (rapid focused improvement events); **value-stream mapping**.
- **Lean Six Sigma** combines lean's speed/waste focus with Six Sigma's variation/defect focus; lean tools work well in services (hospitals, banks) too.

PART D · CAPABILITY & SPC

10. Process Capability: Cp, Cpk and the Flyrock Tires Case

Slides Day 5 (17 Oct) & Day 6 (18 Oct) · Textbook Chapter 8 — Process Capability Measurement

In simple words
Stable means the process behaves predictably (only random, common-cause variation). **Capable** means that predictable behaviour fits inside the customer's specification limits. A process can be stable but incapable (consistently bad) or capable on paper but unstable (unpredictable). You want both.

10.1 Voice of the process vs voice of the customer

In simple words
The customer sets specification limits (what they will accept). The process has its own natural spread (what it actually produces). Capability compares the two. Control limits describe the process, not the customer — never mix them up.

Everyday example
A school bus must reach school between 7:50 and 8:00 (customer's limits). The bus naturally arrives between 7:45 and 8:05 (process's spread). The process is not capable even if it is perfectly predictable.

- **Process capability** = the ability of a process to *consistently* meet requirements — its natural behaviour after all causes of instability are removed. It is the **voice of the process**.
- **Specification (tolerance) limits** USL/LSL come from the customer/design — the **voice of the customer**.
- **Control limits** UCL/LCL come from the process data (±3σ of the plotted statistic) — they tell you about **stability**, not about the customer. Never confuse them.
- Improvement is needed if the voice of the process is inferior to the voice of the customer.

The quality-assurance loop (Day 5 slide 51 / Day 6 slide 19)

In simple words
First make the process stable (remove special causes), then check whether it is capable (meets specs). If stable but not capable, management must change the system.

Everyday example
A tailor's stitching machine sometimes jams (special cause) — fix the machine first. Once it runs smoothly, check whether its stitch length is within the customer's requirement; if not, buy a more precise machine (system change).

1. Observe and document the process.
2. Check for variations using **control charts**.
3. If **out of control** → assess the special causes of variation → remove/address them → check again.
4. If **stable** → ask: is it within tolerance (capable)? If YES → keep monitoring. If NO → devise and implement an improvement (attack common causes — a management/system change) → check

again.

10.2 Capability indices

In simple words

Cp compares the spec width with the process spread (how capable it could be if centred). Cpk also accounts for how far the mean is from the centre (how capable it actually is). Above 1 = fits inside specs; below 1 = some output outside.

Everyday example

Parking a car in a garage 3 m wide when the car needs 2 m of 'wobble space': Cp = 3/2 = 1.5. If you park too close to the left wall, the real margin on that side is small — that smaller side is what Cpk measures.

```
Cp = (USL − LSL) / 6σ − potential capability (ignores centring)
          CPU = (USL − μ) / 3σ CPL = (μ − LSL) / 3σ
Cpk = min(CPU, CPL) = min[(μ − LSL)/3σ , (USL − μ)/3σ] − actual
                            capability
    (If the process is centred, Cp = Cpk. Cpk ≤ Cp always.)
```

Cpk value	Meaning (slide)
> 1	Process is better than specifications (the ±3σ spread fits inside with room to spare)
= 1	Process just meets specifications (±3σ touches a limit)
between 0 and 1	Process does not meet specs (part of the spread falls outside)
= 0	Does not meet specs — the mean coincides with one tolerance limit (50% out on that side)
< 0	Does not meet specs — the mean is outside the limits (both limits on the same side of the mean)

Exam tip

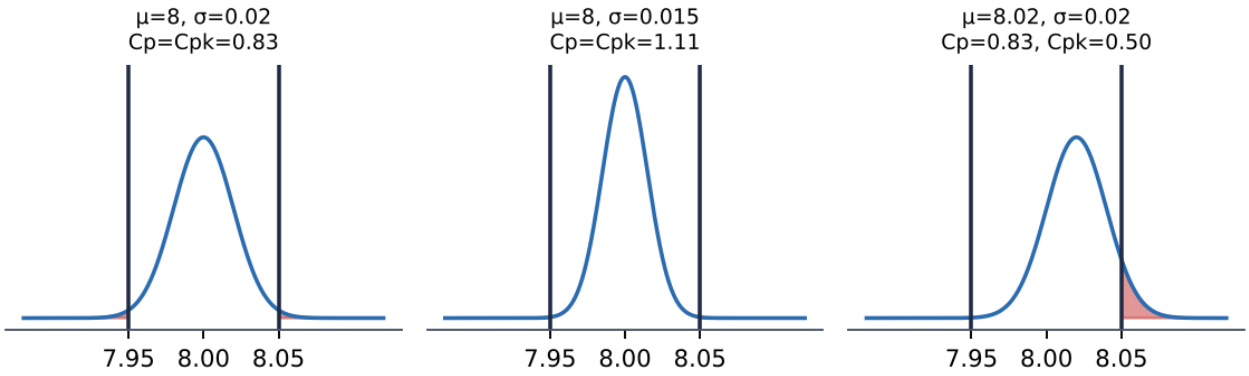
Use **Cp** only for a **centred** process (or to show potential if you centred it). Use CPU or CPL alone for one-sided specs (e.g., 'at most 72 hours').

Common targets: Cpk ≥ 1.33 (4σ to nearest limit) for existing processes; Cpk ≥ 1.67 for new/critical; Six Sigma = Cp 2.0, Cpk 1.5 (after a 1.5σ shift).

Capability indices are meaningful **only if the process is stable** (in statistical control).

Problem 1 — Potato chips (Day 5, slides 52–54) [Day 5]

Spec: thickness 8 ± 0.05 mm (LSL 7.95, USL 8.05). The machine has σ = 0.02 mm. (a) Is it good enough? (b) What if σ is reduced to 0.015? (c) What if (with σ = 0.02) the mean shifts to 8.02?



(a) ±3σ spread	8 ± 3(0.02) = 7.94 to 8.06 — **wider** than the tolerance 7.95–8.05 → some chips will be out of spec.
(a) Cpk	Cpk = min[(8 − 7.95)/0.06, (8.05 − 8)/0.06] = min(0.833, 0.833) = **0.83** (< 1). Out-of-spec ≈ 2 × P(z > 2.5) = 2 × 0.0062 = **1.24%**.
Why 3σ?	±3σ covers 99.73% of output, so if the ±3σ band fits inside the tolerance almost everything conforms.
(b) σ = 0.015	±3σ = 7.955 to 8.045 → fits inside. Cpk = 0.05/0.045 = **1.11** → capable (≈0.09% out).
(c) mean 8.02	Cpk = min[(8.02 − 7.95)/0.06, (8.05 − 8.02)/0.06] = min(1.17, 0.50) = **0.50** → much worse (≈6.7% out, mostly above USL). A shift in the mean leads to higher wastage even though σ hasn't changed.

Answer: No — with σ = 0.02 the machine is not capable (Cpk 0.83). Reducing σ to 0.015 makes it capable (1.11). Causes of poor performance: **higher σ (inconsistency)** and **shift in mean**. Both must be controlled.

Careful — check against the slide
The slide shows Cpk = 0.81 for part (a); the exact value is 0.05/0.06 = 0.833. Either way it is below 1.

Problem 2 — Amy's team selection (Day 5, slide 55) [Day 5]

To be selected, a person must score between 30 and 70 points (target 50). Amy has mean 50, σ = 5. Will she get in? What is her capability? If lack of training shifts her mean to 45, what is her new capability?

Limits	LTL = 30, UTL = 70, 3σ = 15.
Original Cpk	min[(50 − 30)/15, (70 − 50)/15] = min(1.333, 1.333) = **1.33** → yes, she is comfortably capable.
After shift	min[(45 − 30)/15, (70 − 45)/15] = min(1.0, 1.667) = **1.0** → just meets the requirement; any more drift and she risks falling below 30.

Answer: Cpk = 1.33 originally; after the mean shifts to 45, Cpk = 1.0 (Cp is still 1.33 — spread unchanged, only centring got worse).

Textbook Examples 8.9 & 8.11 — U-bolts [Ch 8]

Mean 10.7171, s = 0.0868, specs 10.55–10.90. Find % nonconforming and Cp, CPU, CPL, Cpk.

Below LSL	NORM.DIST(10.55, 10.7171, 0.0868, TRUE) = 0.0271
Above USL	1 − NORM.DIST(10.90, 10.7171, 0.0868, TRUE) = 0.0176
Total	0.0447 → **4.47%** nonconforming
Indices	Cp = 0.35/(6 × 0.0868) = **0.67**; CPU = (10.90 − 10.7171)/0.2604 = **0.70**; CPL = (10.7171 − 10.55)/0.2604 = **0.64**; Cpk = **0.64**

Example 8.13 — raising Cp with fixed spec width 8: Cp = 8/(6σ) → for Cp 1.33, 1.66, 2.00, 2.33 the required σ is 1.0, 0.8, 0.67, 0.57. **Example 8.12** — Cpk is an estimate: with n = 45 and Cpk = 1.15, a 95% CI is (0.89, 1.41) — the true value might be below 1; with n = 400 it is (1.06, 1.24).

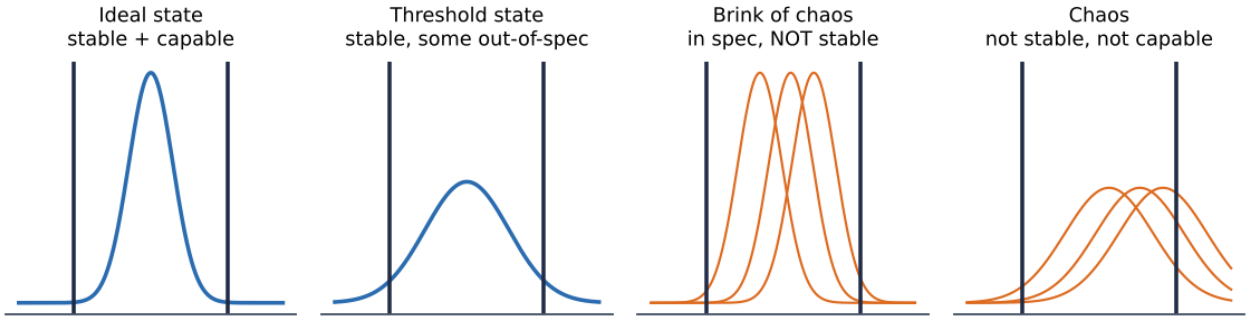
10.3 Types of processes

In simple words

Combining stability and capability gives four states: ideal (both), threshold (stable but some defects), brink of chaos (no defects yet but unpredictable), and chaos (neither).

Everyday example

A student who always scores 85–90 is ideal; always 30–40 against a pass mark of 40 is threshold (predictable but failing); scoring 50–95 unpredictably is on the brink of chaos.



State	Stable (in control)?	Conformance	Typical Cpk	Action
Ideal state	Yes	100% within spec	> 1	Monitor; continue improving
Threshold state	Yes	Some nonconforming	close to 1 / < 1	Reduce common-cause variation or re-centre (system change)
Brink of chaos	No	100% (for now)	close to 1 (> 1 on a sample)	Remove special causes before they push output out of spec
State of chaos	No	Some nonconforming	meaningless	First stabilise (remove special causes), then improve capability

Concept check (Day 6, slides 4–6) [Day 6]

Based on a sample study the Cpk of a process was calculated as more than 1. While operating, the process is found to be not stable. Then...

Answer: The Cpk value **cannot be trusted** — capability is defined only for a stable process. The process is on the 'brink of chaos': conforming now, but special causes can push it out of spec at any time. First remove the special causes, then recompute capability. (The slide's picture: USL 10.5 > UCL 10.3, LCL 9.7 > LSL 9.5 — points at 10.4 are *out of control* but still *within spec*.)

10.4 More from the textbook: capability studies, performance indices and pre-control

In simple words

A capability study is a planned data-collection exercise. Pp/Ppk use the overall long-term spread; Cp/Cpk use short-term (within-subgroup) spread. Pre-control is a simple traffic-light check using the specification limits.

Everyday example

Pre-control at a juice counter where a glass should hold 290–310 ml: green 295–305, yellow 290–295 or 305–310, red outside 290–310. Two yellows in a row → adjust the dispenser; any red → stop.

- **Process capability study** steps: choose a representative machine/segment; define process conditions; select a representative operator; provide standard materials; specify gauging; plan sampling; collect data; plot (histogram/control chart); compute indices; analyse and act. Types: peak-performance study, process characterisation, component variability.
- **Histograms alone can mislead** (Example 8.10): overall data looked centred at 0.762 vs target 0.75, but a run chart showed the mean shifted at sample 17 (0.738 → 0.789) — the process was never actually centred. Always look at data over time.
- **Process performance indices Pp, Ppk**: same formulas as Cp, Cpk but using the *overall* (long-term) standard deviation of all data rather than the within-subgroup σ (R-bar/d_2). Use Cp/Cpk for a stable process; Pp/Ppk describe actual past performance.
- **Pre-control** (Example 8.14, spec 3–7 g): divide the tolerance into zones — **green** = middle half (4–6), **yellow** = outer quarters (3–4 and 6–7), **red** = outside spec (< 3 or > 7). Rules: start when 5 consecutive units are green; then sample two consecutive units periodically — both green, or one green and one yellow → continue; two yellows in the same zone → adjust; yellows in opposite zones → stop and investigate (variation increased); any red → stop. Simple, but based on specs, not on process statistics.

10.5 Case: Six Sigma Quality at Flyrock Tires (Day 6, slides 7–16)

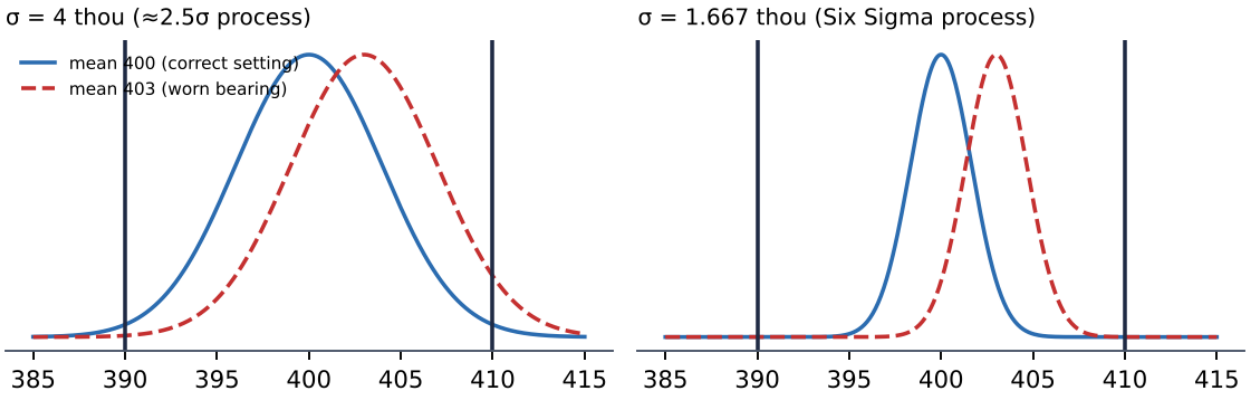
In simple words

The case compares a normal process (σ = 4) with a Six Sigma process (σ = 1.667): the better process makes almost no defects and its control chart catches problems almost instantly.

Everyday example

Like two smoke alarms: the cheap one takes 4 hours on average to notice a fire, the better one notices within the first check. Plus the better house is far less likely to catch fire at all.

Data: extruder output thickness — mean 400 thou, σ = 4 thou. Tyre thickness specification 400 ± 10 thou (LSL 390, USL 410). Operators sample 10 sheets every hour.



Flyrock questions 1–3: the original process (σ = 4)

In simple words
Use z-scores to find the fraction outside the 390–410 spec; use σ/√n for the control limits on sample averages; then find how likely a shifted process is to show up on the chart.

Everyday example
Q2's key idea: averages of 10 sheets vary much less than single sheets (σ/√10), so the control limits on the chart are tighter than the spec limits.

Q1 [Flyrock]

If the extruder setting is accurate, what proportion of rubber extruded will be within specifications?

z	z = (410 − 400)/4 = 2.5 (and −2.5 at LSL).
Tail	Appendix A: P(z > 2.5) = 0.0062 each side → total out = 2 × 0.0062 = 0.0124.
Result	Within spec = 1 − 0.0124 = **98.76%**; i.e., **124 of every 10,000 sheets are defective**.

Q2 [Flyrock]

Douglas asks operators to take 10 sheets every hour and use the average to decide whether the process is in control. With three-sigma control limits, what UCL and LCL should she specify?

Key idea	We plot the **sample mean**, whose standard deviation (standard error) is σ/√n, not σ.
Standard error	$\sigma_{\bar{x}} = 4/\sqrt{10} =$ **1.2649**.
Limits	UCL = 400 + 3(1.2649) = **403.795**; LCL = 400 − 3(1.2649) = **396.205**.

Careful — check against the slide
The slide prints 'UCL = 10 + 3 × 1.2649'; it means 400 + 3 × 1.2649 = 403.795 (the answer shown is correct).

Q3 [Flyrock]

A worn bearing shifts the mean to 403 (setting still 400). What proportion is defective? With the Q2 limits, what is the probability that an hourly sample is out of control? On average how many hours pass before the worn bearing is detected?

Defects above USL	$z_1 = (410 - 403)/4 = 1.75 \rightarrow P = 0.0401$.
Defects below LSL	$z_2 = (403 - 390)/4 = 3.25 \rightarrow P = 0.000577$. (Slide typo: '403 − 290'.)
Proportion defective	0.0401 + 0.000577 = **0.0406** → about **406 per 10,000** (vs 124 before).
Sample within limits?	Sample means now ~ Normal(403, 1.265). P(396.205 < $\bar{x}$ < 403.795) = NORM.DIST(403.795, 403, 1.265, 1) − NORM.DIST(396.205, 403, 1.265, 1) = **0.7351**.
Detection probability	P(out of control in an hour) = 1 − 0.7351 = **0.2649**.
Average time to detect	1/0.2649 = **3.77 ≈ 4 hours** (average run length). Meanwhile ~4% of output is defective (the slide estimates about 1,600 defective sheets by then).

Flyrock questions 4–6: the Six Sigma process (σ = 1.667)

In simple words

Repeat the same three calculations with the smaller σ — defects nearly vanish, control limits tighten, and a shift of 3 thou is detected almost immediately.

Everyday example

Halving the wobble of a process both reduces defects and makes any new problem stand out clearly, like spotting one grey hair in black hair.

Q4 [Flyrock]

Now extrusion is a Six Sigma process: σ = 1.667 thou. What proportion is within specification?

z	$z = (410 - 400)/1.667 = 6.0 \rightarrow$ the limits are ±6σ away.
Result	P(within) = 0.999999998 → only about **2 defects per billion** (the slide quotes 0.99999999901 ≈ 4 per billion — same idea: practically zero).

Q5 [Flyrock]

With samples of 10 each hour, what control limits for the Six Sigma process?

Standard error	$1.667/\sqrt{10} =$ **0.527**.
Limits	UCL = 400 + 3(0.527) = **401.58**; LCL = 400 − 3(0.527) = **398.42**. Control limits tighten because the process σ has reduced.

Q6 [Flyrock]

For the Six Sigma process, the worn bearing again shifts the mean to 403. Proportion defective? Probability a sample is out of control? Hours to detection?

Defective	Within spec = NORM.DIST(410, 403, 1.667, 1) − NORM.DIST(390, 403, 1.667, 1) = 0.999987 → ≈ **0.0013%** defective (≈13 per million).
Sample within limits	NORM.DIST(401.581, 403, 0.527, 1) − NORM.DIST(398.419, 403, 0.527, 1) = **0.0036** → less than 1% chance a sample stays inside.
Detection	P(signal) = 0.9964 → detected essentially at the **very next hourly sample** (average run length ≈ 1.0 hour).

Key takeaways

Effect of quality improvement: defects fall from 124 per 10,000 to a few per billion.

Power of control charts: they detect a mean shift — slowly (≈4 h) for a 2.5σ process, almost instantly for a 6σ process.

Effect of Six Sigma: even when something goes wrong (worn bearing), defects stay tiny and the problem is caught at once.

PART D · CAPABILITY & SPC

11. Measurement System Evaluation (Textbook Chapter 8)

Metrology · accuracy & precision · calibration · repeatability & reproducibility (gauge R&R;)

In simple words
Before trusting any chart, make sure the ruler is good. If the measuring device or the people using it vary a lot, you will 'see' variation that isn't really in the product.

Metrology, accuracy, precision and calibration

In simple words
Metrology is the science of measurement. Accuracy = close to the true value on average; precision = repeated readings close to each other; calibration = adjusting an instrument against a known standard.

Everyday example
A kitchen scale that always shows 50 g extra is precise but inaccurate — calibrate it (set zero) and it becomes both. A scale whose readings jump around is imprecise; calibration won't fix that.

- **Metrology**: the science of measurement. A measurement system = instruments + people + methods + environment.
- **Accuracy** = closeness of the average measurement to the true value (bias). **Precision** = closeness of repeated measurements to each other (low spread).
- **Calibration**: comparing an instrument against a known standard and correcting it, traceable to national standards.
- **Repeatability** (equipment variation, EV): variation when the *same* operator measures the *same* item repeatedly with the same gauge.
- **Reproducibility** (appraiser variation, AV): variation in the averages when *different* operators measure the same items.

Textbook Example 8.6 — Accuracy vs precision [Ch 8]

True dimension 0.250 in. Instrument A reads 0.248; B reads 0.259. Repeated readings: A = 0.248, 0.246, 0.251; B = 0.259, 0.258, 0.259.

Accuracy	Relative error A = (0.250 − 0.248)/0.250 = 0.8%; B = (0.259 − 0.250)/0.250 = 3.6% → **A is more accurate**.
Precision	B's readings are tightly clustered (0.258–0.259); A's spread 0.246–0.251 → **B is more precise**.

Answer: A = accurate but not precise; B = precise but biased (like John the shooter) — B can be fixed by calibration.

Gauge R&R; study (Example 8.7)

In simple words

A gauge R&R; study checks how much of the observed variation comes from the measuring system: repeatability (same person, same gauge, repeated) and reproducibility (different people). If R&R; is a big share of total variation, fix the measurement first.

Everyday example

Three nurses measure the same patients' blood pressure twice each. If readings differ a lot between nurses, the problem is technique (reproducibility); if the same nurse gets different readings, the problem is the device (repeatability).

Ten parts measured twice by each of three operators (gasket thickness, spec 0.50–1.00 mm). Operator averages 0.830, 0.774, 0.829 → $\bar{x}_D$ = 0.830 − 0.774 = 0.056. Average ranges 0.037, 0.034, 0.017 → R-bar = 0.0293. Range control limit D_4R-bar = 3.27 × 0.0293 = 0.096 (all ranges below → no special causes in measuring).

$$\text{EV (repeatability)} = K_1 \times \text{R-bar} = 4.56 \times 0.0293 = 0.134$$

$$\text{AV (reproducibility)} = \sqrt{[(\bar{x}_D \times K_2)^2 - EV^2/(n \cdot r)]} = \sqrt{[(0.056 \times 2.70)^2 - 0.134^2/20]} = 0.147$$

$$R\&R; = \sqrt{(EV^2 + AV^2)} = 0.199 \quad PV = R_p \times K_3 = 0.525 \times 1.62 = 0.851 \quad TV = \sqrt{(R\&R;^2 + PV^2)} = 0.873$$

$$\%R\&R; \text{ of total variation} = 0.199/0.873 \approx 23\%$$

Exam tip

Guidelines: %R&R; < 10% → measurement system acceptable; 10–30% → marginal (may be acceptable depending on cost/importance); > 30% → unacceptable, fix the gauge/method. The textbook concludes this gauge is at best marginal and its variation should be reduced.

PART D · CAPABILITY & SPC

12. Statistical Process Control — Foundations

Slides Day 5 (slides 58–72), Day 6 & Day 7 · Textbook Chapters 6 & 8

In simple words
Instead of inspecting every finished item (expensive and too late), take small random samples **while** the process runs and plot them on a chart with limits. If points stay inside the limits with no strange pattern, leave the process alone. If not, stop and look for the special cause.

12.1 Ensuring a process continuously meets requirements

In simple words
Instead of checking every finished item (expensive and too late), take small samples while the process runs and plot them over time. That is Statistical Process Control.

Everyday example
A cook tastes the dal while it is cooking (SPC) rather than waiting for customers to complain after eating (inspection).

- **Reactive model** (inspect output, fix after) vs **proactive model** (control the process as it runs).
- **Census** (check everything) vs **sampling** (check a few).
- **Statistical Quality Control / SPC**: testing a random sample of output from a process to determine whether the process is producing items within a pre-selected range.
- What to measure: **attributes** (counts: good/bad, number of defects) and **variables** (measurements: length, time, weight). Method: **control charts**.

Control chart = a graph of the performance of a process (feature) over time, arranged to emphasise process variation. Examples on the slide: number of aeroplanes leaving within 30 seconds of scheduled time, number of good burgers in a batch, average number of satisfied customers per day, average marks of students in different evaluations — all are a **statistic tracked over time with a centre line and limits**.

Uses of control charts (slide)	
Detect when something is wrong with the process	Establish what the process is inherently capable of achieving
Diagnose the causes of abnormal behaviour	Monitor and control the process — **management by exception**

12.2 Statistics you need (Day 5 slides 63–71, Textbook Ch 6)

In simple words
Different data need different distributions: binomial for pass/fail counts, Poisson for numbers of defects, normal for measurements, exponential for time to failure. Sample averages are approximately normal (Central Limit Theorem), with spread $\sigma/\sqrt{n}$.

Everyday example
Binomial: how many of 20 eggs in a tray are cracked. Poisson: how many potholes on a 1 km road. Normal: weight of 1 kg sugar packets. Exponential: hours until a bulb fails.

- Experiment, outcome, sample space; probability = likelihood of an outcome; **random variable** assigns a value to each outcome; probability distribution = probability it takes each value.
- **Discrete distributions**: **Binomial** — pass/fail, defective-rate sampling (basis of p, np charts and OC curves); **Poisson** — defects per unit/area/time (basis of c, u charts); **Hypergeometric** — acceptance sampling without replacement from a small lot.
- **Continuous distributions**: **Normal** — many measurements aggregate to normal (Central Limit Theorem); standard normal & z-scores; **Exponential / Weibull** — time-to-failure & reliability.
- **Statistical methodology** (Ch 6, p. 269): data collection → organisation & presentation (frequency distributions, histograms) → descriptive measures (central tendency, dispersion) → statistical inference (hypothesis testing, ANOVA, experimental design) → predictive statistics (correlation, regression).
- **Statistical inference**: drawing conclusions about a population from sample data — confidence intervals, hypothesis testing, experimental design. **Predictive statistics**: predictions from historical data.
- **Sampling methods**: simple random, stratified, systematic, cluster, judgement. Population parameters (μ, p, σ, N, ρ) vs sample statistics ($\bar{x}$, $\hat{p}$, s, n, r).
- **Sampling distribution**: the distribution of a statistic over all possible samples of a fixed size. **Central Limit Theorem**: sample means are approximately normal (even if the population isn't), with mean μ and standard error σ/√n — the distribution narrows as n grows (n = 5 vs n = 30 in the slide).

12.3 Control limits

In simple words
Control limits are the mean ± 3 standard deviations of what you plot. If only common causes act, 99.73% of points fall inside, so a point outside almost certainly signals a special cause.

Everyday example
Your daily phone screen time is usually 3 ± 0.5 h (limits 1.5–4.5 h). A day with 7 hours is outside the limits — something special happened (exam over, new game).

```
Control limits = centre line ± 3 × (standard deviation of the plotted
                                    statistic)
  ≈ 99.73% of points fall inside if only common causes are present
```

- Points outside ±3σ are so unlikely (0.27%) under common causes that we assume a special cause.
- A point beyond the UCL/LCL = 'possible problem, investigate'. A stable chart = 'normal behaviour'.
- Control limits are calculated from the process; they are **not** specification limits.

12.4 Which control chart? (depends on the type of data)

In simple words

Pick the chart by asking: am I counting or measuring? If counting — whole bad units (p, np) or number of flaws (c, u)? If measuring — subgroup size small (x-bar & R), large (x-bar & s) or one at a time (I-MR)?

Everyday example

% of late trains per day → p. Number of typing errors per page → c. Errors per page when pages differ in length → u. Weight of 5 biscuits packs every hour → x-bar & R. One daily electricity meter reading → I-MR.

Data type	What is counted / measured	Subgroup size	Chart
Attribute — defectives (classification)	Fraction defective	Constant or variable n	**p-chart**
Attribute — defectives	Number defective	Constant n	**np-chart**
Attribute — defects (count)	Number of defects per unit, same area of opportunity	Constant	**c-chart**
Attribute — defects	Defects per unit, area of opportunity varies	Variable	**u-chart**
Variable (measurement)	Mean and range	Small, n = 2–9 (typically 4–5)	**x-bar & R**
Variable	Mean and standard deviation	n ≥ 10	**x-bar & s**
Variable	Individual values	n = 1	**Individuals & moving range (I-MR)**

PART D · CAPABILITY & SPC

13. Attribute Control Charts: p, np, c and u — every class example solved

Slides Day 6 (18 Oct) & Day 7 (25 Oct) · Textbook Chapter 8

In simple words

Attribute data is 'yes/no' or 'how many' data: a form has an error or not; a cloth has 3 flaws. **Defective** (classification: p, np charts) = the whole unit is bad. **Defect** (count: c, u charts) = one flaw; a unit may have several.

Chart	Plots	Centre line	Control limits	When
p	fraction defective $p_i = d_i/n$	p-bar = Σd / Σn	p-bar ± 3√[p-bar(1 − p-bar)/n]	Defectives; n can vary
np	number defective d_i	n·p-bar	n·p-bar ± 3√[n·p-bar(1 − p-bar)]	Defectives; constant n
c	defects per unit c_i	c-bar = Σc / k	c-bar ± 3√c-bar	Defects; same area of opportunity
u	defects per unit $u_i = c_i/a_i$	u-bar = Σc / Σa	u-bar ± 3√(u-bar/a_i)	Defects; area of opportunity varies

Exam tip

If a calculated LCL is negative, set **LCL = 0** (you can't have a negative proportion or count).

Subgroup size for p-charts should be large enough that the average number of defectives per subgroup is at least ~2 (rule of thumb: if you expect 1% defective, take n ≈ 400). Use at least **25 subgroups** for p and np charts.

p or np charts: the area of opportunity is **discrete** and a single defect makes the unit defective. c and u charts: the area of opportunity is **continuous** (a piece of cloth, a computer program, a washing machine, an answer sheet) and many defects can occur in it.

Steps to create a p-chart (slide): (1) select random samples of size n; (2) count defectives in each sample; (3) compute each p_i, then p-bar, σ_p, UCL and LCL; (4) draw the chart and interpret.

13.1 The p-chart (fraction defective)

In simple words

A p-chart tracks the proportion of defective items in each sample over time. Its limits widen for small samples and narrow for large ones.

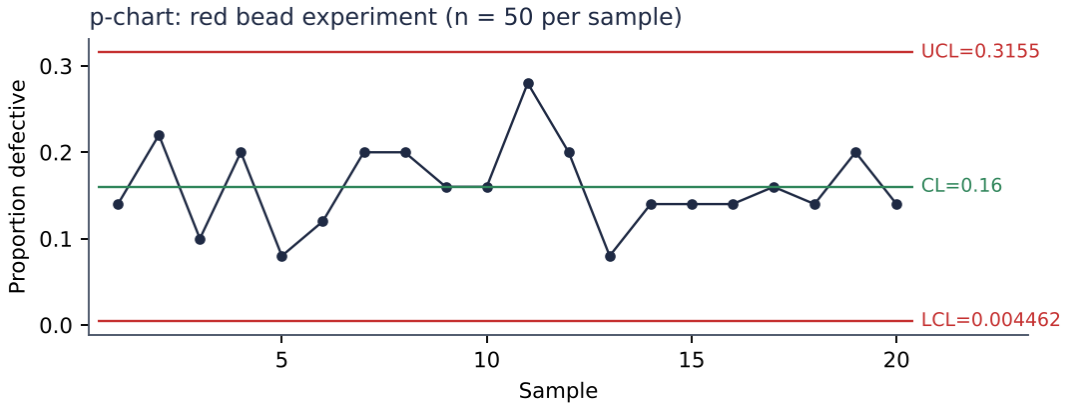
Everyday example

A call centre checks 100 calls a day and plots the fraction handled wrongly. If normally 5% ± a little, a day with 15% wrong is a signal to investigate.

Example 1 — Red bead experiment (Day 6, slides 28–31) [Day 6]

20 samples of 50 beads; red beads per sample: 7, 11, 5, 10, 4, 6, 10, 10, 8, 8, 14, 10, 4, 7, 7, 7, 8, 7, 10, 7. Was the variation random or were there assignable causes?

p_i	Fraction defective: 7/50 = 0.14, 11/50 = 0.22, … , 14/50 = 0.28 (sample 11) …
p-bar	Total red = 160; total beads = 20 × 50 = 1,000 → p-bar = **0.16**.
σp	√(0.16 × 0.84 / 50) = √0.002688 = **0.0518**.
Limits	UCL = 0.16 + 3(0.0518) = **0.3155**; LCL = 0.16 − 3(0.0518) = **0.0045**.
Check points	Highest p = 0.28, lowest = 0.08 → all inside; no runs or trends.



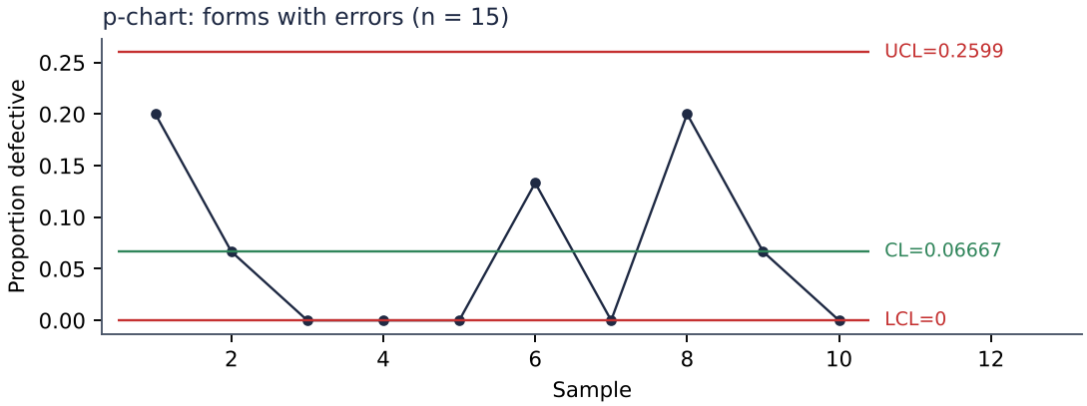
Answer: The process is **in statistical control** — all variation is common-cause (random). Differences between 'workers' are produced by the system; rewarding or punishing them is pointless (Deming).

Careful — check against the slide
The slide shows 'Sigma p = (0.16 × 0.84)/50 = 0.015' — the square root is missing. √(0.002688) = 0.0518, which gives the UCL 0.3155 / LCL 0.0045 shown on the slide's chart.

Example 2 — Forms with errors (Day 6, slides 32–35) [Day 6]

10 samples of 15 forms; forms with errors: 3, 1, 0, 0, 0, 2, 0, 3, 1, 0. Draw the p-chart. If the customer does not accept more than 2 errors per 15, is any change required?

p_i	0.20, 0.07, 0, 0, 0, 0.13, 0, 0.20, 0.07, 0
p-bar	10/150 = **0.0667**
σp	√(0.0667 × 0.9333 / 15) = **0.0645**
Limits	UCL = 0.067 + 3(0.0645) = **0.2599** (slide 0.2605); LCL = 0.067 − 0.1935 = −0.1265 → **0**
Stability	All points within limits → process is **under control**, no special causes.
Capability	Customer tolerance = 2/15 = **0.1333**. Samples 1 and 8 (0.20) exceed it → the process is stable but **not good enough**.

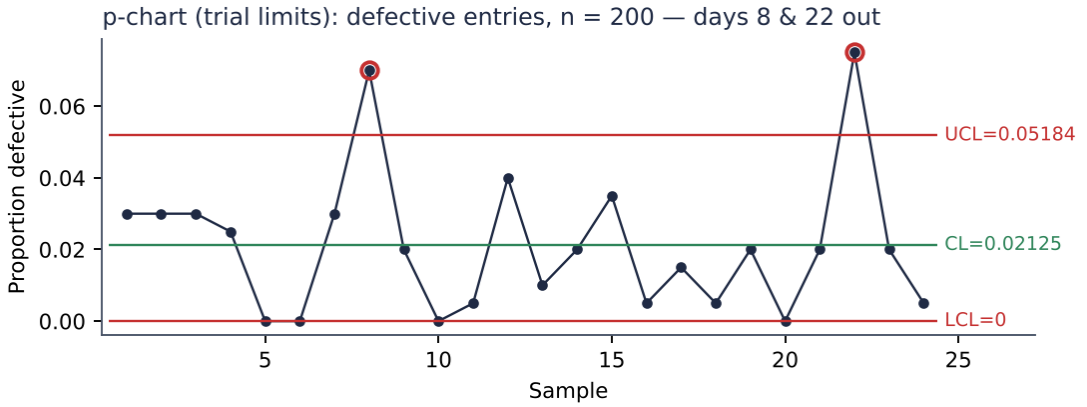


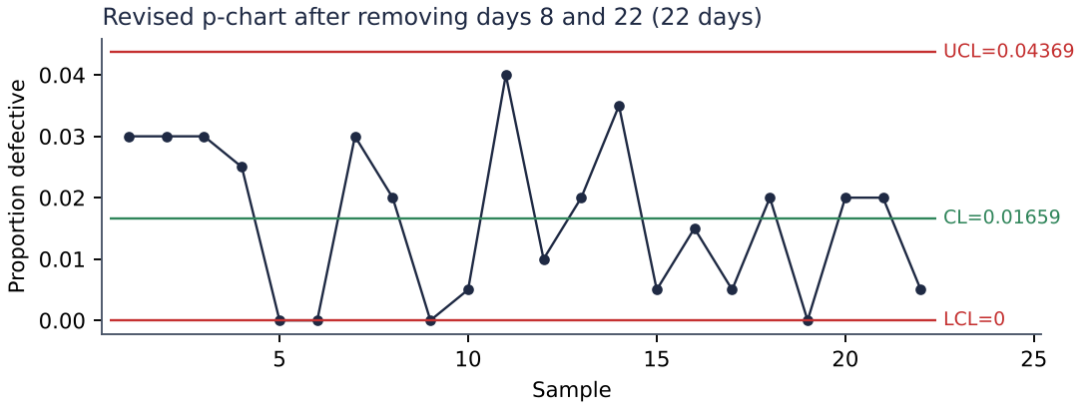
Answer: Stable ≠ capable. Since there are no special causes, the fix must address **common causes** — a system change such as better **training**, clearer forms, checklists.

Example 3 — Process not stable: defective entries (Day 6, slides 36–40) [Day 6]

24 days, 200 entries inspected each day; defectives: 6, 6, 6, 5, 0, 0, 6, 14, 4, 0, 1, 8, 2, 4, 7, 1, 3, 1, 4, 0, 4, 15, 4, 1. Draw the p-chart and act.

Trial limits	Total defectives = 102; p-bar = 102/(24 × 200) = **0.02125**; σp = √(0.02125 × 0.97875/200) = **0.0102** (slide: 0.01).
	UCL = 0.02125 + 3(0.0102) = **0.0518**; LCL negative → **0**.
Out of control	Day 8: 14/200 = 0.070 > 0.0518; Day 22: 15/200 = 0.075 > 0.0518 → **special causes**.
Act	Investigate days 8 and 22, find and remove the special causes (rework those entries).
Revise	Drop days 8 & 22: total = 73 over 22 × 200 = 4,400 → p-bar = **0.01659**; σp = √(0.01659 × 0.98341/200) = **0.0090**; UCL = **0.0437**; LCL = −0.0105 → **0**.
Check	All remaining points (max 8/200 = 0.04) are inside → revised chart is stable; use it to monitor future days.



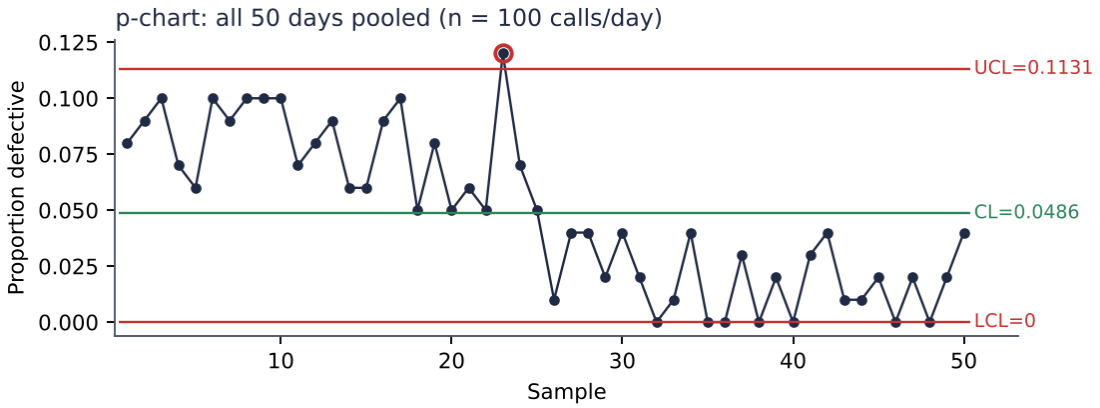


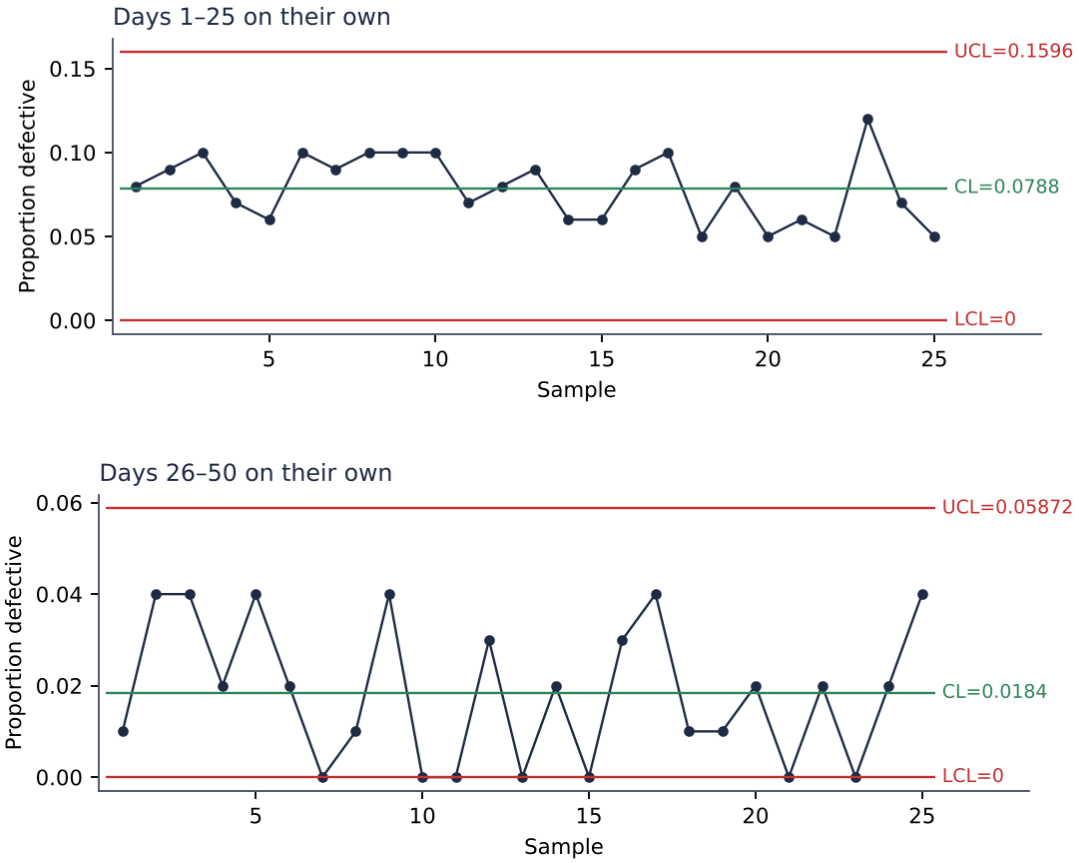
What can you infer (old vs revised)? The revised chart has a lower centre line and tighter limits — it represents the true common-cause performance and will catch future special causes faster.

Example 4 — Call centre: is the process under control? (Day 6 slides 43–46, Day 7 slides 57–59) [Day 6/7]

Defective calls per 100 calls sampled, for 50 days: days 1–25: 8, 9, 10, 7, 6, 10, 9, 10, 10, 10, 7, 8, 9, 6, 6, 9, 10, 5, 8, 5, 6, 5, 12, 7, 5; days 26–50: 1, 4, 4, 2, 4, 2, 0, 1, 4, 0, 0, 3, 0, 2, 0, 3, 4, 1, 1, 2, 0, 2, 0, 2, 4.

Pooled chart	p-bar = 243/5,000 = **0.0486**; σp = √(0.0486 × 0.9514/100) = 0.0215; UCL = **0.1131**; LCL → 0.
Read it	Day 23 (0.12) is above the UCL, and — more importantly — days 1–25 are all **above** the centre line and days 26–50 almost all **below** it: a run of 8+ points on one side → **out of control (shift)**.
Divide the data	Days 1–25: p-bar = 0.0788, UCL = **0.1596**. Days 26–50: p-bar = 0.0184, UCL = **0.0587**. Each half on its own is in control.
Meaning	Something changed around day 25 (e.g., new training, new software) — the process **improved**. The pooled chart mixed two different processes.





Answer: Not in control as a single process — there was a sustained shift. Compute separate limits for 'before' and 'after'; adopt the new (better) limits going forward. Lesson: look for **patterns**, not just points outside limits, and form subgroups from homogeneous conditions.

13.2 Control chart fail points: rational subgrouping

In simple words
Group data so that each subgroup contains only items made under the same conditions. Then variation between subgroups reveals special causes; mixing conditions inside a subgroup hides them.

Everyday example
If you average tea temperatures from two different stalls in one sample, a problem at one stall gets hidden by the other. Sample each stall separately.

Subgrouping is the most common failure point. Choose subgroups so that **homogeneous conditions exist within a subgroup** — variation within a subgroup should be only common cause, so that differences *between* subgroups reveal special causes.

Example 5 — McDonald's service time (Day 6, slides 48–49) [Day 6]

5 observers record service time at 5 counters, sampling 6 consecutive customers every 2 hours from 9 am to 9 pm. How many data points? How should they be subgrouped?

Data points	Sampling times 9, 11, 1, 3, 5, 7 → 6 times (the slide counts 6). Data = 5 counters × 6 customers × 6 times = **180**.

Option A — variation over time (per counter)	Subgroup = the 6 consecutive customers at one counter at one time → 6 points per subgroup, 6 subgroups per counter → **5 control charts**, one per counter.
Option B — across counters and time	Subgroup = 6 customers; 5 such subgroups (counters) at each time period → one chart summarising time & counter variation.
Option C — across customers and time	Subgroup = 5 points (one per counter) for the same customer position; 6 subgroups per time period.

Answer: 180 data points. The right subgrouping depends on the question you want answered (time effect, counter effect, or both) — put only the sources of variation you want to treat as 'common' inside a subgroup.

One more p-chart: burrs

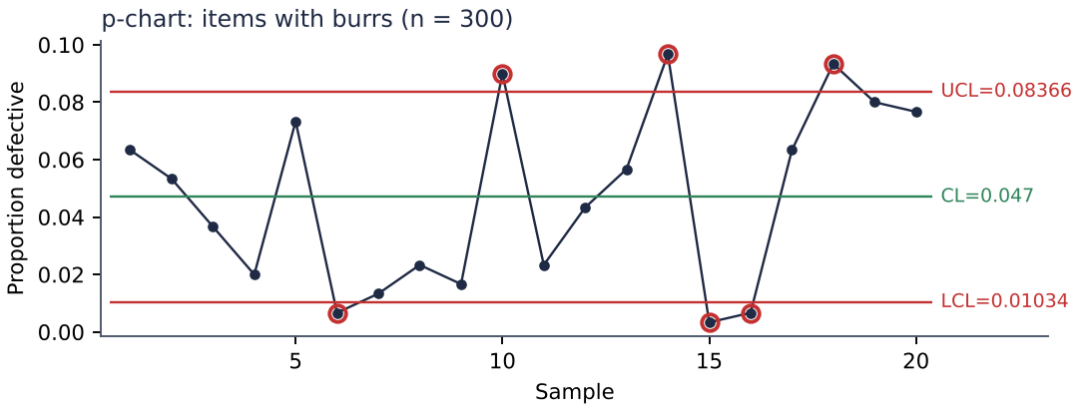
In simple words
Same p-chart method; this example shows a process that is clearly out of control, with points both above and below the limits.

Everyday example
Points below the lower limit are 'too good to be true' — either something genuinely improved that day or the inspector missed defects. Both need checking.

Example 6 — Burrs (Day 6, slides 50–51) [Day 6]

Samples of 300 items; number with burrs in 20 samples: 19, 16, 11, 6, 22, 2, 4, 7, 5, 27, 7, 13, 17, 29, 1, 2, 19, 28, 24, 23.

p_i	0.0633, 0.0533, 0.0367, 0.0200, 0.0733, 0.0067, 0.0133, 0.0233, 0.0167, 0.0900, 0.0233, 0.0433, 0.0567, 0.0967, 0.0033, 0.0067, 0.0633, 0.0933, 0.0800, 0.0767
p-bar	Total 282/6,000 = **0.0470**
σp	√(0.047 × 0.953/300) = **0.0122**
Limits	UCL = 0.047 + 3(0.0122) = **0.0837**; LCL = 0.047 − 0.0366 = **0.0103**
Out-of-control points	Above UCL: samples 10 (0.090), 14 (0.097), 18 (0.093). Below LCL: samples 6 (0.0067), 15 (0.0033), 16 (0.0067).



Answer: The process is **out of control**. Investigate high points (what went wrong?) and also the low points (what went unusually right — or was inspection faulty?).

13.3 The np-chart (number defective)

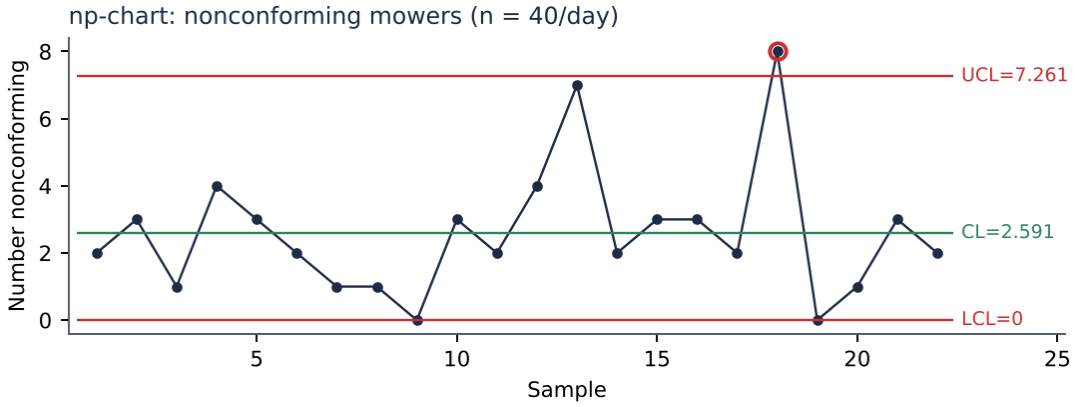
In simple words
Same idea as the p-chart, but you plot the count of defective units instead of the fraction. Only works when every sample is the same size.

Everyday example
Each day check 40 helmets; plot how many failed (e.g., 2, 3, 1 …). Easier for workers to read than '0.05, 0.075, 0.025'.

Example 7 — np-chart: lawn mowers (Day 6, slides 52–56) [Day 6]

22 days, 40 mowers inspected each day; nonconforming: 2, 3, 1, 4, 3, 2, 1, 1, 0, 3, 2, 4, 7, 2, 3, 3, 2, 8, 0, 1, 3, 2.

p-bar	Total 57/(22 × 40 = 880) = **0.0648**
Centre line	n·p-bar = 40 × 0.0648 = **2.59**
σ	√[n·p-bar(1 − p-bar)] = √(2.591 × 0.9352) = **1.557**
Limits	UCL = 2.59 + 3(1.557) = **7.26**; LCL = 2.59 − 4.67 → **0**
Check	Day 18 (8 nonconforming) > 7.26 → out of control. Day 13 (7) is close but inside.



Careful — check against the slide
The slide's spreadsheet uses 'Error = 1.6096' (= √2.59, the Poisson shortcut) giving UCL 7.42. The np-chart formula √[n·p-bar(1 − p-bar)] gives 1.557 and UCL 7.26 — which matches the Minitab chart on the slide. Day 18 is out either way.

How is np different from p? The pattern is identical; np plots the **count** of nonconforming units (each point = number of bad mowers that day), which is easier for shop-floor staff to understand, but needs a **constant** subgroup size. The p-chart plots the fraction and works with varying n.

13.4 The c-chart (defects per unit)

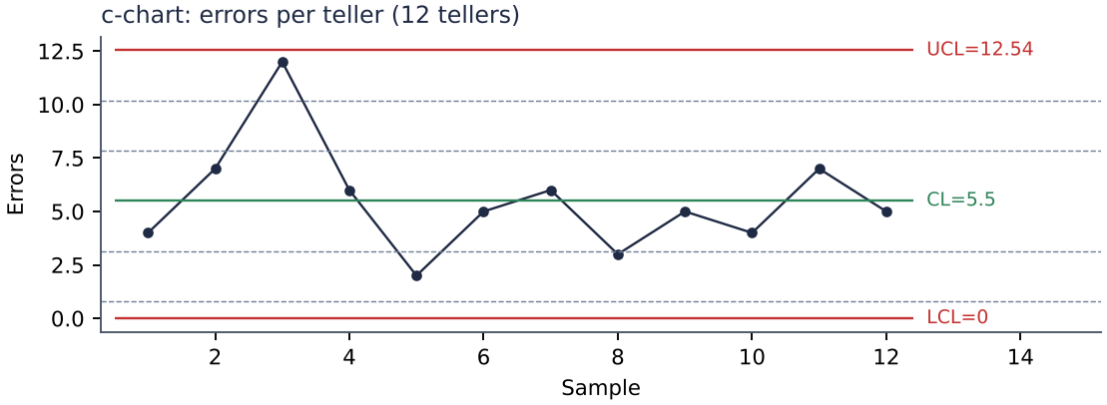
In simple words
A c-chart counts the number of defects (flaws) in units of the same size, where one unit can have many defects.

Everyday example
Count the scratches on each car door coming off the paint line, or the errors made by each bank teller in a day.

Example 8 — c-chart: bank teller errors (Day 6, slides 58–61) [Day 6]

Errors by 12 tellers: Ali 4, Chinmaya 7, Gopal 12, Joseph 6, Lakshmi 2, Matthew 5, Mary 6, Pranav 3, Soman 5, Susan 4, Tamara 7, Victor 5.

c-bar	Total 66 / 12 = **5.5**
σ	√5.5 = **2.345**
Limits	UCL = 5.5 + 3(2.345) = **12.54**; LCL = 5.5 − 7.04 → **0**
Zones	±1σ: 7.85 / 3.15; ±2σ: 10.19 / 0.81; ±3σ: 12.54 / 0
Check	Gopal (12) is inside the UCL but in zone A (beyond 2σ). Single point in zone A is not a signal by itself, but watch him / check his workload.



Answer: Process in control; no teller is statistically 'worse' — differences are within system variation. (Using the chart to blame Gopal would be a Deming 'red bead' mistake.)

13.5 The u-chart (defects per unit, varying size)

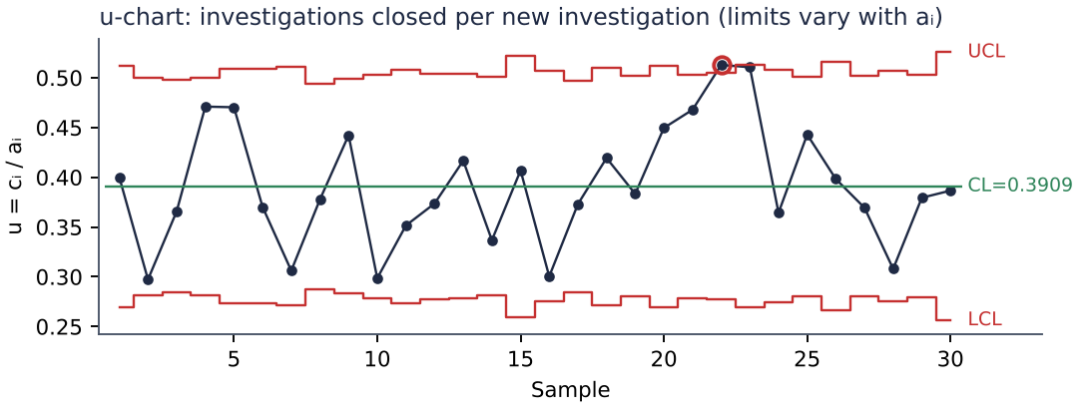
In simple words
A u-chart is a c-chart for units of different sizes: plot defects per unit of size, and let the control limits change with each sample's size.

Everyday example
Typos per page in books of different lengths — a 300-page book naturally has more typos than a 100-page book, so compare typos per page.

Example 9 — u-chart: investigations (Day 6, slides 62–65) [Day 6]

Over 30 days, a_i = new investigations (area of opportunity, varies daily) and c_i = number closed. Σa = 8,036, Σc = 3,141.

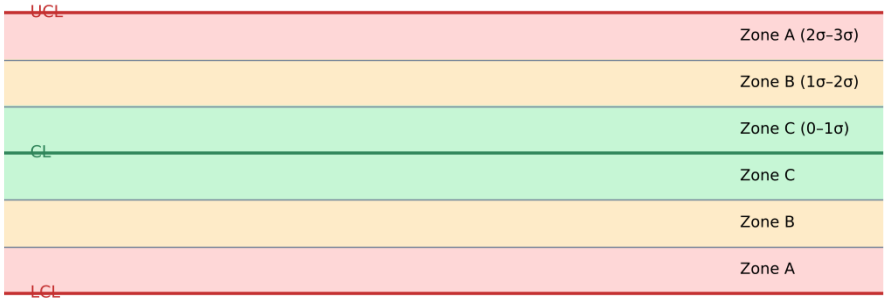
u-bar	3,141 / 8,036 = **0.3909**
Day-wise limits	UCL/LCL = $0.3909 \pm 3\sqrt{(0.3909/a_i)}$. Day 1: a = 240 → σ = √(0.3909/240) = 0.0404 → limits **0.512 / 0.270**; u_1 = 96/240 = 0.40 (inside).
Larger a_i → tighter limits	Day 8 (a = 331): limits 0.494 / 0.288. Day 30 (a = 194): 0.526 / 0.256.
Signal	Day 22: u = 139/271 = **0.513** > UCL 0.505 → out of control (unusually high closure rate — find out why and replicate it?).



13.6 Reading control charts: zones and out-of-control rules

In simple words
Besides points outside the limits, unusual patterns also signal special causes: runs on one side, steady trends, clusters near the limits, or points hugging the centre.

Everyday example
Eight days in a row your bus is later than average — no single day is extreme, but the run tells you something changed (new route, road work).



A process is **out of control** if (slide rules):

1. A point lies outside the control limits.
2. Two out of three consecutive points fall in zone A or beyond, on the same side.
3. Four out of five consecutive points fall in zone B or beyond, in the same direction.
4. Eight or more consecutive points lie on the same side of the centre line.

5. Eight or more consecutive values move upward or downward (trend).
6. There is an unusually small number of runs on either side of the centre line.
7. Thirteen consecutive points lie in zone C on either side of the centre line (too little variation — 'hugging').

Key takeaways

Rules of thumb for a NORMAL (in-control) pattern: no point beyond the limits; most points (≈2/3) near the centre line; a few — but not too many — near the limits; no non-random patterns; neither very high nor very low variability.

Limitations of attribute charts: if the % defective is very small you need huge samples; 'error of complementing causes' — special causes from more than one source can offset each other (masking).

Slido concept checks on p-charts (Day 7, slides 55–56) [Day 7]

'The p-chart of a process is given… based on it we could conclude…'

How to answer any such question: (1) any point beyond UCL/LCL? (2) any run of 8 on one side, trend of 8, 2-of-3 in zone A, 4-of-5 in zone B, 13 hugging the centre? (3) If none → 'process is stable (only common causes); whether it is acceptable depends on customer requirements (capability)'. (4) If a point is below the LCL on a p-chart → special cause, but a favourable one (or a measurement problem) — still investigate.

PART D · CAPABILITY & SPC

14. Variable Control Charts: x-bar & R, x-bar & s, Individuals & MR

Slides Day 7 (25 Oct) · Textbook Chapter 8

In simple words

For measurements (time, length, weight) we watch two things: the **average** (is the process drifting or shifting?) and the **spread** (is it becoming more erratic?). That is why variable charts come in pairs. Always read the spread chart (R or s) first — if it is out of control, the x-bar limits are not reliable.

- x-bar chart: measures the tendency of the **mean to shift or drift**. R / s chart: measures the tendency of the process to behave **more or less randomly** (spread).
- **Control procedure:** sample size small (4–5) to save cost and ensure consistency within a subgroup; x-bar & s when subgroups ≥ 10; number of samples ~25–30; sampling frequency = trade-off between cost of sampling and benefit of adjusting; limits usually 3σ but can change with the 'tightness' of control needed.

x-bar & R: CL = x-double-bar ; UCL/LCL = x-double-bar ± A_2·R-bar
R chart: CL = R-bar ; UCL = D_4·R-bar ; LCL = D_3·R-bar ($D_4 = 1 + 3d_3/d_2$)
If σ is known: x-double-bar ± 3σ/√n ; estimate σ = R-bar/d_2 ($\text{R-bar} = d_2\sigma$, $\sigma_R = d_3\sigma$)
x-bar & s: x-double-bar ± A_3·s-bar ; s chart: UCL = B_4·s-bar, LCL = B_3·s-bar ; σ = s-bar/c_4
Individuals: x-bar ± E_2·MR-bar ($E_2 = 3/d_2 = 2.66$) ; MR chart: UCL = D_4·MR-bar = 3.267·MR-bar, LCL = 0
MR-bar = $\Sigma|x_i - x_{i-1}| / (k - 1)$

n	A_2	D_3	D_4	d_2	A_3	B_3	B_4
2	1.880	0	3.267	1.128	2.659	0	3.267
3	1.023	0	2.574	1.693	1.954	0	2.568
4	0.729	0	2.282	2.059	1.628	0	2.266
5	0.577	0	2.114	2.326	1.427	0	2.089
6	0.483	0	2.004	2.534	1.287	0.030	1.970
7	0.419	0.076	1.924	2.704	1.182	0.118	1.882
8	0.373	0.136	1.864	2.847	1.099	0.185	1.815
9	0.337	0.184	1.816	2.970	1.032	0.239	1.761
10	0.308	0.223	1.777	3.078	0.975	0.284	1.716
15	0.223	0.347	1.653	3.472	0.789	0.428	1.572

Control-chart constants (Annexure B / textbook Appendix). The slide rounds D_3 to 0.08, 0.14, 0.18, 0.22 for n = 7–10.

14.1 The x-bar and R charts

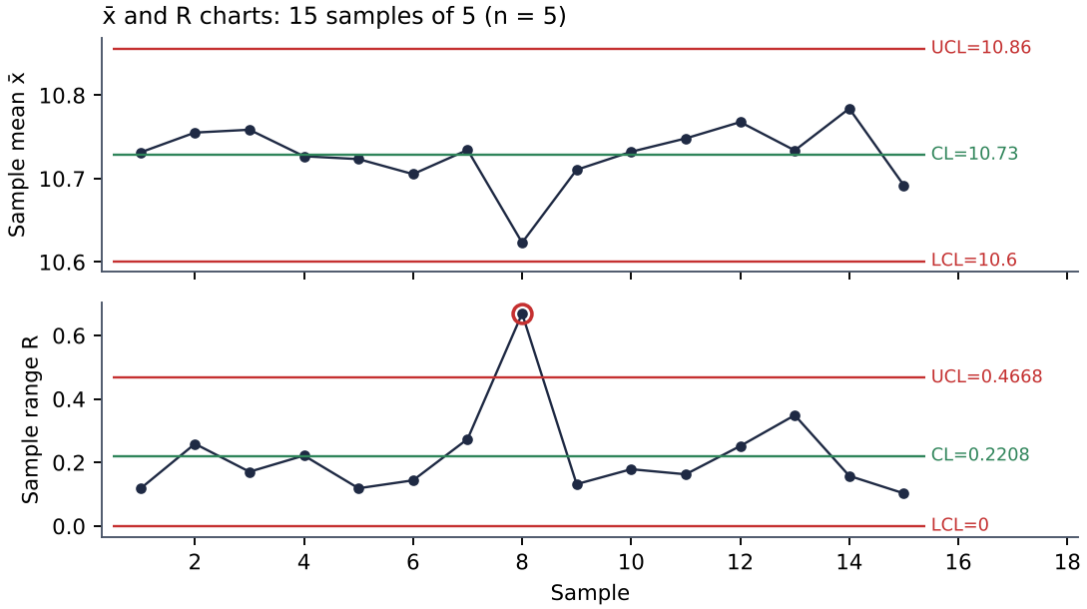
In simple words
For measurements in small samples (2–9 items), plot each sample's average (x-bar chart — is the centre shifting?) and range (R chart — is the spread changing?). Read the R chart first.

Everyday example
Every hour weigh 5 chips packets: average 52 g, range 3 g. If the average creeps up to 55 g, the filler is drifting (x-bar chart); if the range jumps to 10 g, the filler has become erratic (R chart).

Example 1 — x-bar and R charts (Day 7, slides 75–80) [Day 7]

15 samples of 5 observations each (values about 10.6–10.9, shown on the slide). Construct x-bar and R charts.

Step 1: means & ranges	For each sample compute x-bar (average of 5) and R (max − min). E.g., sample 1: x-bar = 10.731, R = 10.798 − 10.68 = 0.118. Sample 8: x-bar = 10.623, R = 10.779 − 10.11 = 0.669.
Step 1 (cont.)	Grand mean x-double-bar = **10.728**; average range R-bar = **0.2204**.
Step 2: constants	n = 5 → A_2 = 0.577, D_3 = 0, D_4 = 2.114.
Step 3–4: x-bar chart	UCL = 10.728 + 0.577 × 0.2204 = 10.728 + 0.127 = **10.855**; LCL = 10.728 − 0.127 = **10.601**. All 15 means lie inside (sample 8 = 10.623 is close to LCL).
Step 5–6: R chart	UCL = 2.114 × 0.2204 = **0.466**; LCL = 0 × 0.2204 = **0**. Sample 8's range 0.669 > 0.466 → **out of control**.
Interpret	Sample 8 contains a reading of 10.11 — likely a measurement/recording error or a special cause. Investigate, remove sample 8 if the cause is found, and recompute the limits.

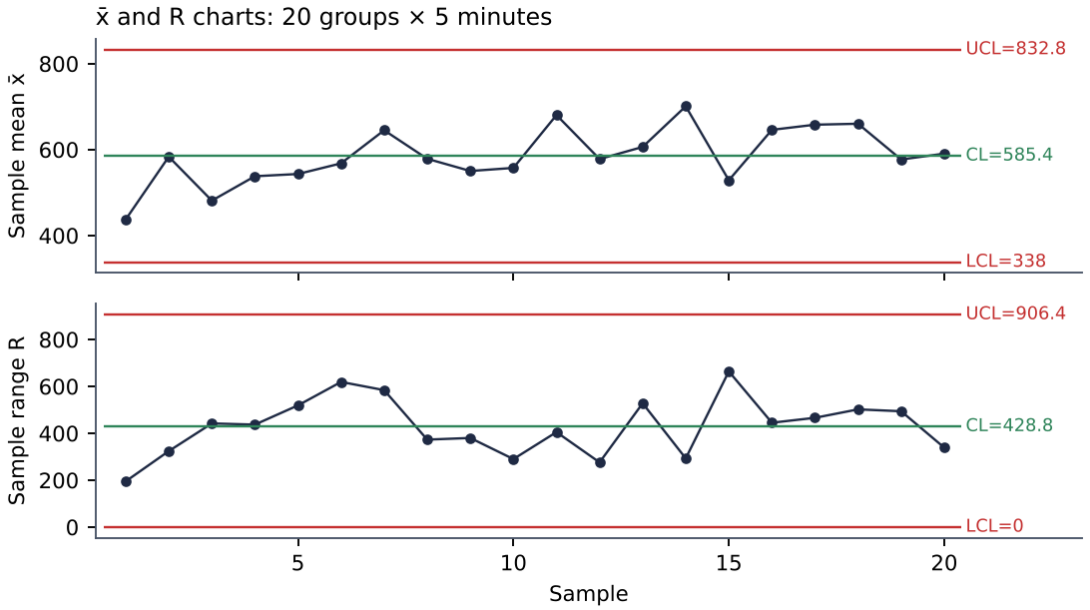


Careful — check against the slide
The slide's R-chart box shows UCL = (2.114)(0.2204) = 0.5247 and the x-bar box shows 10.871; the arithmetic actually gives 0.466 and 10.855 (0.5247 and 10.8715 match a Minitab chart with R-bar = 0.2481). The conclusion — sample 8 out of control on the R chart — is unchanged.

Example 2 — x-bar and R: 20 groups of 5 minutes (Day 7, slides 81–82) [Day 7]

20 groups × 5 readings (e.g., group 1: 474, 386, 528, 333, 465).

Means & ranges	Group 1: x-bar = 437.2, R = 528 − 333 = 195 … ; x-double-bar = **585.4**, R-bar = **428.75**.
x-bar limits	585.4 ± 0.577 × 428.75 = 585.4 ± 247.4 → **UCL 832.8, LCL 338.0**.
R limits	UCL = 2.114 × 428.75 = **906.4**; LCL = 0.
Check	All means (437–701) and ranges (195–664) are inside; no obvious runs → **in control** (but variation within groups is large; capability may be poor).



14.2 The x-bar and s charts

In simple words

Same as x-bar & R, but uses the standard deviation of each sample instead of the range. Use when samples are large (10 or more), because the range wastes information in big samples.

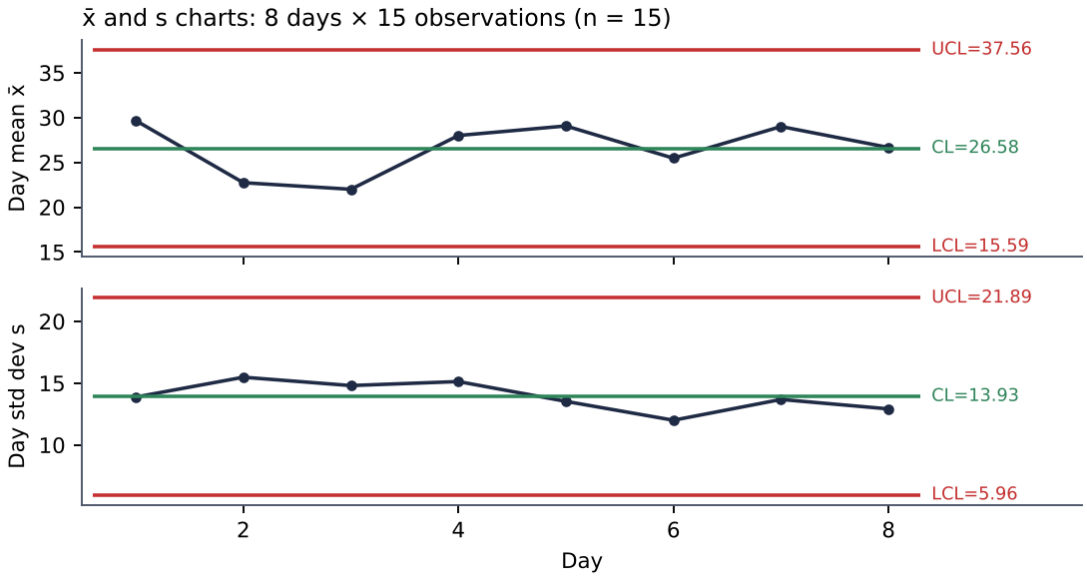
Everyday example

A school plots each class's average test score and its standard deviation (40 students per class) every month.

Example 3 — x-bar and s charts (Day 7, slides 83–88) [Day 7]

8 days × 15 observations per day (n = 15 ≥ 10 → use s).

Day statistics	Day means: 29.67, 22.73, 22.00, 28.00, 29.07, 25.47, 29.00, 26.67 → x-double-bar = **26.575**. Day std devs: 13.88, 15.48, 14.80, 15.13, 13.52, 12.00, 13.68, 12.92 → s-bar = **13.926**.
Constants n = 15	A_3 = 0.789, B_3 = 0.428, B_4 = 1.572.
x-bar chart	26.575 ± 0.789 × 13.926 = 26.575 ± 10.99 → **UCL 37.56, LCL 15.59**.
s chart	UCL = 1.572 × 13.926 = **21.89**; LCL = 0.428 × 13.926 = **5.96**.
Check	All day means and std devs are inside → **in control**.



14.3 Individuals and moving-range (I-MR) charts

In simple words

When you get only one measurement at a time (one batch, one day), plot each value and the change from the previous value (moving range).

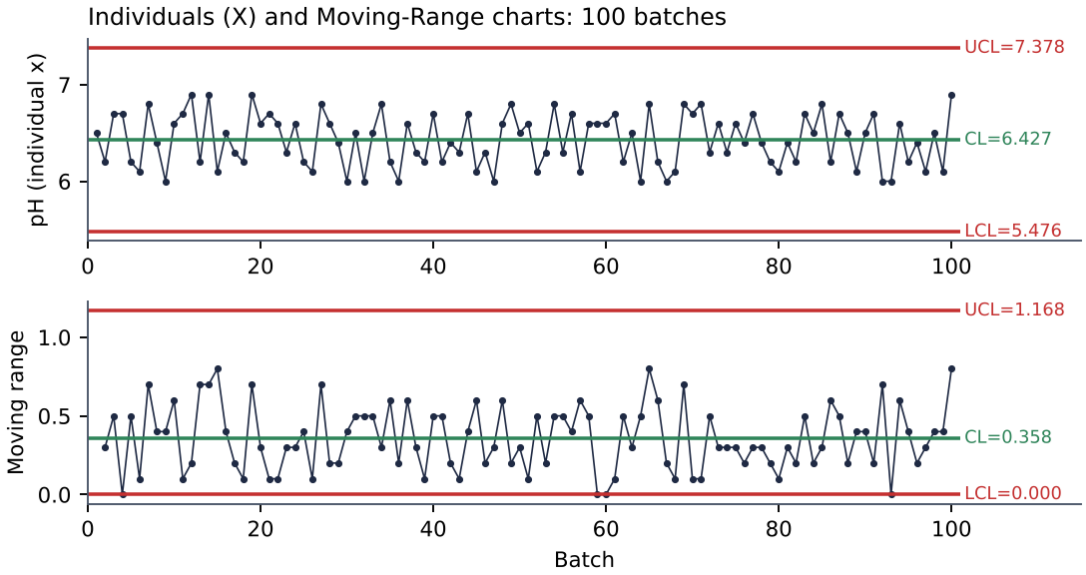
Everyday example

Your monthly electricity bill: one value per month. Plot the bill (X chart) and the month-to-month change (MR chart) to see if a jump is unusual.

Example 4 — Individuals and moving-range chart: pH of 100 batches (Day 7, slides 89–94) [Day 7]

Only one measurement per batch (cost/destructive testing, no rational subgroups). pH values for 100 batches given.

Moving ranges	$MR_i = \|x_i - x_{i-1}\|$, e.g., $\|6.2 - 6.5\| = 0.3$, $\|6.7 - 6.2\| = 0.5$ … (99 values). $\Sigma MR = 35.4$ → MR-bar = 35.4/99 = **0.3576**.
Mean	$\Sigma x = 642.7$ → x-bar = **6.427**.
MR chart	UCL = D_4 × MR-bar = 3.267 × 0.3576 = **1.168**; LCL = 0 ($D_3 = 0$ for n = 2).
X chart	UCL = 6.427 + 2.66 × 0.3576 = **7.378**; LCL = 6.427 − 0.951 = **5.476**.
Check	pH ranges 6.0–6.9, max MR 0.8 → **in control**.



- **Limitations of I-MR charts:** less sensitive; need ~100 observations (vs 25–30 subgroups); consecutive MRs are correlated, so run rules are less reliable; limits can be inflated; if more than 2/3 of MR points lie below the centre line, use the **median** MR (a few large ranges inflate the mean).
- **Revising control limits:** after a process change (improvement, new customer requirement, strategy), use trial limits for a new process until it stabilises; remove out-of-control points whose causes are found and recompute; tighten limits after each kaizen.

14.4 Special cases: periodic vs persistent disturbances

In simple words
A periodic disturbance shifts whole samples now and then, so the chart looks wildly out of control. A persistent disturbance hides inside every sample, inflating the range, widening limits and making a bad process look in control — the more dangerous case.

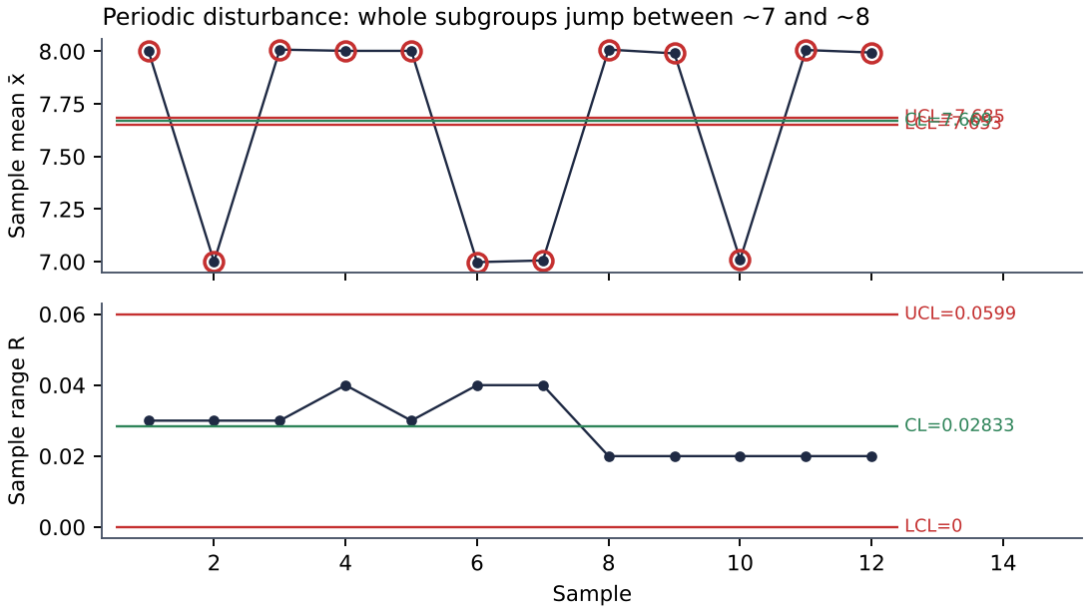
Everyday example
Periodic: the morning shift's samples all run 1 minute slower than the evening shift's. Persistent: every sample mixes one fast and one slow counter, so each sample's average looks normal and the problem never shows.

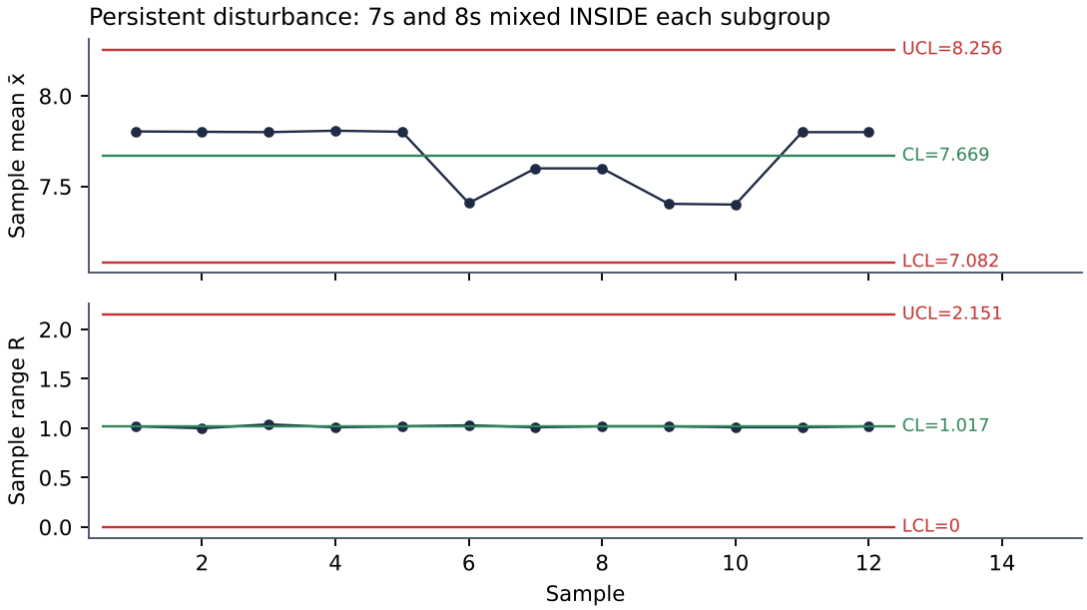
	Periodic disturbance	Persistent disturbance
How it acts	Intermittently — whole subgroups shift (**between-group** variation)	Continuously — inside every subgroup (**within-group** variation)
Effect on limits	Makes control limits **too narrow** for the subgroup statistics → many points out	Makes control limits **too wide** → everything looks 'in control'
Danger	Easy to see	**Dangerous** — hidden

Example 5 — Customer service time, 5 customers per sample, 12 samples (Day 7, slides 98–103) [Day 7]

Two data sets with the same numbers: in the first, each sample is all ≈8 or all ≈7; in the second, each sample mixes 7s and 8s.

Periodic data	x-double-bar = 7.669, R-bar = 0.028 → x-bar limits 7.669 ± 0.577 × 0.028 = **7.653 to 7.685**. The means are ≈8.00 or ≈7.00 → almost every point is out. R chart is fine. → A periodic special cause (e.g., two shifts/servers alternating).
Persistent data	Same grand mean 7.669 but R-bar ≈ 1.02 → x-bar limits 7.669 ± 0.587 = **7.08 to 8.26**; all means (7.40–7.81) inside; R chart flat ≈ 1. Looks perfectly in control!
Truth	In the second case every subgroup mixes two populations (e.g., two counters/servers, one ~7 min and one ~8 min). The chart hides it.





Answer: Learning: a chart pattern within limits is **not enough**; within-group variation is hard to find. Use rational subgroups and learn to **read chart patterns**.

14.5 Types of chart patterns and the Preston caselet

In simple words
Learn to read the shape of the chart, not only the limits: sudden shifts, gradual drifts, trends, cycles, freaks, clusters and mixtures each point to different causes.

Everyday example
A cycle that repeats every 7 points in daily data often means a day-of-week effect (e.g., Monday rush). A slow upward trend often means tool wear.

Pattern	What it looks like	Typical cause
Natural	No points beyond limits, no runs, most points near centre but no pattern	Only common causes
Sudden shift in level	Dramatic change, easily detectable	New operator, material lot, machine setting
Gradual shift	Slow continuous change	Continuous improvement or degradation
Trend	Steady increase or decrease	Tool wear, fatigue, temperature drift
Cycles	Repeating waves of low and high points	Periodic special causes (shifts, days of week, rotation)
Wild patterns / freaks	One or a few subgroups very different	Data error, sudden one-off event
Grouping / bunching	Clusters of similar points	Batch-related causes
Too few/many near centre line (hugging / stratification)	Unnaturally low variation	Mixtures inside subgroups; wrong limits
Multi-universe / mixtures	Points near both limits, few at centre (stable mixture) or proportions changing over time (unstable)	Two or more distributions (machines, suppliers) mixed

Concept-check caselet — Preston Ltd. (Day 7, slides 3-5) [Day 7]

Preston makes special inkjet paper, uses SPC and investigates any point beyond the control limits, and **revises the control chart every week based on the previous week's samples**. Sales are going up but so are rejects. All points lie within control limits and Cpk > 1. How can a process that is stable and capable produce out-of-spec items? Is anyone faking data?

Suspect 1 — moving limits	Recalculating limits every week from last week's data lets the limits **follow** a slowly drifting mean. A gradual shift/trend is absorbed into the new limits, so points never cross them — the chart is 'stable' relative to itself, not to the original target (picture: USL 10.5, UCL 10.3, mean drifting to 10.2, LSL 9.5).
Suspect 2 — capability index	Cpk computed from within-subgroup σ (short-term) or on a recent sample can look > 1 while long-term performance (Ppk, overall spread including drift) is worse.
Suspect 3 — subgrouping / mixtures	Higher sales may mean more machines, shifts or suppliers; mixing them inside subgroups inflates R-bar and widens limits (persistent disturbance).
How to find out	Freeze control limits from a stable baseline period; apply run/trend rules; plot the raw data and histograms against specs; compute Pp/Ppk; stratify by machine/shift/material; verify the measurement system (gauge R&R;).

Answer: Not necessarily faking — most likely the frequent revision of control limits (and/or poor subgrouping) hides a drift. Control limits must be revised only after a deliberate, verified process change.

PART D · CAPABILITY & SPC

15. The 7 QC Tools, 5W2H and the Gati Case

Slides Day 7 (25 Oct) · Textbook Chapter 9 (DMAIC tools)

In simple words
Seven simple pictures solve most quality problems (Ishikawa: about 95%). They help you **organise, display and analyse** process data to find areas of improvement.

Tool	What it is	Helps you to…	DMAIC phase
Flow chart / process chart	Representation of the steps in a process	Define and analyse the process	Define
Check sheet	Organised form to record data — tally marks of how often a problem occurs	Summarise and display data	Measure
Run chart	Data (often the median) plotted in time order	Discover patterns over time	Measure
Scatter diagram	Plot of two variables against each other	Identify relationships / indicator variables	Analyse
Histogram	Bar graph of frequency data (measurement on x, frequency on y)	See the distribution / variation	Measure / Analyse
Pareto chart	Bar graph of categorised frequency data in descending order + cumulative %	Identify and prioritise the vital few	Analyse
Cause-and-effect (fishbone) diagram	Identifies all possible causes of a specific effect, by category	Create and categorise a list of causes	Analyse
Control chart	Statistic over time with control limits	Measure consistency; identify out-of-control points	Control

15.1 Flow chart and check sheet

In simple words
A flow chart shows the steps of a process; a check sheet is a simple form to tally how often each problem occurs.

Everyday example
Flow chart: the steps of getting a driving licence. Check sheet: a teacher ticks each type of mistake ('spelling', 'grammar', 'punctuation') while correcting 50 essays.

- **Flow chart / process chart**: boxes and arrows showing every step of the process in order (used in Define to agree what the process actually is).
- **Check sheet** (slide): Problem A: IIII IIII II … — tally each occurrence as it happens; later feeds a Pareto chart.

15.2 Run chart and histogram

In simple words
A run chart plots data in time order to reveal trends and patterns. A histogram shows how values are distributed — their centre, spread and shape.

Everyday example
Run chart: your daily steps over a month (do weekends drop?). Histogram: how many students scored 0–10, 10–20 ... 90–100 in an exam.

- **Run chart** (slide): % of students attending sessions on each day of the week (term average) — reveals day-of-week patterns. A control chart is a run chart plus control limits.
- **Histogram** (slide): ice-cream dispensing machine — weights of servings; shows centre, spread and shape (normal, skewed, bimodal = two machines/operators mixed).

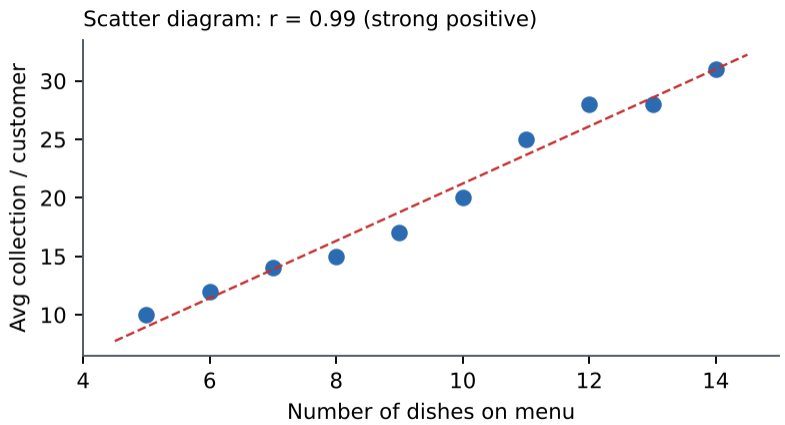
15.3 Scatter diagram

In simple words
A scatter diagram plots one variable against another to see whether they move together (correlation). Correlation alone does not prove cause.

Everyday example
Hours studied vs marks scored: points rising together show a positive relationship. Ice-cream sales and drowning cases also rise together — but both are caused by hot weather.

Scatter diagram — restaurant (Day 7, slides 13–14) [Day 7]

Number of dishes on the menu vs average collection per customer over 10 days: (12, 28), (10, 20), (8, 15), (11, 25), (13, 28), (5, 10), (6, 12), (14, 31), (7, 14), (9, 17).



Plot	x = number of dishes, y = average collection.
Pattern	Points rise together in an almost straight line.
Correlation	r ≈ **0.99** (strong positive); fitted line ≈ y = −3.26 + 2.45x — each extra dish ≈ +2.45 per customer.
Caution	Correlation is not causation — confirm with an experiment before expanding the menu (complexity has costs).

15.4 Pareto charts

In simple words
Sort problem categories from biggest to smallest and add a cumulative % line. Usually a few categories (the 'vital few') cause about 80% of the problem — fix those first.

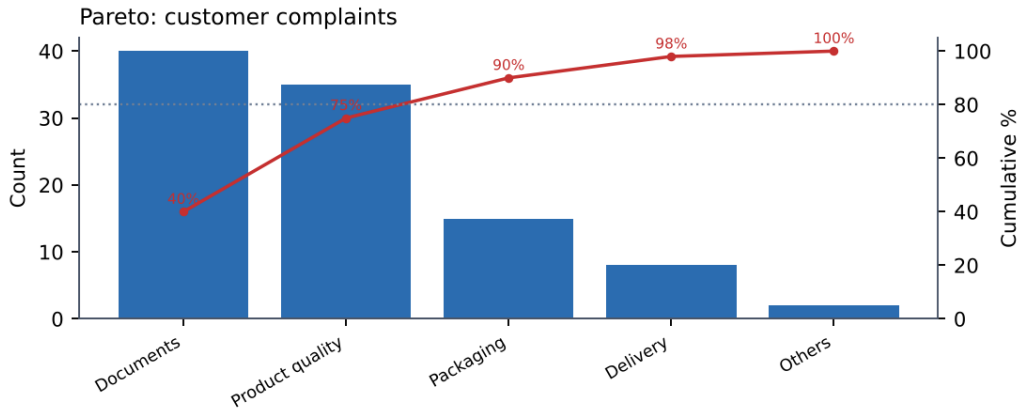
Everyday example
Of 200 hostel complaints, 90 are about Wi-Fi, 60 about food, 20 about water, and 30 spread over 10 other issues. Fixing Wi-Fi and food solves 75% of complaints.

Vilfredo Pareto (1848–1923): 20% of the population held 80% of the wealth. Juran applied it to quality: **a few causes account for most defects** (the 'vital few' vs the 'useful many'); different defects also incur different costs. Steps: (1) choose categories of defects; (2) collect and classify data; (3) arrange in descending order; (4) draw bars + cumulative % line; (5) concentrate on the tall bars. Slide example: causes of poor quality — poor design 64%, defective parts 13%, operator errors 10%, machine calibration 6%, defective materials 3%, surface abrasions 2%, wrong dimensions 2%.

Illustration — customer complaints (Day 7, slides 20–21) [Day 7]

Complaints: Documents 40, Product quality 35, Packaging 15, Delivery 8, Others 2.

Category	Count	%	Cumulative %
Documents	40	40%	40%
Product quality	35	35%	75%
Packaging	15	15%	90%
Delivery	8	8%	98%
Others	2	2%	100%

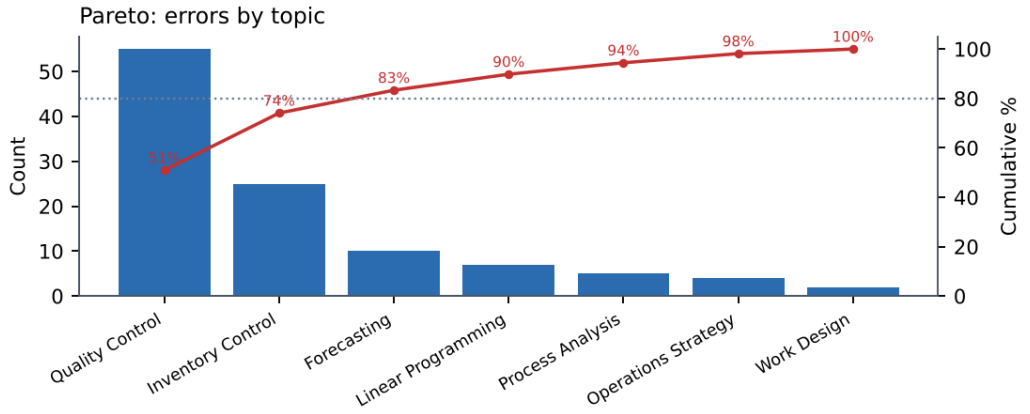


Answer: Documents and product quality together cause 75% of complaints → the sigma-improvement project should focus on these two first.

Concept check — exam errors (Day 7, slides 22–23) [Day 7]

Errors by topic: Operations Strategy 4, Forecasting 10, Quality Control 55, Work Design 2, Process Analysis 5, Inventory Control 25, Linear Programming 7. Prepare a Pareto chart.

Step 1: sort descending	QC 55, Inventory 25, Forecasting 10, LP 7, Process Analysis 5, Ops Strategy 4, Work Design 2 (total 108).
Step 2: cumulative %	50.9%, 74.1%, 83.3%, 89.8%, 94.4%, 98.1%, 100%.
Step 3: plot	Causes on the x-axis, occurrences on the y-axis, cumulative % line on a second axis.



Answer: Quality Control and Inventory Control account for ~74% of errors → re-teach these two topics first.

15.5 Cause-and-effect (fishbone / Ishikawa) diagrams

In simple words

Write the problem at the fish's head, draw main bones for categories (people, methods, machines, materials, environment, measurement), brainstorm causes on each bone, and keep asking 'why?' to reach root causes.

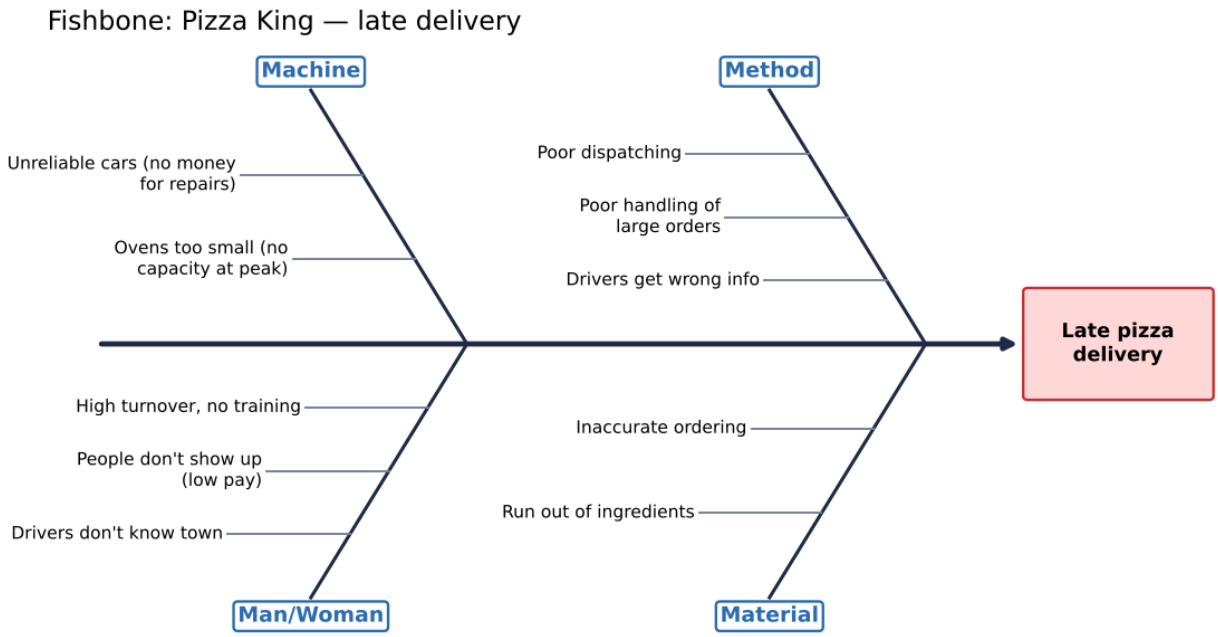
Everyday example

Problem: 'Late for class'. People — overslept; Method — no fixed routine; Machine — alarm phone battery dead; Environment — traffic. 'Why overslept?' → 'Slept at 2 am' → 'Watching videos' → root cause.

- Why: problems are complex and often unmanageable → 'divide and conquer' into small chunks; gives clarity of thought (problem structuring); helps **delineate problems from symptoms**.
- Typical bone categories: **Methods, Materials, Environment, People, Equipment** (also the 4 Ms/5 Ms: Man, Machine, Method, Material, Measurement; for services: Policies, Procedures, People, Place).
- **Steps (slide):** define the problem and put it in the 'head' on the right → identify main cause categories and draw the bones → brainstorm subordinate causes and attach them → ask 'Why?' repeatedly to reach **root causes** (look at frequency and relations) → propose solutions to the root causes.
- Illustration: Staff performance → factors: staff recruitment, job structure, career development & training, communication & teamwork → causes/actions: train managers, org chart, improve job spec, set targets, provide career path, appraisals, briefing groups, newsletter.

Illustration — Pizza King late deliveries (Day 7, slides 36–41) [Day 7]

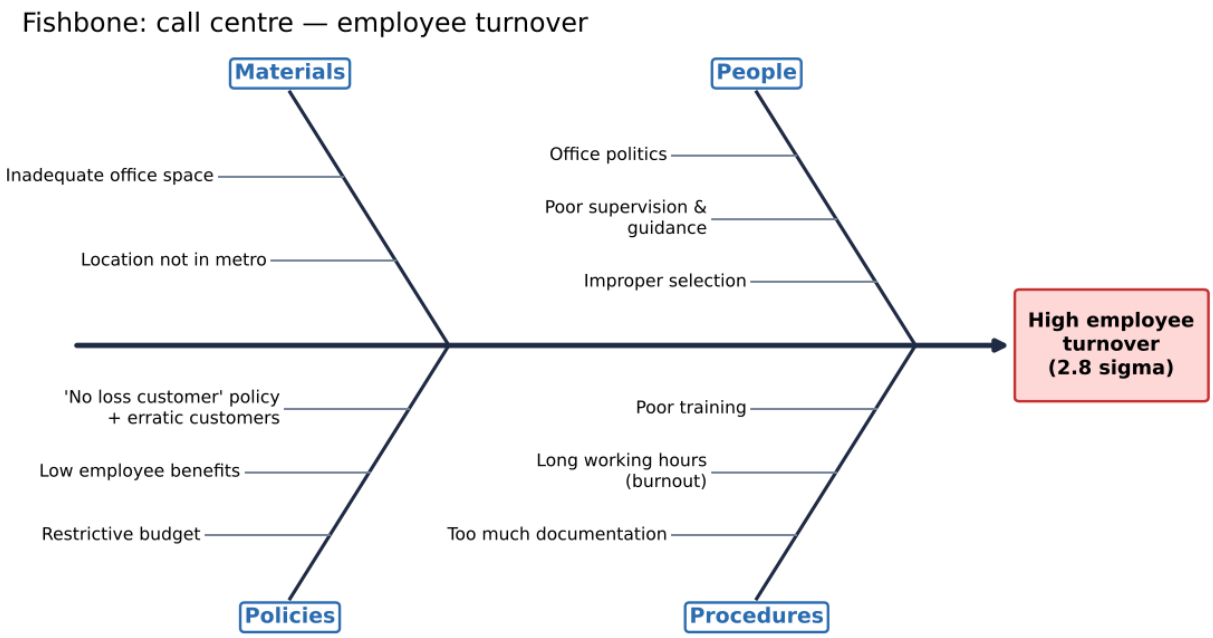
Many complaints about late delivery. Use a fishbone diagram to find root causes.



Asking 'why' repeatedly: drivers get lost ← don't know town ← high turnover ← low pay; unreliable cars ← no money for repairs / drivers' old cars; ovens too small ← no capacity for peak periods; poor dispatching ← poor training, rushed; run out of ingredients ← inaccurate ordering. Root causes such as **high turnover/low pay** and **lack of training** appear on several bones — attack them first.

Concept check 2 — call-centre turnover (Day 7, slides 42–44) [Day 7]

A call centre works at 2.8 sigma; employee turnover is 60% (industry 20%). Causes from brainstorming: inadequate office space, restrictive budget, low benefits, improper selection, location not in a metro, long working hours, office politics, poor supervision & guidance, erratic customers & 'no-loss customer' policy, poor training, too much documentation. Prepare a fishbone.



15.6 5W2H and the is–is not matrix

In simple words
5W2H is a checklist of questions (who, what, when, where, why, how, how much) to understand and challenge a process. The is–is not matrix compares where the problem occurs with where it could occur but does not, to narrow down causes.

Everyday example
Is–is not: printing errors happen on Printer 2 but not Printer 1, in the afternoon but not the morning, with recycled paper but not new paper → suspect the recycled paper in humid afternoons.

5W2H	Typical questions	Improvement questions
WHO?	Who does this? Who should be involved but is not? Who is involved but shouldn't be? Who has to approve?	Should someone else do it? Could fewer people do it? Could approvals be eliminated?
WHAT?	What is done? What is essential?	Does every step have to be done? Are steps omitted?
WHEN?	When is it started, ended, repeated?	Different time? Shorter cycle time? Less frequently?
WHERE?	Where is this activity done?	Can it be done elsewhere?
WHY?	Why do we do this?	Can it be eliminated? Can another group do it? Outsourced?
HOW?	How is this done?	Is there a better way?
HOW MUCH?	How much does it cost?	How much less could it cost?

Is–is not matrix: refine the 5W2H answers by comparing what the problem **is** with what it **is not** (where/when/who it happens vs where/when/who it could happen but doesn't). The distinctive features point to possible causes.

15.7 Case: Gati — achieving quality excellence in service delivery (Day 7)

In simple words
Gati shows the full sequence: check stability with a p-chart, check capability with Cpk, then use a Pareto chart and fishbone to pick the root causes to improve.

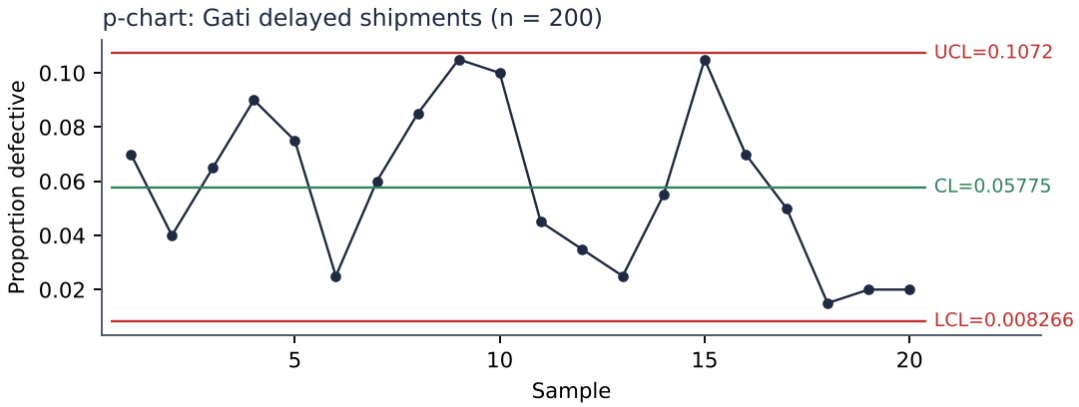
Everyday example
Like a doctor who first checks the patient is stable, then measures how far from healthy they are, then treats the two or three main causes rather than every symptom.

How to proceed: (1) Is the process stable? → (2) remove assignable causes → (3) is it capable? → (4) improve by addressing root causes.

Step 1 — Stability (p-chart of delayed shipments) [Day 7]

20 samples of 200 shipments; delayed: 14, 8, 13, 18, 15, 5, 12, 17, 21, 20, 9, 7, 5, 11, 21, 14, 10, 3, 4, 4.

p-bar	231 / 4,000 = **0.05775**
σp	√(0.05775 × 0.94225/200) = **0.0165**
Limits	UCL = 0.05775 + 3(0.0165) = **0.1072**; LCL = 0.05775 − 0.0495 = **0.0083**
Check	Highest p = 21/200 = 0.105 (< 0.1072); lowest 3/200 = 0.015 (> 0.0083) → **in control** (the last few samples are low — worth watching for an improvement shift).



Step 2 — Capability [Day 7]

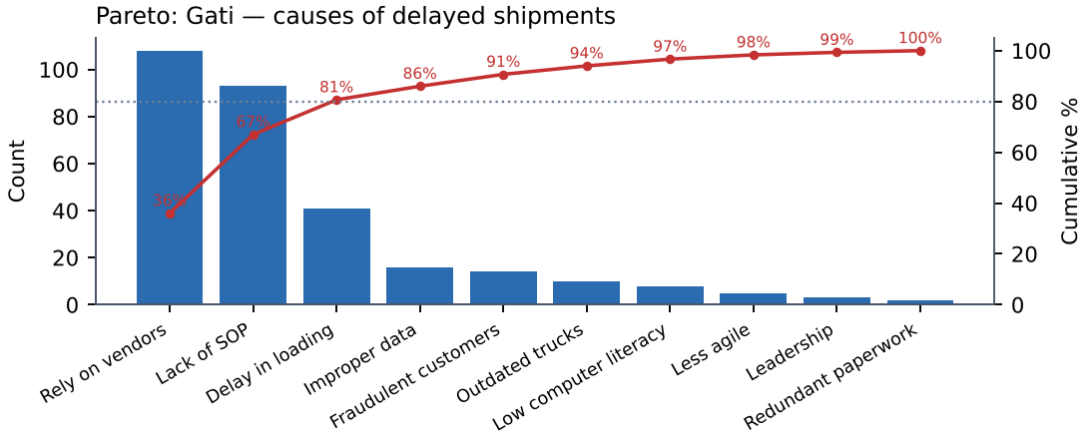
Delivery time: mean 38.5 hours, σ = 7.8 hours, upper tolerance (promise) UTL = 72 hours (one-sided).

Cpk (one-sided)	CPU = (UTL − mean)/3σ = (72 − 38.5)/(3 × 7.8) = 33.5/23.4 = **1.43**.
Defect %	P(X > 72) = 1 − NORM.DIST(72, 38.5, 7.8, 1) = 1 − 0.999991 = **0.0009%**.

Answer: The process is **stable and capable** against a 72-hour promise. Then what is the issue? The target itself — competitors/customers now expect faster delivery, and ~5.8% of shipments are still 'delayed' by Gati's own standard. Raise the target (tighter UTL) and improve.

Step 3 — Improve: which root causes to address? [Day 7]

Causes of complaints: rely on vendors 108, lack of SOP 93, delay in loading 41, improper data 16, fraudulent customers 14, outdated trucks 10, low computer literacy 8, less agile 5, leadership 3, redundant paperwork 2 (total 300).



Pareto	Rely on vendors 36%, + lack of SOP → 67%, + delay in loading → 80.7%.
Fishbone (slide)	Categories: **Nature** (accidents, unscheduled halts & slow traffic, bad weather, IT system failure); **Operational** (low maintenance, reliance on vendors, outdated small trucks, delay in loading/unloading, lack of SOP); **Staff** (inadequate incentive policy, low motivation, festivals, unpunctuality & absenteeism, poor health, low computer literacy, lack of training, less agile); **Culture/management** (traditional principles, orthodox, insouciant attitude, fixed incentives, excessive advertisement); **Reporting** (incorrect shipment, fraudulent customers, redundant paperwork, improper data entry).

Answer: Attack the vital few: reduce dependence on vendors (own fleet / vendor SLAs), write and train on SOPs, and speed up loading — these three cover ~81% of complaints. Then control with p-charts.

PART D · CAPABILITY & SPC

16. Acceptance Sampling and OC Curves

Slides Day 7 (slides 108–116)

In simple words
When a lot (batch) arrives, you can't test everything. You test a small sample and accept or reject the whole lot. Sampling always carries risk: a good lot may be rejected, or a bad lot accepted. The OC curve shows these risks.

- **Acceptance sampling**: deciding to accept or reject a lot based on the performance of a sample. Different from SPC: done **before or after** the process (incoming/outgoing), while SPC is **in-process**.
- Elements of design: how much to inspect (n), how often, and the acceptance criterion (c). Notation: **N = lot size, n = sample size, c = acceptance number** (accept if defectives ≤ c). Works for attribute and variable measures.

16.1 Types of sampling plans

In simple words
Single plans decide from one sample; double plans allow a second sample if the first is unclear; multiple and sequential plans take more, smaller steps. More steps usually mean fewer items inspected on average.

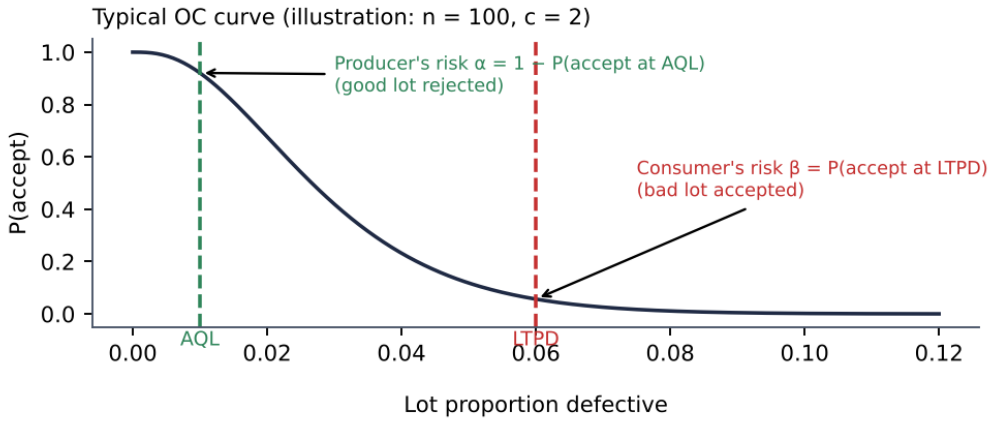
Everyday example
A shopkeeper checks 10 mangoes from a crate of 200 and rejects the crate if more than 1 is rotten (single plan). If exactly 2 are rotten, she checks 10 more before deciding (double plan).

Plan	How it works	Example
Single sampling	One sample; decide	Draw 10 cookies from a lot of 500; reject if more than 4 are bad → N = 500, n = 10, c = 4
Double sampling	If the first sample is inconclusive, take a second sample	$n_1 = 50$: accept if ≤ 1 bad, reject if ≥ 4; else take $n_2 = 50$ and accept if total ≤ 4
Multiple sampling	Several successive samples with accept/reject/continue rules	Military-standard plans
Sequential sampling	Inspect one item at a time; decide after each	Cumulative count plotted against accept/reject lines

16.2 The Operating Characteristic (OC) curve

In simple words
The OC curve shows the chance of accepting a lot for every possible defect level. Producer's risk = a good lot is rejected; consumer's risk = a bad lot is accepted.

Everyday example
Like a security guard checking a few bags at a mall entrance: sometimes he stops an innocent person (producer's risk) and sometimes misses someone carrying a prohibited item (consumer's risk).



- OC curve = probability of accepting a lot vs the lot's true proportion defective.
- **AQL (Acceptable Quality Level)**: percentage of defects the customer is willing to accept (a 'good' lot). **Producer's risk α** = probability of **rejecting a good lot** (typically 5%; e.g., 95% chance of accepting a lot with AQL defectives).
- **LTPD (Lot Tolerance Percent Defective)**: upper limit of % defective the customer will tolerate (a 'bad' lot). **Consumer's risk β** = probability of **accepting a bad lot** (typically 10%).
- Slide example: when the lot has 25% defectives, the chance of acceptance is around 18%. Between AQL and LTPD is the indifference zone.
- A steeper OC curve (larger n) discriminates better between good and bad lots.

$$P(\text{accept}) = P(X \le c) = \sum_{x=0}^{c} C(n, x) \cdot p^x \cdot (1-p)^{n-x}$$
(binomial, when N is large relative to n)

16.3 Building an OC curve and the buyer's risk

In simple words
Use the binomial formula (or table) to find P(defectives ≤ c) for a range of defect levels and plot it. Read the buyer's risk off the curve at the LTPD.

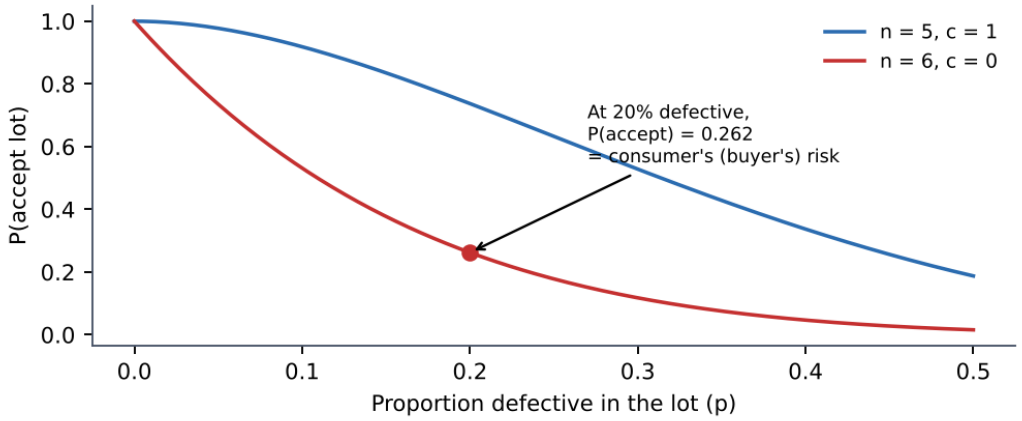
Everyday example
With n = 6 and c = 0, if a lot is 20% defective, the chance all 6 sampled items are good is $0.8^6 = 0.26$ — one in four bad lots slips through.

Developing an OC curve (Day 7, slide 114) [Day 7]

From a lot of 1,000 items, a sample of 5 is drawn; accept if no more than 1 defect is found (N = 1000, n = 5, c = 1).

Formula	$P(\text{accept}) = P(0) + P(1) = (1-p)^5 + 5p(1-p)^4$.
p = 0.05	$0.95^5 + 5(0.05)(0.95^4) = 0.7738 + 0.2036 =$ **0.9774**
p = 0.10	0.5905 + 0.3281 = **0.9185**
Others (cumulative binomial table)	p = 0.15 → 0.8352; 0.20 → 0.7373; 0.25 → 0.6328; 0.30 → 0.5282; 0.35 → 0.4284; 0.40 → 0.3370; 0.45 → 0.2562; 0.50 → 0.1875.
Plot	Plot P(accept) against p to get the OC curve (blue line below).

Careful — check against the slide
The slide's text says 'N = 1000 n = 10 c = 1' and lists 0.9974 at p = 0.05, but the curve title and table use n = 5: the correct value is 0.9774.



Issue of sampling (Day 7, slides 115–116) [Day 7]

A sample of 6 items is taken from a lot of 100; the lot is rejected if even 1 defective is found (N = 100, n = 6, c = 0). (a) What does acceptance tell you? (b) If the Lot Tolerance Percentage Defective is 20%, what is the buyer's risk?

OC values	P(accept) = $(1 - p)^6$: p = 0.05 → 0.735; 0.10 → 0.531; 0.15 → 0.377; 0.20 → 0.262; 0.25 → 0.178; 0.30 → 0.118; 0.35 → 0.075; 0.40 → 0.046.
(b) Buyer's risk	At LTPD = 20%: P(accept) = 0.8^6 = **0.2621** → a 26% chance of accepting a lot that is 20% defective (hypergeometric exact for N = 100: 0.252).
(a) Interpretation	Even with c = 0 ('zero tolerance'), an accepted lot is **not** guaranteed defect-free; with a tiny sample, bad lots are often accepted, and a lot with only 5% defectives is rejected 26% of the time (producer's risk).

Answer: Buyer's (consumer's) risk ≈ **26%**. Acceptance with n = 6, c = 0 does not mean the lot has zero defectives — sampling plans need adequate n to give low α and β. Better still: build quality into the process (SPC, Six Sigma) instead of relying on inspection (Deming point 3).

APPENDIX

17. One-page Formula Sheet & Quick Revision

Everything numeric from the course in one place

Topic	Formula / rule
Normal distribution	z = (x − μ)/σ; ±1σ = 68.27%, ±2σ = 95.45%, ±3σ = 99.73%; P(X ≤ x) = NORM.DIST(x, μ, σ, TRUE)
Standard error of the mean	σ_x̄ = σ/√n (used for x-bar control limits, e.g., Flyrock: 4/√10 = 1.265)
DPU / DPMO	DPU = defects/units; DPMO = defects/(units × opportunities) × 10^6
Sigma level	NORM.S.INV(1 − DPMO/10^6) + 1.5 (Six Sigma, shifted); without '+1.5' = long-term z
DPMO from sigma	(1 − NORM.DIST(σ-level, 1.5, 1, TRUE)) × 10^6 → 6σ = 3.4; 5σ = 233; 4σ = 6,210; 3σ = 66,807; 2σ = 308,537
Yield	TY = e^(−NPU); RTY = $Y_1 \times Y_2 \times \ldots$; p = nonconforming/inspected; NPU = nonconformances/units
Capability	Cp = (USL − LSL)/6σ; CPU = (USL − μ)/3σ; CPL = (μ − LSL)/3σ; Cpk = min(CPU, CPL)
Taguchi loss	L(x) = k(x − T)²; k = loss at tolerance ÷ (tolerance)²; EL = k(σ² + D²); smaller-better kx²; larger-better k/x²
Economic tolerance	Adjust if k·d² > cost of adjusting → d = √(cost/k) (tape example: 0.075)
Reliability	λ = failures/unit-hours; θ (MTTF/MTBF) = 1/λ; R(T) = e^(−λT); series Rs = ΠR_i; parallel Rs = 1 − $\Pi(1 - R_i)$
QFD priority	Technical importance = Σ(customer importance × relationship score 9/3/1)
NPS	% promoters (9–10) − % detractors (0–6)
Queue / Little's law	W = 1/(μ − λ); L = λW; Lq = $\lambda^2/[\mu(\mu - \lambda)]$; PCE = value-added time / total lead time; takt = available time / demand
p-chart	p-bar = Σd/Σn; limits p-bar ± 3√[p-bar(1 − p-bar)/n]; LCL < 0 → 0
np-chart	n·p-bar ± 3√[n·p-bar(1 − p-bar)]
c-chart	c-bar ± 3√c-bar
u-chart	u-bar = Σc/Σa; u-bar ± 3√(u-bar/a_i) (limits vary per subgroup)
x-bar & R	x-double-bar ± A_2R-bar; R: D_4R-bar, D_3R-bar (n = 5: A_2 0.577, D_4 2.114, D_3 0)
x-bar & s	x-double-bar ± A_3s-bar; s: B_4s-bar, B_3s-bar (n = 15: A_3 0.789, B_4 1.572, B_3 0.428)
I-MR	x-bar ± 2.66 MR-bar; MR UCL = 3.267 MR-bar; MR-bar = $\Sigma\lvert x_i - x_{i-1}\rvert/(k - 1)$
Average run length	Expected samples to detect a shift = 1/P(point out of limits) (Flyrock: 1/0.2649 = 3.77 h)
Acceptance sampling	P(accept) = Σ_0^c C(n,x)$p^x(1 - p)^{n-x}$; α = P(reject | AQL); β = P(accept | LTPD)
Pareto	Sort descending, cumulative %, focus on the categories reaching ~80%

Quick revision: one-line answers to the concept checks

Question	Answer
ISI-marked product vs ISO-certified company — which is better?	Can't say: ISI = product certification; ISO = process/system certification
Funnel experiment — lowest variation?	Rule 1: leave a stable process alone (tampering increases variation)
Common-pool game — learning?	Individually optimal ≠ collectively optimal → collaboration over competition; break silos
Sam (5 hits, scattered) vs John (1 hit, tight cluster)?	John — consistent; just re-centre him. Low variation is what matters
Aravind's secret?	Superior operations model removing waste → ~7× doctor productivity

Question	Answer
A-B-C line averaging 40/hr — enough for 300/day?	On average yes (320), but variability (starving/blocking) makes it fall short
When to start a Six Sigma project?	When voice of process (capability) is inferior to voice of customer (CTQs)
Cpk > 1 on a sample but process unstable?	Cpk is not valid — stabilise first (brink of chaos)
Stable process with errors above customer tolerance?	Stable ≠ capable → fix common causes via system change (training)
Preston: in control, Cpk > 1, yet rejects rising?	Weekly re-computed limits absorb drift; freeze limits, use run rules, check Ppk & subgrouping
n = 6, c = 0 plan; lot accepted → ?	Not proof of zero defects; at 20% defective, 26% chance of acceptance (buyer's risk)
Why 3.4 DPMO for Six Sigma?	±6σ spec with a 1.5σ long-term mean shift → 4.5σ to nearest limit → 3.4 ppm
Why 3σ control limits?	99.73% of common-cause points fall inside — a point outside is very likely a special cause

Good luck with the quiz and the end-term exam!